

Exhibit 14

U.S. Patent No. 12,675,407 Infringement Claim Chart

EXHIBIT 14

Infringement Claim Chart for Claims of U.S. Patent No. 12,675,407 (the “’407 Patent”) Based on Accused Products

Netlist contends that at least the following claims of U.S. Patent No. 12,675,407 (the “’407 Patent”) are infringed by the “’407 Accused Products,” which include at least Micron’s MTC20F2085S1RC48BAY,¹ MTC20F2085S1HC88XD1,² HPE’s P56152-001 (MTC20F2085S1RC56BD1), Lenovo 4X77A77030 (MTC20F1045S1RC48BA2), and SuperMicro’s SuperMicro’s MEM-DR516L-ER48 /802C0F222939D318AC (MTC10F1084S1RC48BA1).³ It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that infringe in the same or similar manner as discussed below.

¹On information and belief, Micron DDR5 RDIMM products starting with MTC20F2085S1RC48B, including Micron’s MTC20F2085S1RC48BAY and MTC20F2085S1RC48BR, operate in substantially the same manner with respect to infringement of the Asserted Claims. The common elements in Micron’s part numbering system indicate that the **MTC20F2085S1RC48B**, **MTC20F2085S1RC48BAY** and **MTC20F2085S1RC48BR** are each from Micron Technology (**MT**) and are the same with respect to Product Family (**C**=DDR5 SDRAM), number of DRAM die (**20**), Voltage (**F**=12V), Package Ranks (**2**), Logical Ranks (**0**), Component Config (**x8**), Module Density (**5**=32GB), Module Height (**S**=LP), Die In Package (**1** = Single Die), Module Type (**R**=288 pin/297-pin RDIMM x80), Package Temperature Capability (**C**=Commercial Temp), and DDR5 SDRAM Module Speed (**48B**=DDR5-4800B). *See* Ex. CC-93 at 1 (explaining the codes at the end of the product part number). The only differences in these product numbers are in the final characters of the part string. The two additional characters in Micron’s MTC20F2085S1RC48BAY part string (e.g., -AY) denote printed circuit board and die revisions. *See* Ex. CC-93 at 1 (explaining the codes at the end of the product part number). Similarly, Micron’s MTC20F2085S1RC48BR includes an optional “R” suffix indicating shipment packaging. *See* Ex. CC-93 at 1 (showing “R” code denotes “Retail” shipment packaging); Ex. 51 at 5, 13 (label bearing both “MTC20F2085S1RC48BR” and “MTC20F2085S1RC48B”). On information and belief, differences in die revision, PCB revision, and product shipment packaging are not material to infringement of the Asserted Claims. For example, Micron’s datasheets for DDR5 RDIMM are equally applicable to all DDR5 RDIMM devices. *See* Exs. CC-03, CC-04, CC-07.

²On information and belief, Micron’s MTC20F2085S1HC88XD1R is materially the same as the MTC20F2085S1HC88XD1 with respect to infringement of the Asserted Claims, with the optional “R” suffix indicating shipment packaging. *See* Ex. CC-93 at 1 (showing “R” code denotes shipment packaging); Ex. 53 at 5, 13 (label bearing both “MTC20F2085S1HC88XD1R” and “MTC20F2085S1HC88XD1”). For example, Micron’s product brief for DDR5 MRDIMM is equally applicable to all DDR5 MRDIMM devices. *See* Ex. CC-02.

³Netlist reserves the right to identify different or additional ’407 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
[1.pre] A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines, the memory module comprising:	To the extent the preamble is limiting, each of the '407 Accused Products⁵ includes a memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines. For example: The '407 Accused Products include a “memory module.” <i>See, e.g.,</i>

⁴This analysis is preliminary, and Netlist’s investigation is ongoing. Netlist reserves the right to amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

⁵ Analysis below uses RDIMM format as an example, but based on currently available information MRDIMMs would operate in materially the same way in all relevant parts. See Ex. CC-13 JEDEC Press Release (planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption”).

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Claim Language	'407 Accused Products ⁴
	Ex. 51 to Complaint (Micron RDIMM Importation) at 5. Ex. 51 to Complaint (Micron RDIMM Importation) at 8.

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Claim Language	'407 Accused Products ⁴
	 Ex. 51 to Complaint (Micron MRDIMM Importation) at 5.

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Claim Language	'407 Accused Products ⁴
	Ex. 51 to Complaint (Micron MRDIMM Importation) at 12. Ex. 55 to Complaint (HPE RDIMM Importation at 6).

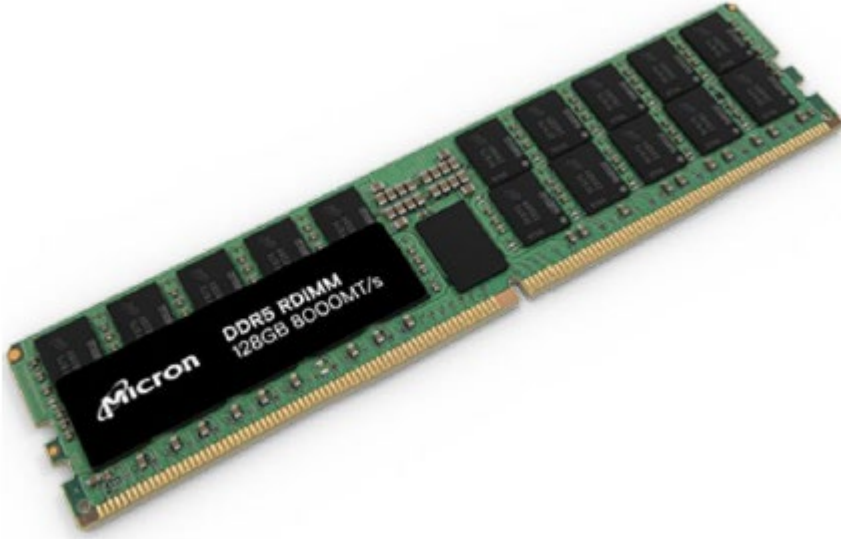
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Claim Language	'407 Accused Products ⁴
	 Ex. 55 to Complaint (Lenovo RDIMM Importation) at 4.

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Claim Language	'407 Accused Products ⁴
	Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6. Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4.

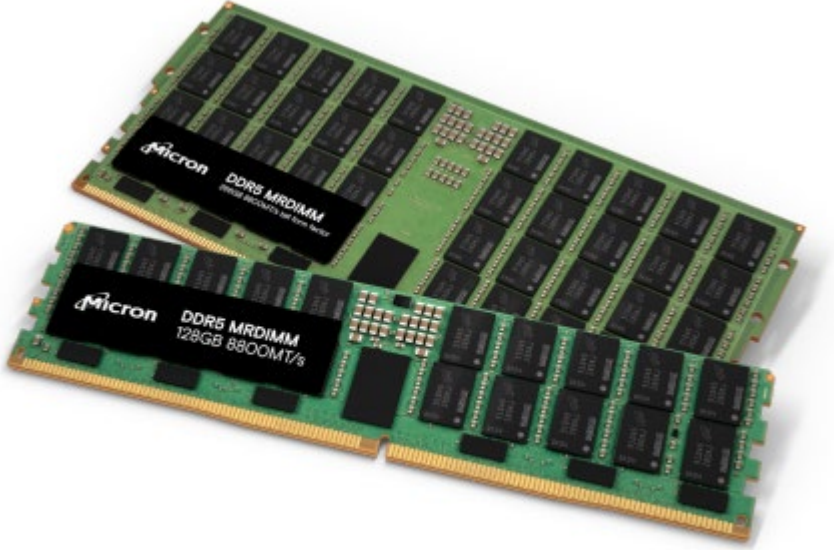
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 Ex. CC-01 <i>High-capacity DDR5 solution from Micron's leading-edge 1β (1-beta) technology</i>, Micron (“DDR5 Brochure”) at 1 (https://assets.micron.com/adobe/assets/urn:aaid:aem:6ffd17ac-e709-469d-9473-a0a904681dd9/renditions/original/as/128gb-ddr5-rdim-product-brief.pdf)

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	Figure 3: 288-Pin DDR5 RDIMM Front view 133.35 ± 0.45 MAX 133.35 ± 0.15 MIN (Area above notch) Module Center See MPN-specific data sheet addendum for component placement 31.25 3.875 TYP Pin 1 62.9 TYP 57.8 TYP Pin 144 133.35 TYP 5.95 TYP Back view See MPN-specific data sheet addendum for component placement 14.6 TYP 8.0 TYP Pin 288 57.8 TYP 4.30 TYP 5.95 TYP See Detail A 62.9 TYP Pin 145 3.0 (4X) TYP 0.5 TYP Detail A 0.85 TYP 2.60 TYP 0.25 TYP 0.60 ± 0.03 1.50 ± 0.05 3.85 ± 0.10 Notes: 1. All dimensions are in millimeters; MAX/MIN or typical (TYP) where noted. 2. Tolerance on all dimensions ±0.15 unless otherwise specified. 3. The dimensional diagram is for reference only. Ex. CC-03 Micron DDR5 SDRAM RDIMM Core Datasheet (“DDR5 RDIMM Datasheet”) at 24, available at https://www.mouser.com/datasheet/2/671/ddr5_rdimm_core-3310292.pdf

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	 Figure 1: MRDIMM low-profile form factor (foreground) and TFF (background) Ex. CC-02 <i>Unlock the power of more cores with MRDIMM</i>, Micron (“MRDIMM Product Brief”) at 1, available at https://assets.micron.com/adobe/assets/urn:aaid:aem:b19be6b5-f4bb-4367-bb0d-d583bdef5bae/renditions/original/as/mrdimm_product_brief.pdf. The memory module of the '407 Accused Products are further operable in a computer system having a memory controller and a system bus. For example, the '407 Accused Products are operable in “server platforms,” and are configured to exchange information with a “host memory controller” over a “command, address and control bus” and a “memory bus.” <i>See, e.g.,</i> Micron is first to enable the market with a monolithic 32Gb-based DDR5 128GB DRAM module that delivers the fastest speed and lowest latency for enabling generations of server platforms to come. Ex. CC-01 (DDR5 Brochure) at 1.

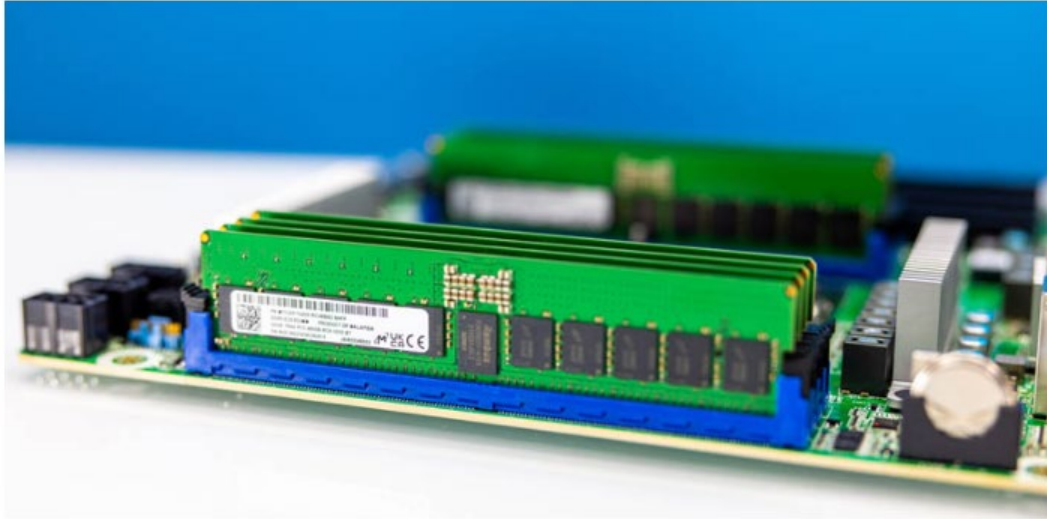
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Claim Language	'407 Accused Products ⁴
	Product Description This specification defines the electrical and mechanical requirements for 287-pin and 288-pin, 1.1V (V_{DD}) double data rate, synchronous DRAM, registered dual in-line memory modules (DDR5 SDRAM RDIMMs). These DDR5 RDIMMs are intended for use as main memory when installed in servers. Some specifications are part number-specific; refer to the module data sheet addendum of the specific Micron part number (MPN) for the complete specification. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. To reduce the electrical load on the host memory controller's command, address, and control bus, Micron's RDIMMs and LRDIMMs utilize a DDR5 registering clock driver (RCD). The RCD presents a single load to the controller while redriving signals to the DDR5 SDRAM devices, which helps enable higher densities and increase signal integrity. The RCD also provides a low-jitter, low-skew PLL that redistributes a differential clock pair to multiple differential pairs of clock outputs. Ex. CC-03 (DDR5 RDIMM Datasheet) at 12; <i>see also id.</i> at 12-13, 21 (discussing operations from the host). Micron memory modules are designed to optimize signal integrity through carefully designed terminations, controlled board impedances, routing topologies, trace length matching, and decoupling. However, good signal integrity starts at the system level. Micron encourages designers to simulate the signal characteristics of the system's memory bus to ensure adequate signal integrity of the entire memory system. Ex. CC-03 (DDR5 RDIMM Datasheet) at 17 (showing Accused Devices communicate over signals with a system memory bus).

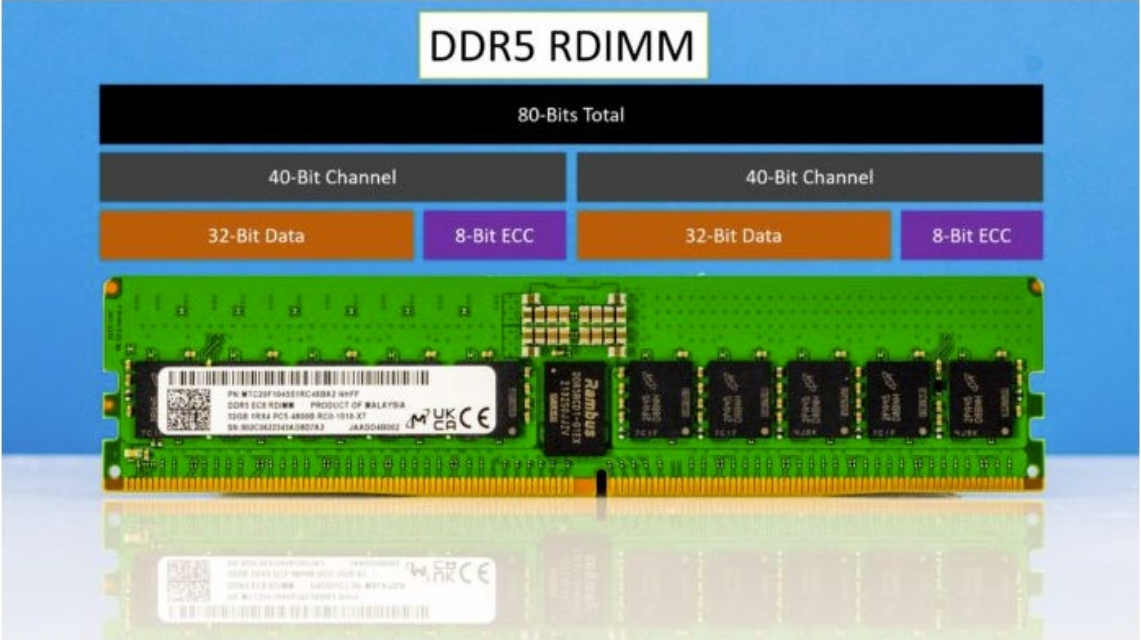
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	Meeting Next-Generation CPU Requirements At the system-level, despite only modest clock rate improvements, the transition to multicore CPU architectures has enabled continuous year-over-year compute performance gains. Signal integrity, power delivery, layout complexity, and other system-level challenges are limiting advances in CPU core frequency. Simultaneously, CPU core counts are continuously increasing, limiting the available memory bandwidth-per-core. New memory architectures beyond DDR4 SDRAM are needed to meet the next-generation bandwidth-per-core requirements. The chart below illustrates the bandwidth trend in real systems and shows how bandwidth-per-core has been relatively constant and is even beginning to trend down (as illustrated in the lower right corner of the chart). Ex. CC-08 Scott Schlachter and Brian Drake, <i>Introducing Micron® DDR5 SDRAM: More Than a Generational Update</i>, Micron (“DDR5 White Paper”) at 2, available at https://assets.micron.com/adobe/assets/urn:aaid:aem:f1706cdb-4b55-4568-84bc-b7ee42f46236/renditions/original/as/ddr5-more-than-a-generational-update-wp.pdf Micron 1β enables generations of server platforms to come Industry’s fastest speed, lowest latency DDR5 based on a 32Gb monolithic die, Micron 1β will be available in 5600MT/s, 6400MT/s, 7200MT/s, and 8000MT/s, enabling future generations of server platforms. Micron’s 1β 16Gb-based modules are available in 16GB, 32GB, and 64GB module densities for general compute applications not requiring larger capacities. Ex. CC-01 (DDR5 Brochure) at 2.

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	 <i>Micron DDR5 RDIMM In ASRock AMD EPYC Genoa Motherboard 2</i> Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 1, available at https://www.servethehome.com/why-ddr5-is-absolutely-necessary-in-modern-servers-micron/3/

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	 Micron DDR5 RDIMM Two 40 Bit Channels Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 2. Furthermore, the Accused Products are operable with a system bus including one or more clock signal lines, control/address (C/A) signal lines and data signal lines. <i>See, e.g.,</i>

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Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram

Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers.

Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (connection to system bus in red); *see also id.* at 6-7 (showing pin assignments for clock signal lines, control/address (C/A) signal lines and data signal lines).

Symbol	Type	I/O Level	Description
CK_t, CK_c	Input	V _{DD}	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.
CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchangeable between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.

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	Ex. CC-03 (DDR5 RDIMM Datasheet) at 8. <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td></tr></table> Ex. CC-03 (DDR5 RDIMM datasheet) at 9. The JEDEC DDR5 RDIMM standard connector pinout and signal description further indicates the system bus of the computer system includes one or more clock signal lines, control/address (C/A) signal lines, and data signal lines. <i>See, e.g.,</i> Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr><tr><td>PAR_A PAR_B</td><td>Parity input</td><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>Data Strobes (positive line of differential pair)</td></tr><tr><td>CK_t</td><td>Clock (true/positive)</td><td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>Data Strobes (negative line of differential pair)</td></tr><tr><td>CK_c</td><td>Clock (complement/negative)</td><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr></table>	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	V _{DD}	Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input/ Output	V _{DD}	Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)	CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)	CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.
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	Ex. CC-88 (JEDEC 305B) at 3 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)). Table 4 — Input/Output Functional Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CK_t, CK_c.</td><td>Input</td><td>VDD</td><td>Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Input</td><td>VDD</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”) and control/address (C/A) signal lines (e.g., “CA[]_A” and “CA[]_B”)). <table><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>VDD</td><td>ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)).	Symbol	Type	I/O Levels	Function	CK_t, CK_c.	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating
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Claim Language	'407 Accused Products ⁴																	
	Table 4 — Input/Output Functional Description (cont'd) <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">VDD</td><td rowspan="2">Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 5.	Symbol	Type	I/O Levels	Function	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
Symbol	Type	I/O Levels	Function															
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.															
DQS[9:0]_A_c DQS[9:0]_B_c																		
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.															
TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.															

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	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label		
	VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c		
	RFU	2	146	VIN_BULK	VSS	75	219	VSS		
	VIN_MGMT	3	147	PCAMP			Key			
	HSCL	4	148	HSA						
	HSDA	5	149	RFU	CA0_B	76	220	RFU / No Pin Present		
	VSS	6	150	RFU	VSS	77	221	CA1_B		
	DQ0_A	7	151	VSS	CA2_B	78	222	VSS		
	VSS	8	152	DQ2_A	VSS	79	223	CA3_B		
	DQ1_A	9	153	VSS	CA4_B	80	224	VSS		
	VSS	10	154	DQ3_A	VSS	81	225	CA5_B		
	DQS0_A_t	11	155	VSS	CA6_B	82	226	VSS		
	DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	VSS	83	227	PAR_B		
	VSS	13	157	DQS5_A_t, TDQS5_A_t	CS0_B_n	84	228	VSS		
	DQ4_A	14	158	VSS	VSS	85	229	CS1_B_n		
	VSS	15	159	DQ6_A	DLBDQ	86	230	VSS		
	DQ5_A	16	160	VSS	DLBDQS	87	231	RFU		
	VSS	17	161	DQ7_A	VSS	88	232	RFU		
	DQ8_A	18	162	VSS	CB4_B	89	233	VSS		
	VSS	19	163	DQ10_A	VSS	90	234	CB6_B		
	DQ9_A	20	164	VSS	CB5_B	91	235	VSS		
	VSS	21	165	DQ11_A	VSS	92	236	CB7_B		
	DQS1_A_t	22	166	VSS	DQS9_B_t, TDQS9_B_t	93	237	VSS		
	DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c		
	VSS	24	168	DQS6_A_t, TDQS6_A_t	VSS	95	239	DQS4_B_t		
	DQ12_A	25	169	VSS	CB0_B	96	240	VSS		
	VSS	26	170	DQ14_A	VSS	97	241	CB2_B		
	DQ13_A	27	171	VSS	CB1_B	98	242	VSS		
	VSS	28	172	DQ15_A	VSS	99	243	CB3_B		
	DQ16_A	29	173	VSS	DQ0_B	100	244	VSS		
	VSS	30	174	DQ18_A	VSS	101	245	DQ2_B		
	DQ17_A	31	175	VSS	DQ1_B	102	246	VSS		
	VSS	32	176	DQ19_A	VSS	103	247	DQ3_B		
	DQS2_A_t	33	177	VSS	DQS0_B_t	104	248	VSS		
	DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c		
	VSS	35	179	DQS7_A_t, TDQS7_A_t	VSS	106	250	DQS5_B_t, TDQS5_B_t		
	DQ20_A	36	180	VSS	DQ4_B	107	251	VSS		
	VSS	37	181	DQ22_A	VSS	108	252	DQ6_B		
	DQ21_A	38	182	VSS	DQ5_B	109	253	VSS		

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Claim Language	'407 Accused Products ⁴
	Ex. CC-88 (JEDEC 305B) at 6 (illustrating clock signal lines (<i>e.g.</i> , “CK_t” and “CK_c”), control/address (C/A) signal lines (<i>e.g.</i> , “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (<i>e.g.</i> , “DQ[]_A” and “DQ[]_B”)).

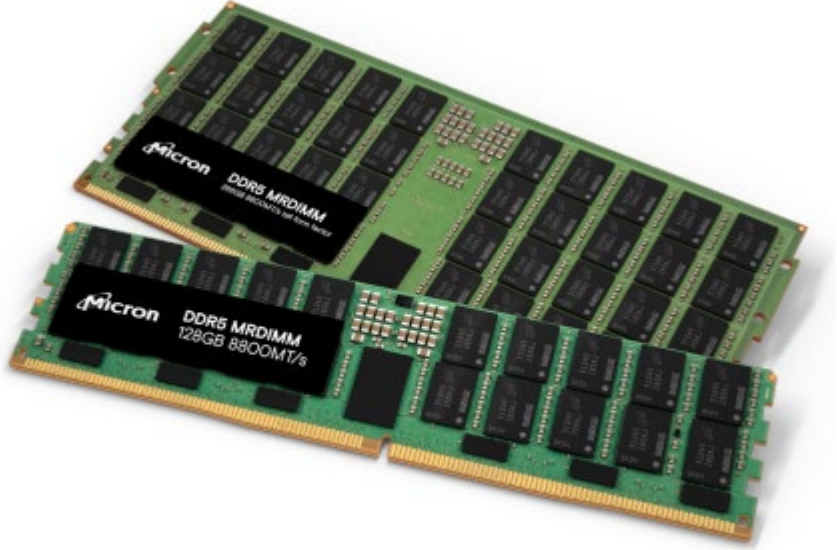
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Claim Language	'407 Accused Products ⁴											
	Front Side		Pin	Pin	Back side		Front Side		Pin	Pin	Back side	
	Pin Label				Pin Label		Pin Label				Pin Label	
	VSS		39	183	DQ23_A		VSS		110	254	DQ7_B	
	DQ24_A		40	184	VSS		DQ8_B		111	255	VSS	
	VSS		41	185	DQ26_A		VSS		112	256	DQ10_B	
	DQ25_A		42	186	VSS		DQ9_B		113	257	VSS	
	VSS		43	187	DQ27_A		VSS		114	258	DQ11_B	
	DQS3_A_t		44	188	VSS		DQS1_B_t		115	259	VSS	
	DQS3_A_c		45	189	DQS8_A_c, TDQS8_A_c		DQS1_B_c		116	260	DQS6_B_c, TDQS6_B_c	
	VSS		46	190	DQS8_A_t, TDQS8_A_t		VSS		117	261	DQS6_B_t, TDQS6_B_t	
	DQ28_A		47	191	VSS		DQ12_B		118	262	VSS	
	VSS		48	192	DQ30_A		VSS		119	263	DQ14_B	
	DQ29_A		49	193	VSS		DQ13_B		120	264	VSS	
	VSS		50	194	DQ31_A		VSS		121	265	DQ15_B	
	CB0_A		51	195	VSS		DQ16_B		122	266	VSS	
	VSS		52	196	CB2_A		VSS		123	267	DQ18_B	
	CB1_A		53	197	VSS		DQ17_B		124	268	VSS	
	VSS		54	198	CB3_A		VSS		125	269	DQ19_B	
	DQS4_A_t		55	199	VSS		DQS2_B_t		126	270	VSS	
	DQS4 A c		56	200	DQS9 A c, TDQS9 A c		DQS2 B c		127	271	DQS7 B c, TDQS7 B c	
	VSS		57	201	DQS9_A_t, TDQS9_A_t		VSS		128	272	DQS7_B_t, TDQS7_B_t	
	CB4_A		58	202	VSS		DQ20_B		129	273	VSS	
	VSS		59	203	CB6_A		VSS		130	274	DQ22_B	
	CB5_A		60	204	VSS		DQ21_B		131	275	VSS	
	VSS		61	205	CB7_A		VSS		132	276	DQ23_B	
	ALERT_n		62	206	VSS		DQ24_B		133	277	VSS	
	VSS		63	207	RESET_n		VSS		134	278	DQ26_B	
	CS0_A_n		64	208	VSS		DQ25_B		135	279	VSS	
	VSS		65	209	CS1_A_n		VSS		136	280	DQ27_B	
	CA0_A		66	210	VSS		DQS3_B_t		137	281	VSS	
	VSS		67	211	CA1_A		DQS3_B_c		138	282	DQS8_B_c, TDQS8_B_c	
	CA2_A		68	212	VSS		VSS		139	283	DQS8_B_t, TDQS8_B_t	
	VSS		69	213	CA3 A		DQ28 B		140	284	VSS	
	CA4_A		70	214	VSS		VSS		141	285	DQ30_B	
	VSS		71	215	CA5_A		DQ29_B		142	286	VSS	
	CA6_A		72	216	VSS		VSS		143	287	DQ31_B	
	VSS		73	217	CK_t		RFU		144	288	VSS	
NOTE 1 Individual Annex standards for RDIMM define pin use where multiple functions are possible.												
NOTE 2 Pin 220 for 8000 Mbps and faster RDIMM raw cards do not have the edge pin present.												

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Claim Language	'407 Accused Products ⁴
	Ex. CC-88 (JEDEC 305B) at 7 (illustrating clock signal lines (<i>e.g.</i>, “CK_t” and “CK_c”), control/address (C/A) signal lines (<i>e.g.</i>, “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (<i>e.g.</i>, “DQ[]_A” and “DQ[]_B”)).
[1.a] a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;	The '407 Accused Products comprise a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus. For example: The '407 Accused Products include a printed circuit board (PCB) having connectors. <i>See, e.g.</i>: Ex. CC-01 (DDR5 Brochure) at 1 (showing green printed circuit board, highlight indicates connectors that provide electrical connections).

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Claim Language	'407 Accused Products ⁴
	 Figure 1: MRDIMM low-profile form factor (foreground) and TFF (background) Ex. CC-02 (MRDIMM Product Brief) at 1.

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Claim Language	'407 Accused Products ⁴
	Ex. 51 to Complaint (Micron RDIMM Importation) at 5. Ex. 51 to Complaint (Micron RDIMM Importation) at 8.

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Claim Language	'407 Accused Products ⁴
	 Ex. 51 to Complaint (Micron MRDIMM Importation) at 5.

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Claim Language	'407 Accused Products ⁴
	Ex. 51 to Complaint (Micron MRDIMM Importation) at 12. Ex. 51 to Complaint (HPE RDIMM Importation at 6).

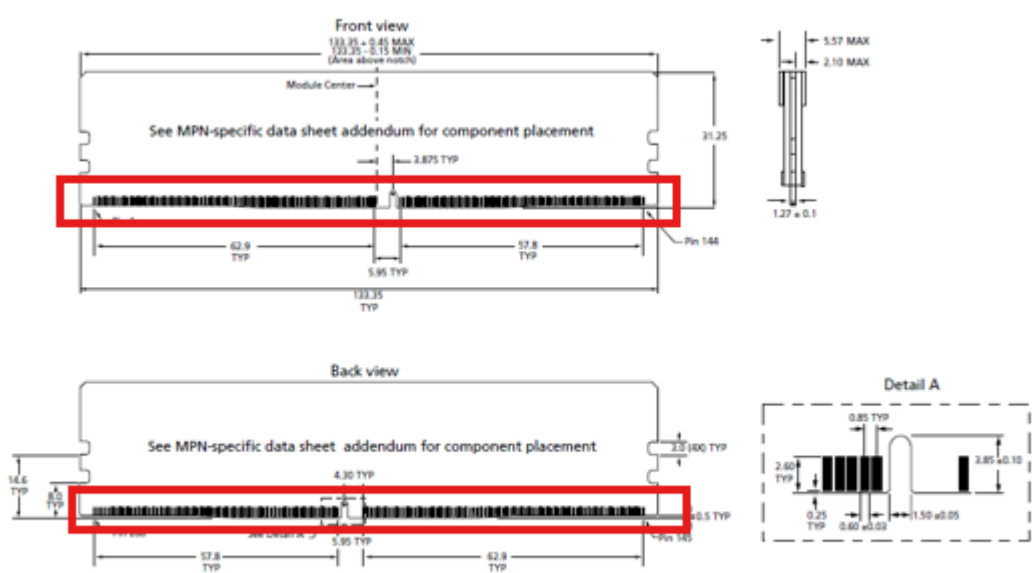
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Claim Language	'407 Accused Products ⁴
	 Ex. 57 to Complaint (Lenovo RDIMM Importation) at 4.

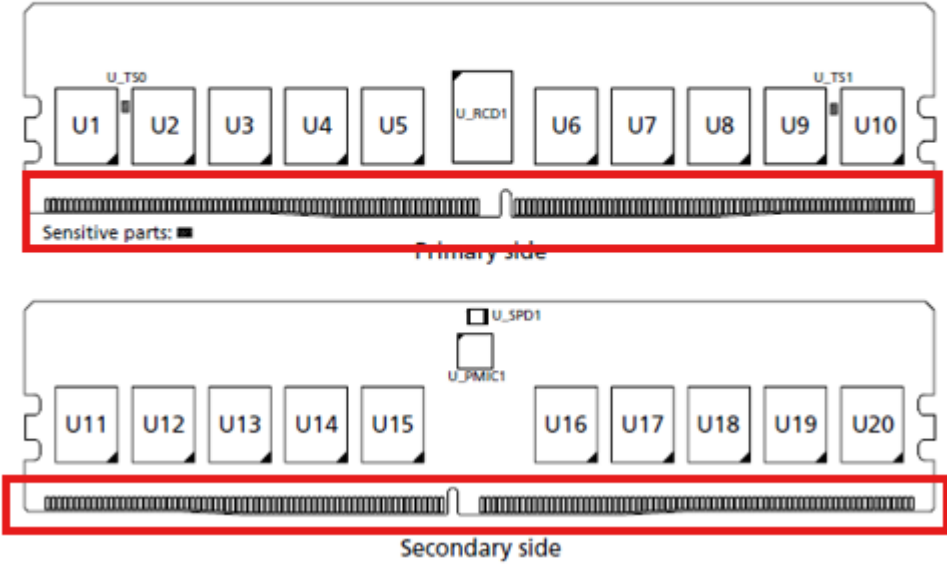
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Claim Language	'407 Accused Products ⁴
	 A photograph of a green Lenovo RDIMM memory module. The module is populated with several black memory chips. Two white labels are visible: one on the left with a barcode and text including 'DDR5 ECC RDIMM', '32GB 1Rx8 PC5-4800B-RCC-1010-XT', and 'MUKCE'; and another on the right with 'Lenovo' branding, 'P) Lenovo P/N: SM37B27362', 'OPT: 4X77A77030 / 4X77A81438', 'Mant.Date: 2023-10-27', and 'FRU: 03KL457'. Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6.  A photograph of a green SuperMicro RDIMM memory module. The module is populated with several black memory chips. Two white labels are visible: one on the left with a barcode and text including 'DDR5 ECC RDIMM', '16GB 1Rx8 PC5-4800B-RDD-1010-XT', and 'MUKCE'; and another on the right with 'SUPERMICRO' branding, 'WARRANTY VOID IF REMOVED', and the part number '802CDF222939D318AC'. Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4.

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Claim Language	'407 Accused Products ⁴
	Figure 3: 288-Pin DDR5 RDIMM  Notes: 1. All dimensions are in millimeters; MAX/MIN or typical (TYP) where noted. 2. Tolerance on all dimensions ± 0.15 unless otherwise specified. 3. The dimensional diagram is for reference only. Ex. CC-03 (DDR5 RDIMM Datasheet) at 24 (showing the edge connections).

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Claim Language	'407 Accused Products ⁴
	Figure 1: 288-Pin DDR5 RDIMM (R/C-E0)  Ex. CC-07 DDR5 SDRAM RDIMM Addendum, MTC20F2085S1RC, Micron, CCM005-80224854-9 (Rev. F, Dec 2022) (“DDR5 RDIMM Addendum”) at 1 (showing edge connections). Furthermore, the connectors provide electrical connections between the memory module and the system bus. For example, the connectors in the Accused Products are assigned to pin assignments for connecting with lines in the system bus, including clock signal lines, control/address (C/A) signal lines and data signal lines. <i>See, e.g.,</i>

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Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (red illustrate the connectors); see also <i>id.</i> at 6-7.

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Pin Assignments

The pin assignment table below is a comprehensive list of all possible pin assignments for DDR5 RDIMM modules. Certain pins may not apply for a specific part number. Refer to the functional block diagram in the module data sheet addendum for a specific MPN.

Table 3: Pin Assignments

288-Pin DDR5 RDIMM Front								288-Pin DDR5 RDIMM Back							
Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol		
1	VIN_BULK	37	V _{SS}	73	V _{SS}	109	DQ5_B	145	VIN_BULK	181	DQ22_A	217	CK_t		
2	RFU	38	DQ21_A	74	PAR_A	110	V _{SS}	146	VIN_BULK	182	V _{SS}	218	CK_c		
3	VIN_MGMT	39	V _{SS}	75	V _{SS}	111	DQ8_B	147	PCAMP	183	DQ23_A	219	V _{SS}		
4	HSCL	40	DQ24_A	76	CA0_B	112	V _{SS}	148	HSA	184	V _{SS}	220	RFU, NPP ¹		
5	HSDA	41	V _{SS}	77	V _{SS}	113	DQ9_B	149	RFU	185	DQ26_A	221	CA1_B		
6	V _{SS}	42	DQ25_A	78	CA2_B	114	V _{SS}	150	RFU	186	V _{SS}	222	V _{SS}		
7	DQ0_A	43	V _{SS}	79	V _{SS}	115	DQ51_B_t	151	V _{SS}	187	DQ27_A	223	CA3_B		
8	V _{SS}	44	DQ53_A_t	80	CA4_B	116	DQ51_B_c	152	DQ2_A	188	V _{SS}	224	V _{SS}		
9	DQ1_A	45	DQ53_A_c	81	V _{SS}	117	V _{SS}	153	V _{SS}	189	DQ58_A_c, TDQ58_A_c	225	CA5_B		
10	V _{SS}	46	V _{SS}	82	CA6_B	118	DQ12_B	154	DQ3_A	190	DQ58_A_t, TDQ58_A_t	226	V _{SS}		
11	DQ50_A_t	47	DQ28_A	83	V _{SS}	119	V _{SS}	155	V _{SS}	191	V _{SS}	227	PAR_B		
12	DQ50_A_c	48	V _{SS}	84	CS0_B_n	120	DQ13_B	156	DQ55_A_c, TDQ55_A_c	192	DQ30_A	228	V _{SS}		
13	V _{SS}	49	DQ29_A	85	V _{SS}	121	V _{SS}	157	DQ55_A_t, TDQ55_A_t	193	V _{SS}	229	CS1_B_n		
14	DQ4_A	50	V _{SS}	86	LBDQ	122	DQ16_B	158	V _{SS}	194	DQ31_A	230	V _{SS}		
15	V _{SS}	51	CB0_A	87	LBDQS	123	V _{SS}	159	DQ6_A	195	V _{SS}	231	RFU		
16	DQ5_A	52	V _{SS}	88	V _{SS}	124	DQ17_B	160	V _{SS}	196	CB2_A	232	RFU		
17	V _{SS}	53	CB1_A	89	CB4_B	125	V _{SS}	161	DQ7_A	197	V _{SS}	233	V _{SS}		
18	DQ8_A	54	V _{SS}	90	V _{SS}	126	DQ52_B_t	162	V _{SS}	198	CB3_A	234	CB6_B		
19	V _{SS}	55	DQ54_A_t	91	CB5_B	127	DQ52_B_c	163	DQ10_A	199	V _{SS}	235	V _{SS}		
20	DQ9_A	56	DQ54_A_c	92	V _{SS}	128	V _{SS}	164	V _{SS}	200	DQ59_A_c, TDQ59_A_c	236	CB7_B		
21	V _{SS}	57	V _{SS}	93	DQ59_B_t, TDQ59_B_t	129	DQ20_B	165	DQ11_A	201	DQ59_A_t, TDQ59_A_t	237	V _{SS}		
22	DQ51_A_t	58	CB4_A	94	DQ59_B_c, TDQ59_B_c	130	V _{SS}	166	V _{SS}	202	V _{SS}	238	DQ54_B_c		
23	DQ51_A_c	59	V _{SS}	95	V _{SS}	131	DQ21_B	167	DQ56_A_c, TDQ56_A_c	203	CB6_A	239	DQ54_B_t		
24	V _{SS}	60	CB5_A	96	CB0_B	132	V _{SS}	168	DQ56_A_t, TDQ56_A_t	204	V _{SS}	240	V _{SS}		
25	DQ12_A	61	V _{SS}	97	V _{SS}	133	DQ24_B	169	V _{SS}	205	CB7_A	241	CB2_B		
26	V _{SS}	62	ALERT_n	98	CB1_B	134	V _{SS}	170	DQ14_A	206	V _{SS}	242	V _{SS}		
27	DQ13_A	63	V _{SS}	99	V _{SS}	135	DQ25_B	171	V _{SS}	207	RESET_n	243	CB3_B		

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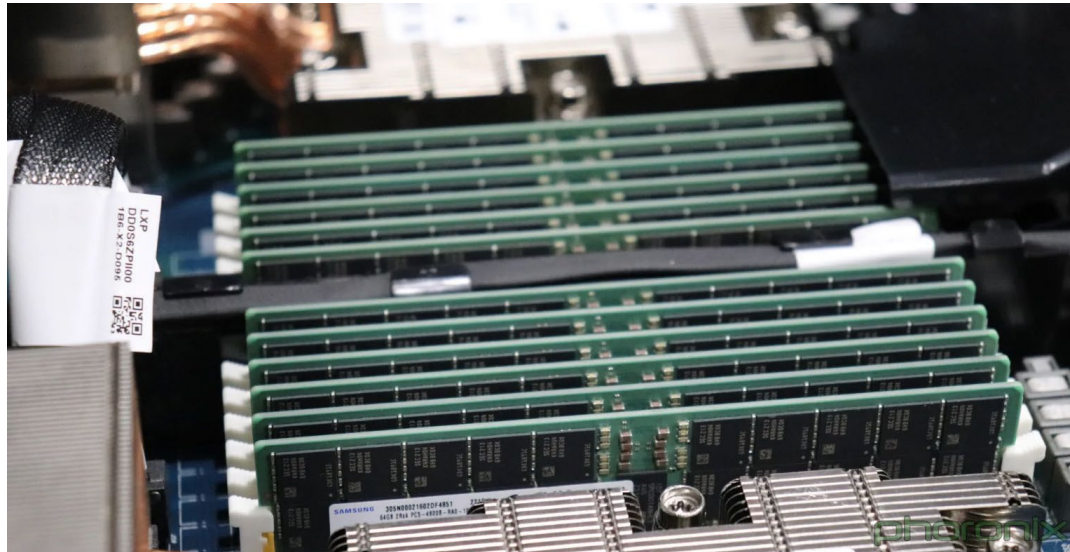
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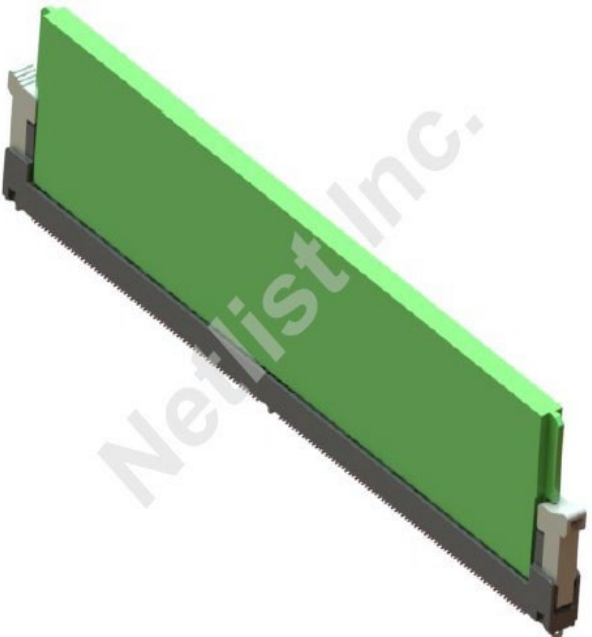
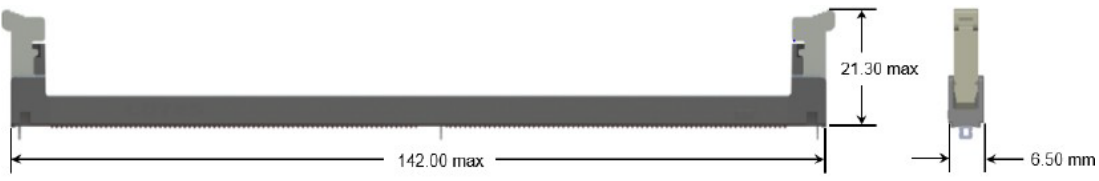
U.S. Patent No. 12,675,407 Infringement Claim Chart

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	Table 3: Pin Assignments (Continued) <table><thead><tr><th colspan="8">288-Pin DDR5 RDIMM Front</th><th colspan="8">288-Pin DDR5 RDIMM Back</th></tr><tr><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th><th>Pin</th><th>Symbol</th></tr></thead><tbody><tr><td>28</td><td>V_{SS}</td><td>64</td><td>CS0_A_n</td><td>100</td><td>DQ0_B</td><td>136</td><td>V_{SS}</td><td>172</td><td>DQ15_A</td><td>208</td><td>V_{SS}</td><td>244</td><td>V_{SS}</td><td>280</td><td>DQ27_B</td></tr><tr><td>29</td><td>DQ16_A</td><td>65</td><td>V_{SS}</td><td>101</td><td>V_{SS}</td><td>137</td><td>DQ53_B_t</td><td>173</td><td>V_{SS}</td><td>209</td><td>CS1_A_n</td><td>245</td><td>DQ2_B</td><td>281</td><td>V_{SS}</td></tr><tr><td>30</td><td>V_{SS}</td><td>66</td><td>CA0_A</td><td>102</td><td>DQ1_B</td><td>138</td><td>DQ53_B_c</td><td>174</td><td>DQ18_A</td><td>210</td><td>V_{SS}</td><td>246</td><td>V_{SS}</td><td>282</td><td>DQ58_B_c, TDQ58_B_c</td></tr><tr><td>31</td><td>DQ17_A</td><td>67</td><td>V_{SS}</td><td>103</td><td>V_{SS}</td><td>139</td><td>V_{SS}</td><td>175</td><td>V_{SS}</td><td>211</td><td>CA1_A</td><td>247</td><td>DQ3_B</td><td>283</td><td>DQ58_B_t, TDQ58_B_t</td></tr><tr><td>32</td><td>V_{SS}</td><td>68</td><td>CA2_A</td><td>104</td><td>DQ50_B_t</td><td>140</td><td>DQ28_B</td><td>176</td><td>DQ19_A</td><td>212</td><td>V_{SS}</td><td>248</td><td>V_{SS}</td><td>284</td><td>V_{SS}</td></tr><tr><td>33</td><td>DQ52_A_t</td><td>69</td><td>V_{SS}</td><td>105</td><td>DQ50_B_c</td><td>141</td><td>V_{SS}</td><td>177</td><td>V_{SS}</td><td>213</td><td>CA3_A</td><td>249</td><td>DQ55_B_c, TDQ55_B_c</td><td>285</td><td>DQ30_B</td></tr><tr><td>34</td><td>DQ52_A_c</td><td>70</td><td>CA4_A</td><td>106</td><td>V_{SS}</td><td>142</td><td>DQ29_B</td><td>178</td><td>DQ57_A_c, TDQ57_A_c</td><td>214</td><td>V_{SS}</td><td>250</td><td>DQ55_B_t, TDQ55_B_t</td><td>286</td><td>V_{SS}</td></tr><tr><td>35</td><td>V_{SS}</td><td>71</td><td>V_{SS}</td><td>107</td><td>DQ4_B</td><td>143</td><td>V_{SS}</td><td>179</td><td>DQ57_A_t, TDQ57_A_t</td><td>215</td><td>CA5_A</td><td>251</td><td>V_{SS}</td><td>287</td><td>DQ31_B</td></tr><tr><td>36</td><td>DQ20_A</td><td>72</td><td>CA6_A</td><td>108</td><td>V_{SS}</td><td>144</td><td>RFU</td><td>180</td><td>V_{SS}</td><td>216</td><td>V_{SS}</td><td>252</td><td>DQ6_B</td><td>288</td><td>V_{SS}</td></tr></tbody></table> Notes: 1. On modules with nominal transfer rates <6400 MT/s, this pin is RFU. On modules with nominal transfer rates ≥6400 MT/s, this pin is NPP (no pin present). Ex. CC-03 (DDR5 RDIMM Datasheet) at 6-7. <table><thead><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr></thead><tbody><tr><td>CK_t, CK_c</td><td>Input</td><td>V_{DD}</td><td>Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>DDR Input</td><td>V_{DD}</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.</td></tr></tbody></table> Ex. 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Ex. CC-03 (DDR5 RDIMM Datasheet) at 9.																							
The Accused Products are configured to fit into a JEDEC-compliant memory slot, as illustrated below:																							
																							

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	Ex. CC-70 (DDR5 Memory Channel Scaling Performance).  Figure 1 DDR5 socket and module (SMT)  Figure 2 DDR5 U/R/LR DIMM Surface Mount (SMT) Connector Socket Outline Ex. CC-73 (JEDEC PS-005B) at 6.

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	Contacting the RDIMM's edge connectors with the corresponding pins in the memory slot forms electrical connections between the RDIMM and the system bus: Figure 6: DDR5 288-pin connector. Figure 7: Device under test (DUT) Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on a DIMM being electrically connected with a corresponding pin on the memory slot). The JEDEC DDR5 RDIMM standard further demonstrates the Accused Products' edge connectors provide electrical connections between the memory module and the system bus because it includes connector pin assignments for clock signal lines, control/address (C/A) signal lines, and data signal lines included in the system bus. <i>See, e.g.,</i>

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	Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr><tr><td>PAR_A PAR_B</td><td>Parity input</td><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>Data Strobes (positive line of differential pair)</td></tr><tr><td>CK_t</td><td>Clock (true/positive)</td><td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>Data Strobes (negative line of differential pair)</td></tr><tr><td>CK_c</td><td>Clock (complement/negative)</td><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3. Table 4 — Input/Output Functional Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CK_t, CK_c.</td><td>Input</td><td>VDD</td><td>Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Input</td><td>VDD</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.</td></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>VDD</td><td>ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4.	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)	CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)	CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	Symbol	Type	I/O Levels	Function	CK_t, CK_c.	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating
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	Ex. CC-88 (JEDEC 305B) at 4. Table 4 — Input/Output Functional Description (cont'd) <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">VDD</td><td rowspan="2">Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 5.	Symbol	Type	I/O Levels	Function	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
Symbol	Type	I/O Levels	Function															
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.															
DQS[9:0]_A_c DQS[9:0]_B_c																		
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.															
TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.															

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Ex. CC-88 (JEDEC 305B) at 6.	Front Side		Pin	Pin	Back side		Front Side		Pin	Pin	Back side	
	Pin Label				Pin Label		Pin Label				Pin Label	
	VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c				
	RFU	2	146	VIN_BULK	VSS	75	219	VSS				
	VIN_MGMT	3	147	PCAMP				Key				
	HSCL	4	148	HSA								
	HSDA	5	149	RFU	CA0_B	76	220	RFU / No Pin Present				
	VSS	6	150	RFU	VSS	77	221	CA1_B				
	DQ0_A	7	151	VSS	CA2_B	78	222	VSS				
	VSS	8	152	DQ2_A	VSS	79	223	CA3_B				
	DQ1_A	9	153	VSS	CA4_B	80	224	VSS				
	VSS	10	154	DQ3_A	VSS	81	225	CA5_B				
	DQS0_A_t	11	155	VSS	CA6_B	82	226	VSS				
	DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	VSS	83	227	PAR_B				
	VSS	13	157	DQS5_A_t, TDQS5_A_t	CS0_B_n	84	228	VSS				
	DQ4_A	14	158	VSS	VSS	85	229	CS1_B_n				
	VSS	15	159	DQ6_A	DLBDQ	86	230	VSS				
	DQ5_A	16	160	VSS	DLBDQS	87	231	RFU				
	VSS	17	161	DQ7_A	VSS	88	232	RFU				
	DQ8_A	18	162	VSS	CB4_B	89	233	VSS				
	VSS	19	163	DQ10_A	VSS	90	234	CB6_B				
	DQ9_A	20	164	VSS	CB5_B	91	235	VSS				
	VSS	21	165	DQ11_A	VSS	92	236	CB7_B				
	DQS1_A_t	22	166	VSS	DQS9_B_t, TDQS9_B_t	93	237	VSS				
	DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c				
	VSS	24	168	DQS6_A_t, TDQS6_A_t	VSS	95	239	DQS4_B_t				
	DQ12_A	25	169	VSS	CB0_B	96	240	VSS				
	VSS	26	170	DQ14_A	VSS	97	241	CB2_B				
	DQ13_A	27	171	VSS	CB1_B	98	242	VSS				
	VSS	28	172	DQ15_A	VSS	99	243	CB3_B				
	DQ16_A	29	173	VSS	DQ0_B	100	244	VSS				
	VSS	30	174	DQ18_A	VSS	101	245	DQ2_B				
	DQ17_A	31	175	VSS	DQ1_B	102	246	VSS				
	VSS	32	176	DQ19_A	VSS	103	247	DQ3_B				
	DQS2_A_t	33	177	VSS	DQS0_B_t	104	248	VSS				
	DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c				
	VSS	35	179	DQS7_A_t, TDQS7_A_t	VSS	106	250	DQS5_B_t, TDQS5_B_t				
	DQ20_A	36	180	VSS	DQ4_B	107	251	VSS				
VSS	37	181	DQ22_A	VSS	108	252	DQ6_B					
DQ21_A	38	182	VSS	DQ5_B	109	253	VSS					

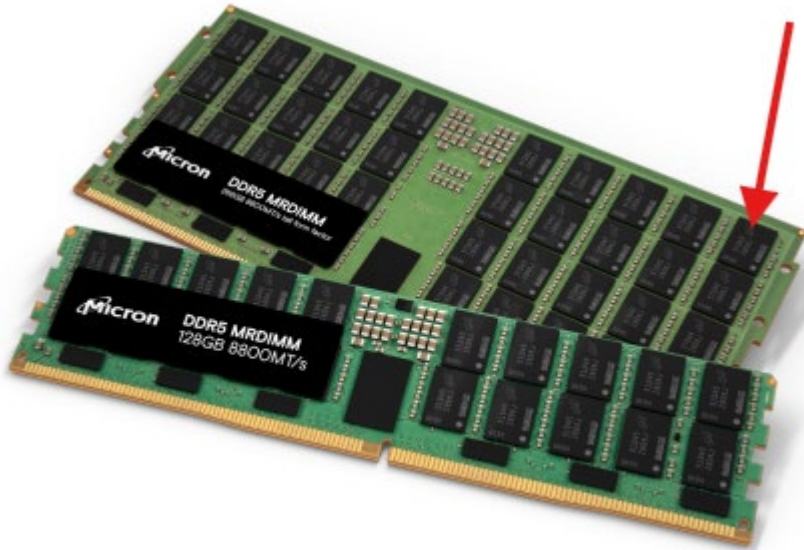
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TDQS8_A_c</td><td>DQS1_B_c</td><td>116</td><td>260</td><td>DQS6_B_c, TDQS6_B_c</td></tr><tr><td>VSS</td><td>46</td><td>190</td><td>DQS8_A_t, TDQS8_A_t</td><td>VSS</td><td>117</td><td>261</td><td>DQS6_B_t, TDQS6_B_t</td></tr><tr><td>DQ28_A</td><td>47</td><td>191</td><td>VSS</td><td>DQ12_B</td><td>118</td><td>262</td><td>VSS</td></tr><tr><td>VSS</td><td>48</td><td>192</td><td>DQ30_A</td><td>VSS</td><td>119</td><td>263</td><td>DQ14_B</td></tr><tr><td>DQ29_A</td><td>49</td><td>193</td><td>VSS</td><td>DQ13_B</td><td>120</td><td>264</td><td>VSS</td></tr><tr><td>VSS</td><td>50</td><td>194</td><td>DQ31_A</td><td>VSS</td><td>121</td><td>265</td><td>DQ15_B</td></tr><tr><td>CB0_A</td><td>51</td><td>195</td><td>VSS</td><td>DQ16_B</td><td>122</td><td>266</td><td>VSS</td></tr><tr><td>VSS</td><td>52</td><td>196</td><td>CB2_A</td><td>VSS</td><td>123</td><td>267</td><td>DQ18_B</td></tr><tr><td>CB1_A</td><td>53</td><td>197</td><td>VSS</td><td>DQ17_B</td><td>124</td><td>268</td><td>VSS</td></tr><tr><td>VSS</td><td>54</td><td>198</td><td>CB3_A</td><td>VSS</td><td>125</td><td>269</td><td>DQ19_B</td></tr><tr><td>DQS4_A_t</td><td>55</td><td>199</td><td>VSS</td><td>DQS2_B_t</td><td>126</td><td>270</td><td>VSS</td></tr><tr><td>DQS4_A_c</td><td>56</td><td>200</td><td>DQS9_A_c, TDQS9_A_c</td><td>DQS2_B_c</td><td>127</td><td>271</td><td>DQS7_B_c, TDQS7_B_c</td></tr><tr><td>VSS</td><td>57</td><td>201</td><td>DQS9_A_t, TDQS9_A_t</td><td>VSS</td><td>128</td><td>272</td><td>DQS7_B_t, TDQS7_B_t</td></tr><tr><td>CB4_A</td><td>58</td><td>202</td><td>VSS</td><td>DQ20_B</td><td>129</td><td>273</td><td>VSS</td></tr><tr><td>VSS</td><td>59</td><td>203</td><td>CB6_A</td><td>VSS</td><td>130</td><td>274</td><td>DQ22_B</td></tr><tr><td>CB5_A</td><td>60</td><td>204</td><td>VSS</td><td>DQ21_B</td><td>131</td><td>275</td><td>VSS</td></tr><tr><td>VSS</td><td>61</td><td>205</td><td>CB7_A</td><td>VSS</td><td>132</td><td>276</td><td>DQ23_B</td></tr><tr><td>ALERT_n</td><td>62</td><td>206</td><td>VSS</td><td>DQ24_B</td><td>133</td><td>277</td><td>VSS</td></tr><tr><td>VSS</td><td>63</td><td>207</td><td>RESET_n</td><td>VSS</td><td>134</td><td>278</td><td>DQ26_B</td></tr><tr><td>CS0_A_n</td><td>64</td><td>208</td><td>VSS</td><td>DQ25_B</td><td>135</td><td>279</td><td>VSS</td></tr><tr><td>VSS</td><td>65</td><td>209</td><td>CS1_A_n</td><td>VSS</td><td>136</td><td>280</td><td>DQ27_B</td></tr><tr><td>CA0_A</td><td>66</td><td>210</td><td>VSS</td><td>DQS3_B_t</td><td>137</td><td>281</td><td>VSS</td></tr><tr><td>VSS</td><td>67</td><td>211</td><td>CA1_A</td><td>DQS3_B_c</td><td>138</td><td>282</td><td>DQS8_B_c, TDQS8_B_c</td></tr><tr><td>CA2_A</td><td>68</td><td>212</td><td>VSS</td><td>VSS</td><td>139</td><td>283</td><td>DQS8_B_t, TDQS8_B_t</td></tr><tr><td>VSS</td><td>69</td><td>213</td><td>CA3_A</td><td>DQ28_B</td><td>140</td><td>284</td><td>VSS</td></tr><tr><td>CA4_A</td><td>70</td><td>214</td><td>VSS</td><td>VSS</td><td>141</td><td>285</td><td>DQ30_B</td></tr><tr><td>VSS</td><td>71</td><td>215</td><td>CA5_A</td><td>DQ29_B</td><td>142</td><td>286</td><td>VSS</td></tr><tr><td>CA6_A</td><td>72</td><td>216</td><td>VSS</td><td>VSS</td><td>143</td><td>287</td><td>DQ31_B</td></tr><tr><td>VSS</td><td>73</td><td>217</td><td>CK_t</td><td>RFU</td><td>144</td><td>288</td><td>VSS</td></tr></table>										Front Side	Pin	Pin	Back side	Front Side	Pin	Pin	Back side	Pin Label			Pin Label	Pin Label			Pin Label	VSS	39	183	DQ23_A	VSS	110	254	DQ7_B	DQ24_A	40	184	VSS	DQ8_B	111	255	VSS	VSS	41	185	DQ26_A	VSS	112	256	DQ10_B	DQ25_A	42	186	VSS	DQ9_B	113	257	VSS	VSS	43	187	DQ27_A	VSS	114	258	DQ11_B	DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c	VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t	DQ28_A	47	191	VSS	DQ12_B	118	262	VSS	VSS	48	192	DQ30_A	VSS	119	263	DQ14_B	DQ29_A	49	193	VSS	DQ13_B	120	264	VSS	VSS	50	194	DQ31_A	VSS	121	265	DQ15_B	CB0_A	51	195	VSS	DQ16_B	122	266	VSS	VSS	52	196	CB2_A	VSS	123	267	DQ18_B	CB1_A	53	197	VSS	DQ17_B	124	268	VSS	VSS	54	198	CB3_A	VSS	125	269	DQ19_B	DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	VSS	DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c	VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t	CB4_A	58	202	VSS	DQ20_B	129	273	VSS	VSS	59	203	CB6_A	VSS	130	274	DQ22_B	CB5_A	60	204	VSS	DQ21_B	131	275	VSS	VSS	61	205	CB7_A	VSS	132	276	DQ23_B	ALERT_n	62	206	VSS	DQ24_B	133	277	VSS	VSS	63	207	RESET_n	VSS	134	278	DQ26_B	CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS	VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B	CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS	VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c	CA2_A	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t	VSS	69	213	CA3_A	DQ28_B	140	284	VSS	CA4_A	70	214	VSS	VSS	141	285	DQ30_B	VSS	71	215	CA5_A	DQ29_B	142	286	VSS	CA6_A	72	216	VSS	VSS	143	287	DQ31_B	VSS	73	217	CK_t	RFU	144	288	VSS
Front Side	Pin	Pin	Back side	Front Side	Pin	Pin	Back side																																																																																																																																																																																																																																																																																																											
Pin Label			Pin Label	Pin Label			Pin Label																																																																																																																																																																																																																																																																																																											
VSS	39	183	DQ23_A	VSS	110	254	DQ7_B																																																																																																																																																																																																																																																																																																											
DQ24_A	40	184	VSS	DQ8_B	111	255	VSS																																																																																																																																																																																																																																																																																																											
VSS	41	185	DQ26_A	VSS	112	256	DQ10_B																																																																																																																																																																																																																																																																																																											
DQ25_A	42	186	VSS	DQ9_B	113	257	VSS																																																																																																																																																																																																																																																																																																											
VSS	43	187	DQ27_A	VSS	114	258	DQ11_B																																																																																																																																																																																																																																																																																																											
DQS3_A_t	44	188	VSS	DQS1_B_t	115	259	VSS																																																																																																																																																																																																																																																																																																											
DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c	DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c																																																																																																																																																																																																																																																																																																											
VSS	46	190	DQS8_A_t, TDQS8_A_t	VSS	117	261	DQS6_B_t, TDQS6_B_t																																																																																																																																																																																																																																																																																																											
DQ28_A	47	191	VSS	DQ12_B	118	262	VSS																																																																																																																																																																																																																																																																																																											
VSS	48	192	DQ30_A	VSS	119	263	DQ14_B																																																																																																																																																																																																																																																																																																											
DQ29_A	49	193	VSS	DQ13_B	120	264	VSS																																																																																																																																																																																																																																																																																																											
VSS	50	194	DQ31_A	VSS	121	265	DQ15_B																																																																																																																																																																																																																																																																																																											
CB0_A	51	195	VSS	DQ16_B	122	266	VSS																																																																																																																																																																																																																																																																																																											
VSS	52	196	CB2_A	VSS	123	267	DQ18_B																																																																																																																																																																																																																																																																																																											
CB1_A	53	197	VSS	DQ17_B	124	268	VSS																																																																																																																																																																																																																																																																																																											
VSS	54	198	CB3_A	VSS	125	269	DQ19_B																																																																																																																																																																																																																																																																																																											
DQS4_A_t	55	199	VSS	DQS2_B_t	126	270	VSS																																																																																																																																																																																																																																																																																																											
DQS4_A_c	56	200	DQS9_A_c, TDQS9_A_c	DQS2_B_c	127	271	DQS7_B_c, TDQS7_B_c																																																																																																																																																																																																																																																																																																											
VSS	57	201	DQS9_A_t, TDQS9_A_t	VSS	128	272	DQS7_B_t, TDQS7_B_t																																																																																																																																																																																																																																																																																																											
CB4_A	58	202	VSS	DQ20_B	129	273	VSS																																																																																																																																																																																																																																																																																																											
VSS	59	203	CB6_A	VSS	130	274	DQ22_B																																																																																																																																																																																																																																																																																																											
CB5_A	60	204	VSS	DQ21_B	131	275	VSS																																																																																																																																																																																																																																																																																																											
VSS	61	205	CB7_A	VSS	132	276	DQ23_B																																																																																																																																																																																																																																																																																																											
ALERT_n	62	206	VSS	DQ24_B	133	277	VSS																																																																																																																																																																																																																																																																																																											
VSS	63	207	RESET_n	VSS	134	278	DQ26_B																																																																																																																																																																																																																																																																																																											
CS0_A_n	64	208	VSS	DQ25_B	135	279	VSS																																																																																																																																																																																																																																																																																																											
VSS	65	209	CS1_A_n	VSS	136	280	DQ27_B																																																																																																																																																																																																																																																																																																											
CA0_A	66	210	VSS	DQS3_B_t	137	281	VSS																																																																																																																																																																																																																																																																																																											
VSS	67	211	CA1_A	DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c																																																																																																																																																																																																																																																																																																											
CA2_A	68	212	VSS	VSS	139	283	DQS8_B_t, TDQS8_B_t																																																																																																																																																																																																																																																																																																											
VSS	69	213	CA3_A	DQ28_B	140	284	VSS																																																																																																																																																																																																																																																																																																											
CA4_A	70	214	VSS	VSS	141	285	DQ30_B																																																																																																																																																																																																																																																																																																											
VSS	71	215	CA5_A	DQ29_B	142	286	VSS																																																																																																																																																																																																																																																																																																											
CA6_A	72	216	VSS	VSS	143	287	DQ31_B																																																																																																																																																																																																																																																																																																											
VSS	73	217	CK_t	RFU	144	288	VSS																																																																																																																																																																																																																																																																																																											
NOTE 1 Individual Annex standards for RDIMM define pin use where multiple functions are possible. NOTE 2 Pin 220 for 8000 Mbps and faster RDIMM raw cards do not have the edge pin present.																																																																																																																																																																																																																																																																																																																		

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
[1.b] memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines;	Ex. CC-88 (JEDEC 305B) at 7. The '407 Accused Products comprise memory devices mounted on the PCB and organized in a plurality of groups, wherein a respective group of the memory devices is configurable to communicate data with the memory controller via a corresponding subset of the data signal lines. For example, The Accused Products have memory devices mounted on the PCB. <i>See, e.g.,</i>: Ex. CC-01 (DDR5 Brochure) at 1 (showing an example of a memory device on the printed circuit board).

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	 Figure 1: MRDIMM low-profile form factor (foreground) and TFF (background) Ex. CC-02 (MRDIMM Product Brief) at 1.  Ex. 51 to Complaint (Micron RDIMM Importation) at 8.

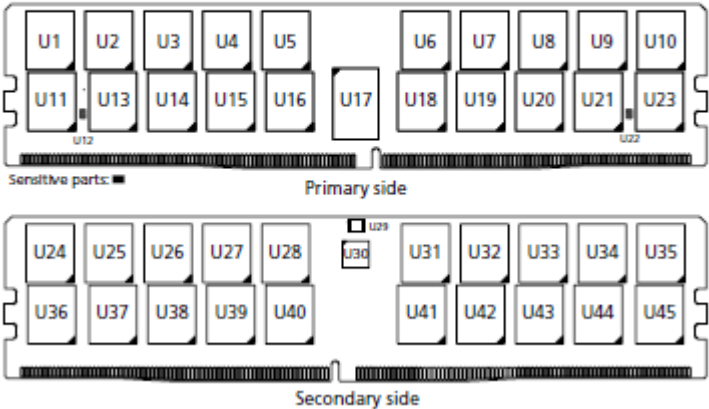
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	Ex. 51 to Complaint (Micron MRDIMM Importation) at 12. Ex. 51 to Complaint (HPE RDIMM Importation at 6).

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	A photograph of a Lenovo RDIMM (Registered Dual In-Line Memory Module) installed in a black plastic tray. The module is green with gold-plated pins. It features two white labels. The left label contains a barcode, the part number PM1TC20P1041S1RC48BA2 HGCC, and other technical specifications. The right label is a Lenovo label with the part number (P) Lenovo P/N: SM37B27362, J1, and other details. A UKCA mark is visible on the module. Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6. A photograph of a SuperMicro RDIMM (Registered Dual In-Line Memory Module) installed in a black plastic tray. The module is green with gold-plated pins. It features two white labels. The left label contains a barcode, the part number PM1TC10P1041S1RC48BA1 HGCC, and other technical specifications. The right label is a SuperMicro label with the part number 802C0F222939D318AC and a warranty notice. A UKCA mark is visible on the module. Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4.

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	General Description High-speed DDR5 SDRAM modules use DDR5 SDRAM devices with four or eight internal memory bank groups. DDR5 SDRAM modules utilizing 4- and 8-bit-wide DDR5 SDRAM devices have eight internal bank groups consisting of four memory banks each, providing a total of 32 banks. 16-bit-wide DDR5 SDRAM devices have four internal bank groups consisting of four memory banks each, providing a total of sixteen banks. DDR5 SDRAM modules benefit from DDR5 SDRAM's use of an 16n-prefetch architecture with an interface designed to transfer two data words per clock cycle at the I/O pins. A single READ or WRITE operation for the DDR5 SDRAM effectively consists of a single 16n-bit-wide, eight-clock data transfer at the internal DRAM core and sixteen corresponding n-bit-wide, one-half-clock-cycle data transfers at the I/O pins. Ex. CC-03 (DDR5 RDIMM Datasheet) at 5. Figure 1: 288-Pin DDR5 RDIMM (R/C-A0)  Ex. CC-07 (DDR5 RDIMM Addendum) at 1 (showing the memory devices); <i>see also id.</i> at 4-6 (“Table-to-Module DQ Map” showing pin mapping for the memory devices).

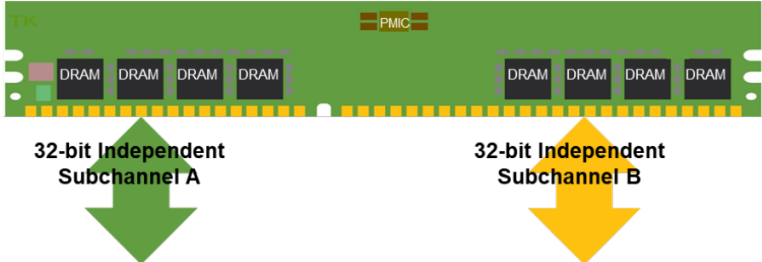
U.S. Patent No. 12,675,407 Infringement Claim Chart

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	The diagram illustrates a DDR5 DIMM with two ranks of memory devices. Rank(s) Left Side A is connected to Independent Channel A via a 40-bit data bus. Rank(s) Left Side B is connected to Independent Channel B via a 40-bit data bus. The DIMM is shown with a green top edge and a yellow bottom edge. The memory devices are represented by black rectangles. The data buses are indicated by blue and orange arrows pointing to the bottom of the DIMM. Figure 4: DDR5 DIMM Ex. CC-08 (DDR5 White Paper) at 4. Furthermore, the memory devices are organized in a plurality of groups, wherein a respective group of the memory devices is configured to communicate data with the memory controller via a corresponding subset of the data signal lines. For example, the Accused Products include memory devices that are organized in a group that communicates data with the memory controller via independent channel A (or subsets of data signal lines in channel A), and a group that communicates with the memory controller via independent channel B (or subsets of data signal lines in channel A). <i>See, e.g.,</i>:

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	In addition to higher data rates and improvements to the I/O circuitry, DDR5 introduces other new protocol features unrelated to data rate that are integral to increasing bandwidth and performance. For example, DDR5 DIMMs feature two 40-bit (32 bits plus ECC) independent channels. When combined with a new default burst length of 16 (BL16) in the DDR5 component, this allows a single burst to access 64B of data (the typical CPU cache line size) using only one of the independent channels, or only half of the DIMM. Providing this ability to interleave accesses from these two independent channels enables tremendous improvements to concurrency, essentially turning an 8-channel system as we know it today into a 16-channel system. The diagram illustrates a DDR5 DIMM layout. It is divided into two main sections: 'Rank(s) Left Side A' and 'Rank(s) Left Side B'. Each side contains a series of memory modules (represented by black squares) connected by green lines. Below the modules, two large double-headed arrows indicate the data paths for 'Independent Channel A' (blue arrow) and 'Independent Channel B' (orange arrow). Each channel is labeled '40 Bits'. Figure 4: DDR5 DIMM Ex. CC-08 (DDR5 White Paper) at 4.

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Claim Language	'407 Accused Products ⁴
	Module Layout and Data Access A key difference of a DDR5 module as compared to a DDR4 module is the presence of subchannels (Figure 1). A standard DDR5 module has two independent subchannels. Each subchannel has up to two physical package ranks. Each DRAM package can be configured in an initiator/target topology to enable additional logical ranks for increased density.  <i>Figure 1: DDR5 1Rx8 UDIMM Module Illustrating Two Independent Subchannels</i> By enabling the new DDR5 subchannel module layout, the default burst length (BL) needed to be increased from 8 to 16; this achieves the same data payload as a DDR4 module per transaction. The doubling of the BL implies a halving of data inputs/outputs (I/Os) required to fulfill the same amount of data for a given system access size. The independent subchannels increase concurrency and support better scheduling from the memory controller. As an example, each subchannel allows for 32 data I/Os with a BL of 16, resulting in 64-byte payloads. A READ operation from the combined subchannels results in an output of 128 bytes. Ex. CC-11 Micron DDR5: Client Module Features, Micron at 4 (“DDR5 Features White Paper”) at 1.

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	Features • DDR5 functionality and operations supported as defined in the component data sheet • 287/288-pin RDIMM • Supports ECC error detection and correction • On-DIMM SPD EEPROM with Hub function and integrated temperature sensor (TS) • Two on-DIMM discrete TS5 temperature sensors • On-DIMM power management integrated circuit (PMIC) • Sideband access with I3C-basic/I2C support • Two independent I/O sub channels for increased bandwidth • DDR command/address bus to RCD • Gold edge contacts • Halogen-free • Fly-by topology • Terminated clock, control and command/address bus Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

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Claim Language	'407 Accused Products ⁴						
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing multiple channels with corresponding subset of data lines). <table><tr><th>Parameter</th><th>Options</th><th>Notes</th></tr><tr><td>DIMM organization</td><td>x80 ECC (EC8), x72 ECC (EC4)</td><td>Two independent 40-bit or 36-bit I/O sub-channels</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 2. Rank Addressing Two chip select inputs per sub-channel (CS0_n and CS1_n) on Micron's modules are used to select a specific rank of DRAM. Each sub-channel has two chip select inputs (CS0x_n and CS1x_n). Ex. CC-03 (DDR5 RDIMM Datasheet) at 12.	Parameter	Options	Notes	DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels
Parameter	Options	Notes					
DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels					

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	<table><tr><td>CA[6:0]_A CA[6:0]_B</td><td>DDR Input</td><td>V_{DD}</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.</td></tr><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>V_{DD}</td><td>Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 8 (showing channels A and B have separate CA Bus and CS commands). <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing channels A and B have separate dedicated data lines). The JEDEC DDR5 RDIMM standard further demonstrates the Accused Products’ edge connectors allow each respective group of the memory devices to be configured to communicate data with the memory controller via a corresponding subset of the data signal lines. <i>See, e.g.,</i>	CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.	CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.
CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.														
CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .														
Symbol	Type	I/O Level	Description														
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.														

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	Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3 (showing each channel has separate CA Bus and CS commands). <table><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 R4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (showing each channel has separate data lines).	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 R4=High. Refer to vendor specific data sheets to determine which DQ is used.
Pin Name	Description	Pin Name	Description														
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B														
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B														
DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 R4=High. Refer to vendor specific data sheets to determine which DQ is used.														

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

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	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr></table> <table><tr><th colspan="10">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td colspan="2"></td><td colspan="2">[15:8]</td><td colspan="2"></td><td colspan="2">[31:24]</td><td colspan="2"></td></tr><tr><td>Bottom</td><td colspan="2">[7:0]</td><td colspan="2"></td><td colspan="2">[23:16]</td><td colspan="2"></td><td colspan="2">CB[7:0]</td></tr></table> <table><tr><th colspan="10">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td colspan="2">[7:0]</td><td colspan="2"></td><td colspan="2">[23:16]</td><td colspan="2"></td><td colspan="2">[31:24]</td></tr><tr><td>Bottom</td><td colspan="2"></td><td colspan="2">[15:8]</td><td colspan="2"></td><td colspan="2">[31:24]</td><td colspan="2">CB[7:0]</td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive CA, CS, and data signals via different subsets of CA, CS, and data signal lines).	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCCK	QACS0	QACA	0	Bottom	Lower	QCCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)										Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top			[15:8]				[31:24]				Bottom	[7:0]				[23:16]				CB[7:0]		1Rx8 Sub Channel 1 (Channel B)										Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top	[7:0]				[23:16]				[31:24]		Bottom			[15:8]				[31:24]		CB[7:0]	
1Rx8 (less than 8000 Mbps)																																																																																																																																																													
DIMM Side	Row	Clock	CS	CA	Rank																																																																																																																																																								
Top	Lower	QACK	QACS0	QACA	0																																																																																																																																																								
Bottom	Lower	QACK	QACS0	QACA	0																																																																																																																																																								
1Rx8 (8000 Mbps and Faster)																																																																																																																																																													
DIMM Side	Row	Clock	CS	CA	Rank																																																																																																																																																								
Top	Lower	QCCK	QACS0	QACA	0																																																																																																																																																								
Bottom	Lower	QCCK	QACS0	QACA	0																																																																																																																																																								
1Rx8 Sub Channel 0 (Channel A)																																																																																																																																																													
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																			
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																			
Top			[15:8]				[31:24]																																																																																																																																																						
Bottom	[7:0]				[23:16]				CB[7:0]																																																																																																																																																				
1Rx8 Sub Channel 1 (Channel B)																																																																																																																																																													
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																			
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																			
Top	[7:0]				[23:16]				[31:24]																																																																																																																																																				
Bottom			[15:8]				[31:24]		CB[7:0]																																																																																																																																																				
[1.c] circuitry mounted on the PCB and configurable to:	The '407 Accused Products comprise circuitry mounted on the PCB. For example, the '407 Accused Products that are RDIMM devices include RCD (Registering Clock Driver) circuitry mounted on the PCB. See, e.g.,																																																																																																																																																												

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 Ex. 51 to Complaint (Micron RDIMM Importation) at 11.

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Claim Language	'407 Accused Products ⁴
	 Ex. 55 to Complaint (HPE RDIMM Importation) at 10.

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Claim Language	'407 Accused Products ⁴
	A close-up photograph of a black integrated circuit (IC) mounted on a green printed circuit board (PCB). The IC is labeled with the following text: "IDT", "5RCD0148HC3", "AVG", "Z2211G", and "R033798 TWN". The chip is surrounded by various electronic components, including resistors and other smaller ICs. Ex. 57 to Complaint (Lenovo RDIMM Importation) at 10. A close-up photograph of a black integrated circuit (IC) mounted on a green printed circuit board (PCB). The IC is labeled with the following text: "Rambus", "DDR5RCD1-G1EX", and "2141A08P82". The chip is surrounded by various electronic components, including resistors and other smaller ICs.

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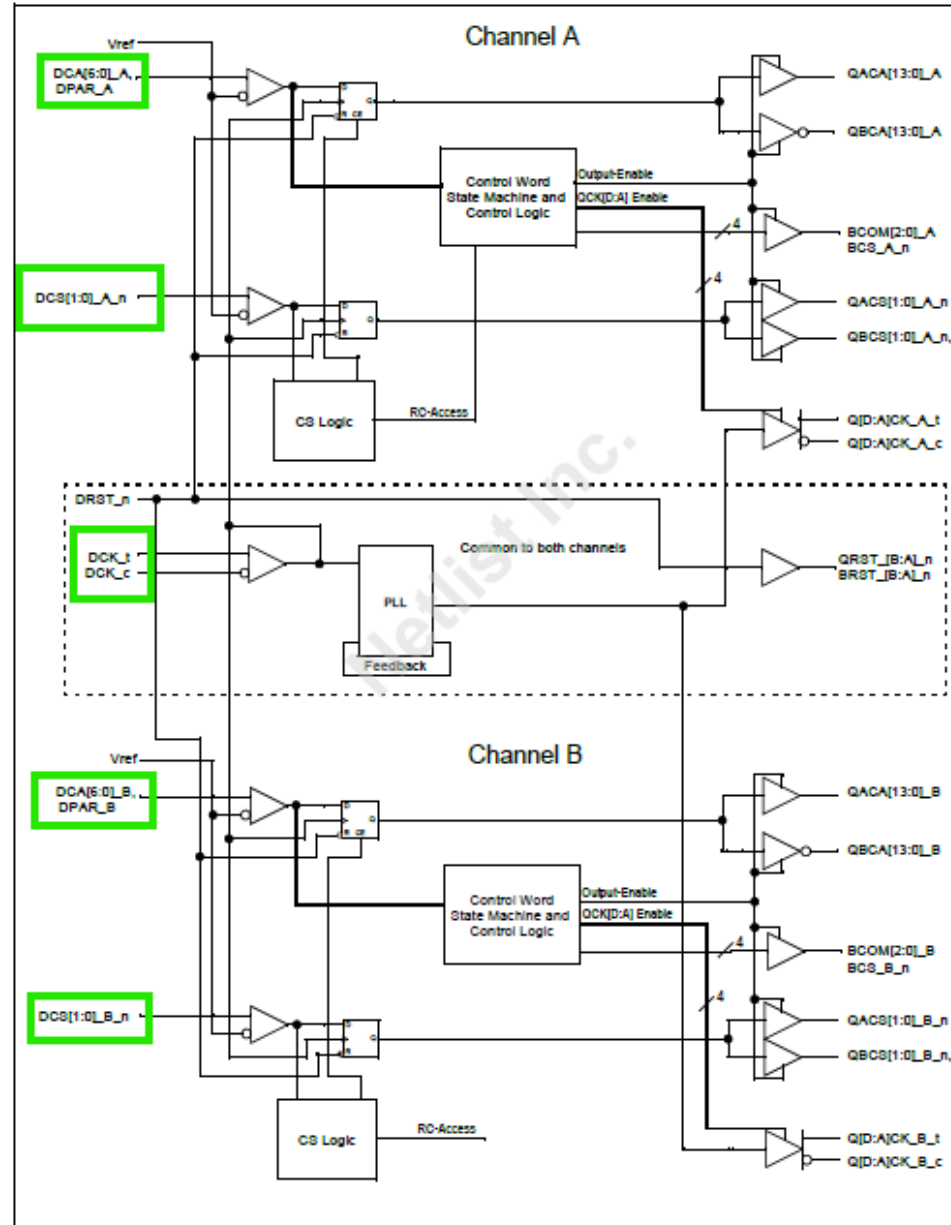
Claim Language	'407 Accused Products ⁴
	Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 10. For example, the '407 Accused Products that are MRDIMM devices include MRCD (Multiplexed Registering Clock Driver) circuitry mounted on the PCB. <i>See, e.g.,</i> Ex. 53 to Complaint (Micron MRDIMM Importation at 11). Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.
[1.c.1] receive from the memory controller a system clock via the one or more clock signal lines and input control and address	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines. The RCD circuitry⁶ mounted on the PCB is configurable to receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) via the C/A signal lines. For example, the RCD receives “CLOCK” and “CA (DDR)” inputs. <i>See, e.g.,</i>

⁶ Analysis for the following limitations uses RCD circuitry in RDIMM products as an example, but MRCDs in MRDIMM products would infringe in materially same way because those products otherwise use common components. *See* Ex. CC-13 JEDEC Press Release (planned features for MRDIMMs include “Utilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption”).

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																								
(C/A) signals via the C/A signal lines;	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-32 (JEDEC 82-512) at 15.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

U.S. Patent No. 12,675,407 Infringement Claim Chart



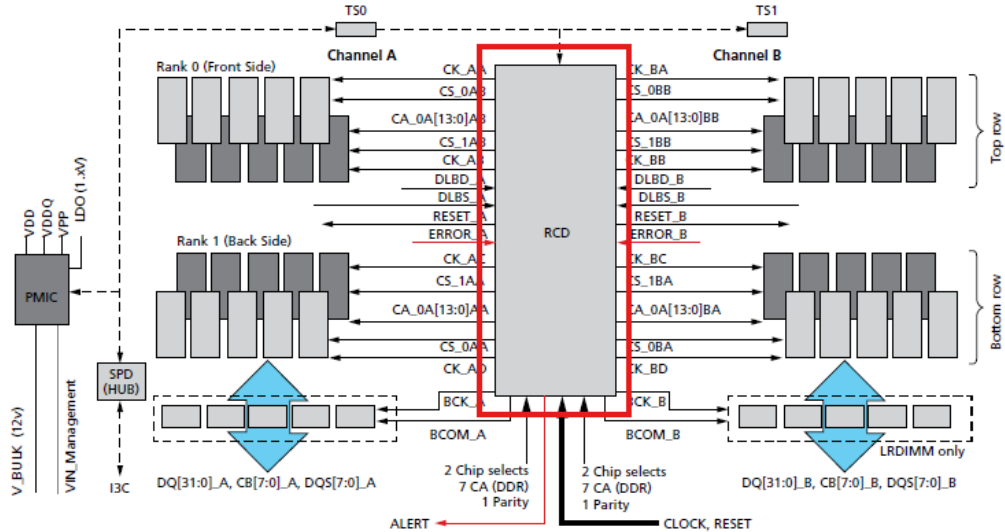
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																								
	Ex. CC-32 (JEDEC 82-512) at 7. <table><tr><th>Signal Group</th><th>Signal Name</th><th>Type</th><th>Description</th></tr><tr><td>Input Control bus</td><td>DCS[1:0]_[B:A]_n</td><td>POD V_{REF} based</td><td>Chip Select inputs to the RCD. Two inputs per DRAM channel.</td></tr><tr><td>Input Address and Command bus</td><td>DCA[6:0]_[B:A]</td><td>POD V_{REF} based</td><td>Command/Address bus inputs to the RCD. Separate sets per channel.</td></tr><tr><td>Parity input</td><td>DPAR_[B:A]</td><td>POD V_{REF} based</td><td>Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.</td></tr><tr><td>Clock inputs</td><td>DCK_t/DCK_c</td><td>POD differential</td><td>Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.</td></tr><tr><td>Feedback</td><td>DPAR_[B:A]</td><td>POD V_{REF} based</td><td>Feedback inputs to RCD</td></tr></table> Ex. CC-32 (JEDEC 82-512) at 4. Features • DDR5 functionality and operations supported as defined in the component data sheet • 287/288-pin RDIMM • Supports ECC error detection and correction • On-DIMM SPD EEPROM with Hub function and integrated temperature sensor (TS) • Two on-DIMM discrete TS5 temperature sensors • On-DIMM power management integrated circuit (PMIC) • Sideband access with I3C-basic/I2C support • Two independent I/O sub channels for increased bandwidth • DDR command/address bus to RCD • Gold edge contacts • Halogen-free • Fly-by topology • Terminated clock, control and command/address bus	Signal Group	Signal Name	Type	Description	Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.	Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.	Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.	Feedback	DPAR_[B:A]	POD V _{REF} based	Feedback inputs to RCD
Signal Group	Signal Name	Type	Description																						
Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.																						
Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.																						
Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.																						
Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.																						
Feedback	DPAR_[B:A]	POD V _{REF} based	Feedback inputs to RCD																						

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Claim Language	'407 Accused Products ⁴
	The DCA, DCS and DCK signals are therefore received via the corresponding C/A, C/S and CK signals lines in the system bus because the DCA, DCS and DCK pins of an RDIMM each mate with a corresponding connector in the memory slot to form electrical connection to the corresponding slot pin and signal traces, as illustrated below: Figure 6: DDR5 288-pin connector. Figure 7: Device under test (DUT) Ex. CC-73 (JEDEC PS-005B) at 13, Figs. 6–7 (annotated) (illustrating an edge connection on a DIMM being electrically connected with a corresponding pin in the memory slot and the corresponding signal trace/line on the system board). Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing the presence of RCD and a command/address bus to RCD).

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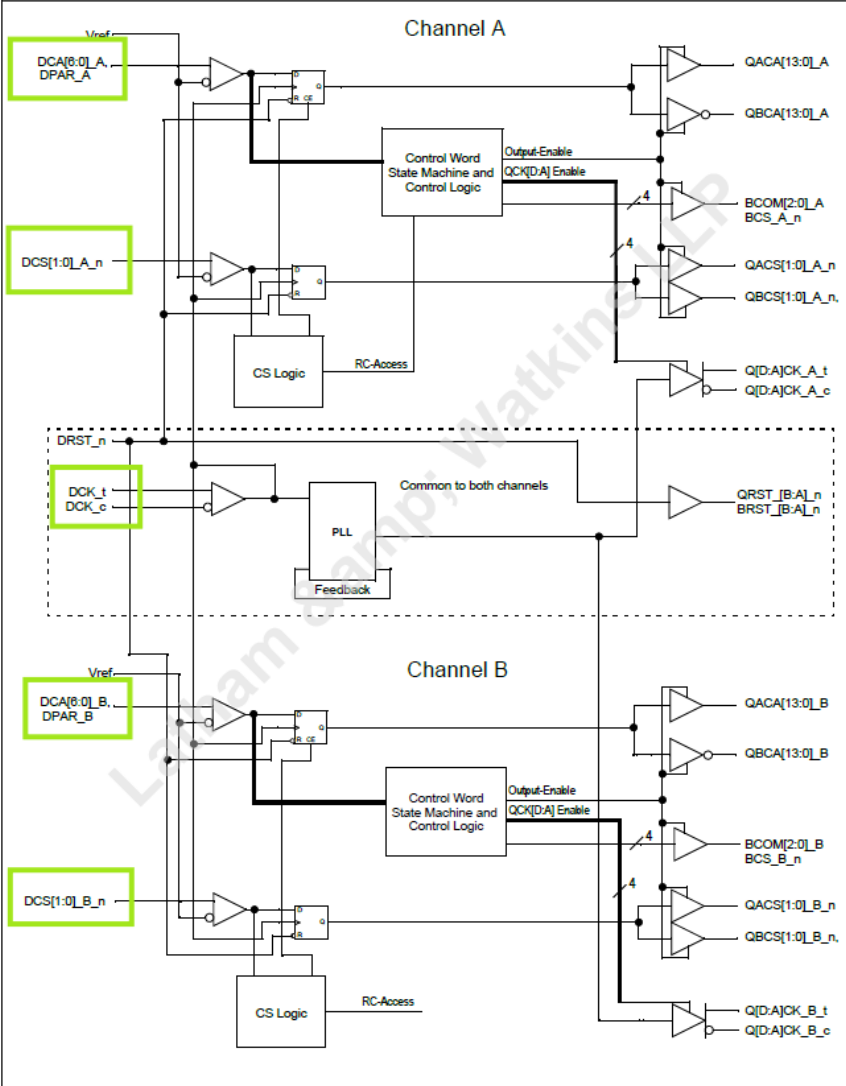
Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing the RCD in red receiving “CLOCK” and “CA (DDR)” inputs). 3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. Ex. CC-32 (JEDEC 82-512) at 6.

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Claim Language	'407 Accused Products ⁴																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

Ex. CC-30 (JEDEC 82-513) at 15.

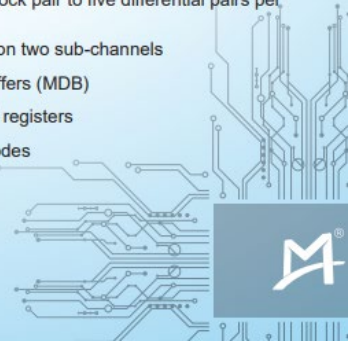
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 The diagram illustrates a dual-channel ADC system with two channels, Channel A and Channel B. Each channel includes a differential input stage (DCA[8:0]_A, DPAR_A and DCS[1:0]_A_n for Channel A; DCA[8:0]_B, DPAR_B and DCS[1:0]_B_n for Channel B), a control word state machine and control logic, and a CS Logic block. A common PLL block is shared between both channels, receiving DCK_t and DCK_c inputs. The system also includes a feedback loop and a common output stage (QD[A]CK_A_t, QD[A]CK_A_c for Channel A; QD[A]CK_B_t, QD[A]CK_B_c for Channel B). The outputs are labeled QACA[13:0]_A, QBCA[13:0]_A, BCOM[2:0]_A, BCS_A_n, QACS[1:0]_A_n, QBSCS[1:0]_A_n, QD[A]CK_A_t, QD[A]CK_A_c, QD[A]CK_B_t, QD[A]CK_B_c, QACA[13:0]_B, QBCA[13:0]_B, BCOM[2:0]_B, BCS_B_n, QACS[1:0]_B_n, QBSCS[1:0]_B_n, QD[A]CK_B_t, QD[A]CK_B_c. The diagram is labeled 'Ex. CC-30 (JEDEC 82-513) at 7.'

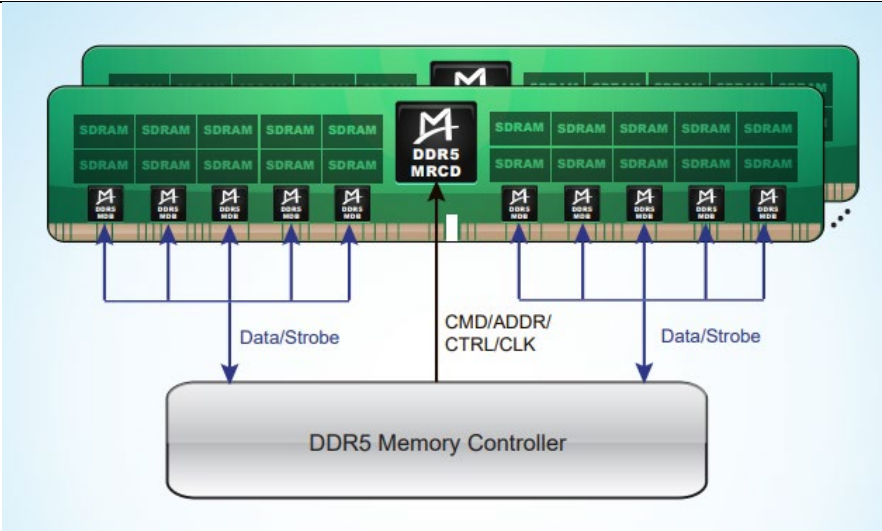
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																				
2.2 Terminal Functions Table 2 — Terminal Functions <table><tr><th>Signal Group</th><th>Signal Name</th><th>Type</th><th>Description</th></tr><tr><td>Input Control bus</td><td>DCS[1:0]_[B:A]_n</td><td>POD V_{REF} based</td><td>Chip Select inputs to the RCD. Two inputs per DRAM channel.</td></tr><tr><td>Input Address and Command bus</td><td>DCA[6:0]_[B:A]</td><td>POD V_{REF} based</td><td>Command/Address bus inputs to the RCD. Separate sets per channel.</td></tr><tr><td>Parity input</td><td>DPAR_[B:A]</td><td>POD V_{REF} based</td><td>Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.</td></tr><tr><td>Clock inputs</td><td>DCK_t/DCK_c</td><td>POD differential</td><td>Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.</td></tr></table> Ex. CC-30 (JEDEC 82-513) at 4. 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. Ex. CC-30 (JEDEC 82-513) at 6. The MRCD circuitry receives address and control (“CA”) signals and system clock signals from the system memory and controls the plurality of memory devices in response to the address and control signals:	Signal Group	Signal Name	Type	Description	Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.	Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.	Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.	
	Signal Group	Signal Name	Type	Description																	
	Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.																	
	Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.																	
	Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.																	
	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.																	

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Claim Language	'407 Accused Products ⁴
	General Description The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis. Application • DDR5 MRDIMM / MCRDIMM Feature • Speed up to 8800MT/s • Two sub-channels, each divided into two pseudo-channels to increase the total bandwidth of the host system • Two pseudo-channels splitting the incoming CA inputs and generating separate CA outputs • Integrated PLL clock driver distributing one differential clock pair to five differential pairs per channel • Parity checking across CA and DPAR inputs separately on two sub-channels • Integrated BCOM bus to control the multiplexed data buffers (MDB) • Output characteristics configurable through control word registers • Sideband bus interface supporting I²C and I³C Basic modes • CA, CS, DFE and BCOM training modes support • Multiple power-saving modes • 1.1V VDD and 1.0V VDDIO voltages • Green package: 240-ball FBGA 

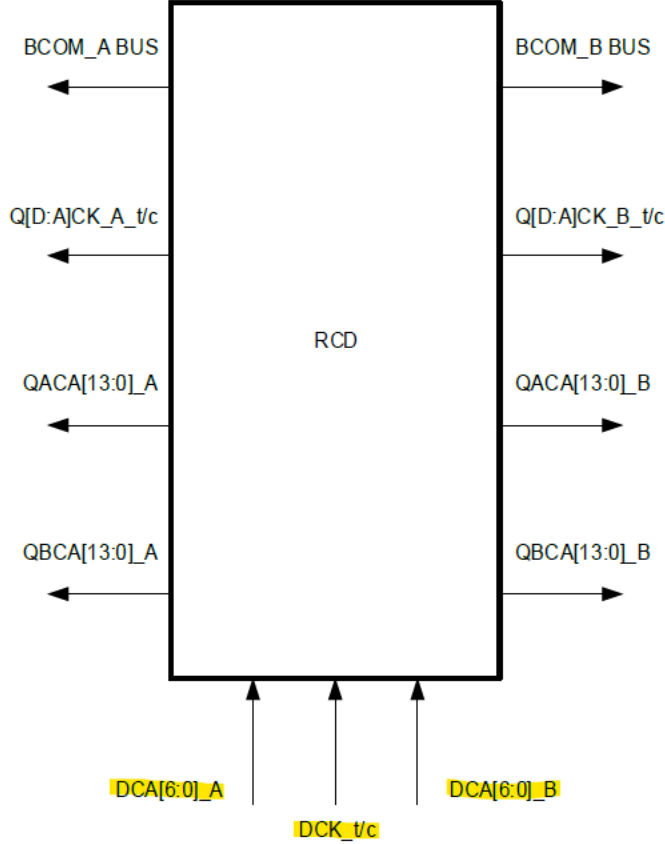
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 The diagram illustrates a DDR5 memory module (M88MR5RCD01) connected to a DDR5 Memory Controller. The module is shown as a green circuit board with multiple SDRAM chips and a central DDR5 MRCD (Memory Refresh Control Driver) chip. The controller is a grey rectangular block below the module. Arrows indicate the flow of data and control signals: 'Data/Strobe' signals flow from the controller to the module, and 'CMD/ADDR/CTRL/CLK' signals flow from the module to the controller. Ex. CC-12 M88MR5RCD01 Flyer, Montage Technology.

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Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

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Claim Language	'407 Accused Products ⁴
	Block Diagram  Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. The JEDEC DDR5 RDIMM standard further demonstrates the RCD circuitry is configurable to receive from the memory controller a system clock via the one or more clock signal lines (<i>e.g.</i>, “CK t” and “CK c” / “DCK t”

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Claim Language	'407 Accused Products ⁴																																				
	and “DCK_c”) and input control and address (C/A) signals via the C/A signal lines (e.g., “CA[]_A” and “CA[]_B”/ “DCA[]_A” and “DCA[]_B” / “CS[]_A_n” and “CS[]_B_n” / “DCS[]_A_n” and “DCS[]_B_n”). See, e.g., Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr><tr><td>PAR_A PAR_B</td><td>Parity input</td><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>Data Strokes (positive line of differential pair)</td></tr><tr><td>CK_t</td><td>Clock (true/positive)</td><td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>Data Strokes (negative line of differential pair)</td></tr><tr><td>CK_c</td><td>Clock (complement/negative)</td><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3 (illustrating circuitry inputs for receiving clock signal lines (e.g., “CK_t” and “CK_c”) and control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”) from the memory controller). Table 4 — Input/Output Functional Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CK_t, CK_c,</td><td>Input</td><td>VDD</td><td>Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Input</td><td>VDD</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating circuitry inputs for receiving clock signal lines (e.g., “CK_t” and “CK_c”) and control/address (C/A) signal lines (e.g., “CA[]_A” and “CA[]_B”) from the memory controller).	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strokes (positive line of differential pair)	CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strokes (negative line of differential pair)	CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.	Symbol	Type	I/O Levels	Function	CK_t, CK_c,	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.
Pin Name	Description	Pin Name	Description																																		
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B																																		
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CK_t, CK_c,	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.																																		
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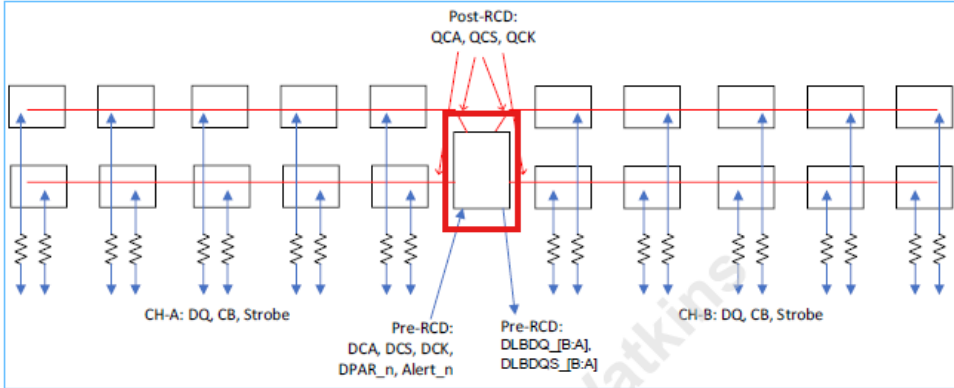
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	<table><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>VDD</td><td>ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4. <table><caption>Table 4 — Input/Output Functional Description (cont'd)</caption><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">VDD</td><td rowspan="2">Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 5.	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating	Symbol	Type	I/O Levels	Function	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.																							
CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating																							
Symbol	Type	I/O Levels	Function																							
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.																							
DQS[9:0]_A_c DQS[9:0]_B_c																										
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.																							
TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.																							

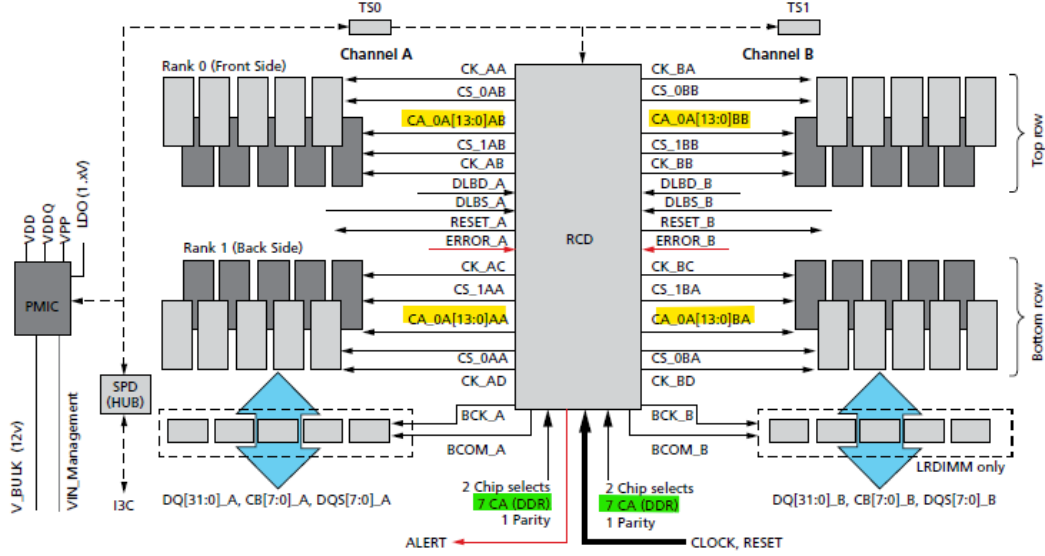
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Claim Language	'407 Accused Products ⁴
	6.1 Signal Groups This standard categorizes DDR5 SDRAM timing-critical signals into specific groups. Figure 2 — (Example) RDIMM 2Rx4 Topology summarizes the high-speed signal groups for RDIMM. Signal groups, except Data and Strobe, implement a fly-by topology. Actual signal names in Annex may vary from individual reference design. The signal groups are: Pre-RCD - Pre-RCD CA and CS group includes: DCA[6:0]_[B:A], DCS[1:0]_[B:A]_n, DPAR_[B:A] - Pre-RCD Clock group includes: DCK_t and DCK_c - Pre-RCD Other group includes: ALERT_n, DRST_n - Pre-RCD Loopback group includes: DLBDQ_[B:A], DLBDQS_[B:A] Post-RCD to SDRAM - Post-RCD CA group includes Q[B:A]CA[13:0]_[B:A] - Post-RCD CS group includes Q[B:A]CS[1:0]_[B:A]_n - Post-RCD Clock group includes Q[D:A]CK_[B:A]_t, Q[D:A]CK_[B:A]_c - Post-RCD Other group includes QRST_[B:A]_n, DERROR_[B:A]_n - Post-RCD Other group includes QLBDQ_[B:A], QLBDQS_[B:A] DQ and DQS - edge connector pin to SDRAM with series resistor SideBand - Host to Hub group: HSDA, HSCL, HSA - Hub local bus group: LSDA, LSCL (star topology from Hub) Ex. CC-88 (JEDEC 305B) at 12 (illustrating circuitry inputs for receiving clock signal lines (<i>e.g.</i>, “DCK_t” and “DCK_c”) and control/address (C/A) signal lines (<i>e.g.</i>, “DCA[]_A,” “DCA[]_B,” “DCS[]_A_n,” and “DCS[]_B_n”) from the memory controller).

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Claim Language	'407 Accused Products ⁴
	6.1 Signal Groups (cont'd)  Figure 2 — (Example) RDIMM 2Rx4 Topology Ex. CC-88 (JEDEC 305B) at 13 (illustrating RCD, in red, receiving “Pre-RCD” control and address signals (“DCA” and “DCS”) and clock signals (“DCK”) from the memory controller).
[1.c.2] generate module C/A signals in response to the input C/A signals;	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to generate module C/A signals in response to the input C/A signals. For example, the RCD generates module C/A signals (yellow) in response to input C/A signals (green):

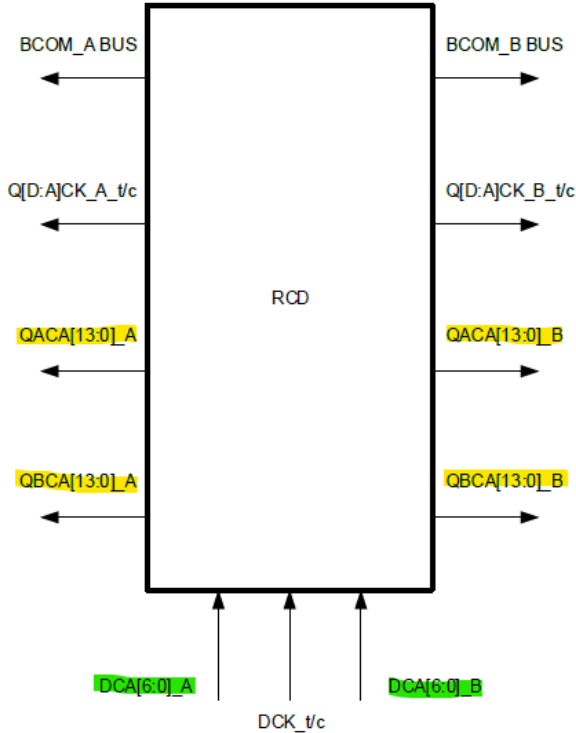
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Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

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	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

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Claim Language	'407 Accused Products ⁴
	Block Diagram  Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. The JEDEC DDR5 specification further demonstrates that the '407 Accused Products generate module C/A signals in response to the input C/A signals.

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	Registering Clock Driver Operation Registered DDR5 SDRAM modules use a registering clock driver device consisting of a register and a phase-lock loop (PLL). The device complies with the JEDEC DDR5 RCD specification. Refer to the JESD82-511 (DDR5RCD01) and JESD82-512 (DDR5RCD02) specifications for more detail. To reduce the electrical load on the host memory controller's command, address, and control bus, Micron's RDIMMs and LRDIMMs utilize a DDR5 registering clock driver (RCD). The RCD presents a single load to the controller while redriving signals to the DDR5 SDRAM devices, which helps enable higher densities and increase signal integrity. The RCD also provides a low-jitter, low-skew PLL that redistributes a differential clock pair to multiple differential pairs of clock outputs. Ex. CC-03 (DDR5 RDIMM Datasheet) at 12. 3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> <pre>graph TD DCA_A[DCA[6:0]_A] -- green --> RCD[RCD] DCA_B[DCA[6:0]_B] -- blue --> RCD RCD -- green --> QACA_A[QACA[13:0]_A] RCD -- green --> QBCA_A[QBCA[13:0]_A] RCD -- blue --> QACA_B[QACA[13:0]_B] RCD -- blue --> QBCA_B[QBCA[13:0]_B]</pre> Figure 6 — RCD DCA and QCA Ports	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

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Ex. CC-32 (JEDEC 82-512) at 15.	<table><tr><th></th><th>POD [B:A]_c</th><th>POD</th><th>DRAM channel</th></tr><tr><td>Output Control bus</td><td>Q[B:A]CS[1:0]_[B:A]_n</td><td>POD</td><td>Chip Select signals to the DRAMs. 2 copies of each signal.</td></tr><tr><td>Output Address and Command bus</td><td>Q[B:A]CA[13:0]_[B:A]</td><td>POD</td><td>Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).</td></tr><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c</td><td>POD differential</td><td>Clock outputs to the DRAMs. Four copies per channel.</td></tr></table>		POD [B:A]_c	POD	DRAM channel	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
	POD [B:A]_c	POD	DRAM channel														
Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.														
Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).														
Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.														
Ex. CC-32 (JEDEC 82-512) at 4.	Host Side DRAM Side The diagram illustrates the electrical interface between a Host Side and a DRAM Side. On the Host Side, signals include DCS[1:0]A_n, DCS[1:0]B_n, DCA_[6:0]A, DCA_[6:0]B, ALERT_n, LBDQ, RSP_A_n, LBDQS, RSP_B_n, PAR_A, PAR_B, HSCL, HSDA, CK_c, CK_t, and RESET_n. On the DRAM Side, signals include QACS[1:0][B:A]_n, QBACS[1:0][B:A]_n, QACA[13:0][B:A], QBACA[13:0][B:A], DERROR_IN_[B:A]_n, DLBD_[B:A], DLBS_[B:A], Q[D:A]CK[A:B]_c, Q[D:A]CK[A:B]_t, and QRST[B:A]. The central block represents the DRAM module, with pins R, E, G, I, S, T, E, R, &, P, S, C, and ZQ. The ZQ pin is connected to Vss.																
Ex. CC-07 (DDR5 RDIMM Addendum) at 8.																	

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	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

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	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
	Clock outputs	Q[D:A]CK_[B:A]_f Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
Ex. CC-30 (JEDEC 82-513) at 4. Furthermore, the Accused Products comply with JEDEC DDR5 RCD specification. <i>See, e.g.,</i> 3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.				

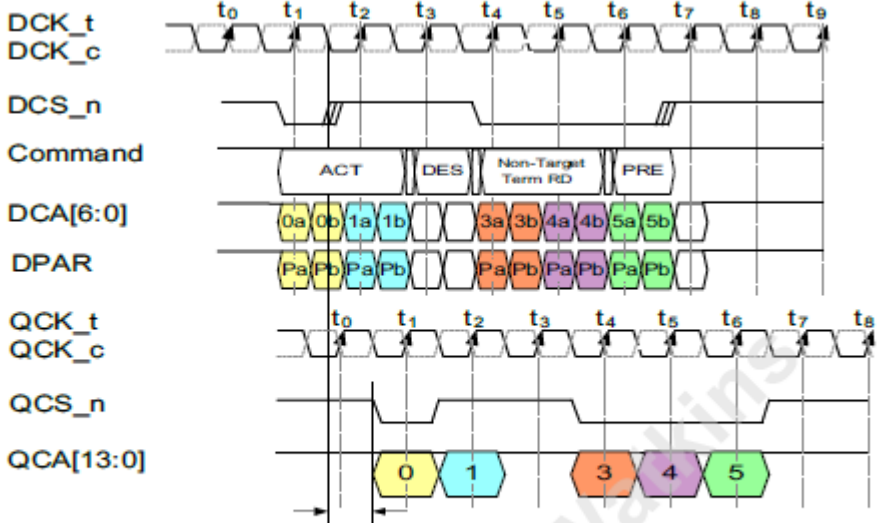
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	Ex. CC-32 (JEDEC 82-512) at 7 (showing the RCD circuitry generates module C/A signals shown in yellow in response to the input C/A signals shown in green).

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	2.2 Terminal Functions Table 2 — Terminal Functions <table><tr><th>Signal Group</th><th>Signal Name</th><th>Type</th><th>Description</th></tr><tr><td>Input Control bus</td><td>DCS[1:0]_[B:A]_n</td><td>POD V_{REF} based</td><td>Chip Select inputs to the RCD. Two inputs per DRAM channel.</td></tr><tr><td>Input Address and Command bus</td><td>DCA[6:0]_[B:A]</td><td>POD V_{REF} based</td><td>Command/Address bus inputs to the RCD. Separate sets per channel.</td></tr><tr><td>Parity input</td><td>DPAR_[B:A]</td><td>POD V_{REF} based</td><td>Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.</td></tr><tr><td>Clock inputs</td><td>DCK_t/DCK_c</td><td>POD differential</td><td>Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.</td></tr><tr><td>Loopback</td><td>DLBD_[B:A] DLBS_[B:A]</td><td>POD V_{REF} based</td><td>Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe</td></tr><tr><td>Reset input</td><td>DRST_n</td><td>CMOS input</td><td>Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.</td></tr><tr><td>Error input</td><td>DERROR_IN_[B:A]_n</td><td>Low voltage swing POD input</td><td>DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.</td></tr><tr><td rowspan="4">Data buffer control and communication outputs</td><td>BCS_[B:A]_n</td><td>POD</td><td>Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.</td></tr><tr><td>BCOM[2:0]_[B:A]</td><td>POD</td><td>Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.</td></tr><tr><td>BRST_[B:A]_n</td><td>CMOS</td><td>Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.</td></tr><tr><td>BCK_[B:A]_t BCK_[B:A]_c</td><td>POD differential</td><td>Differential clock output pair to the data buffer. Separate per DRAM channel.</td></tr><tr><td>Output Control bus</td><td>Q[B:A]CS[1:0]_[B:A]_n</td><td>POD</td><td>Chip Select signals to the DRAMs. 2 copies of each signal.</td></tr><tr><td>Output Address and Command bus</td><td>Q[B:A]CA[13:0]_[B:A]</td><td>POD</td><td>Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).</td></tr><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t</td><td>POD differential</td><td>Clock outputs to the DRAMs. Four copies per channel.</td></tr></table>	Signal Group	Signal Name	Type	Description	Input Control bus	DCS[1:0]_[B:A]_n	POD V _{REF} based	Chip Select inputs to the RCD. Two inputs per DRAM channel.	Input Address and Command bus	DCA[6:0]_[B:A]	POD V _{REF} based	Command/Address bus inputs to the RCD. Separate sets per channel.	Parity input	DPAR_[B:A]	POD V _{REF} based	Command Address input parity is received on the DPAR pin and should maintain even parity across the CA inputs. DPAR is sampled at the rising and falling edges of the input clock.	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.	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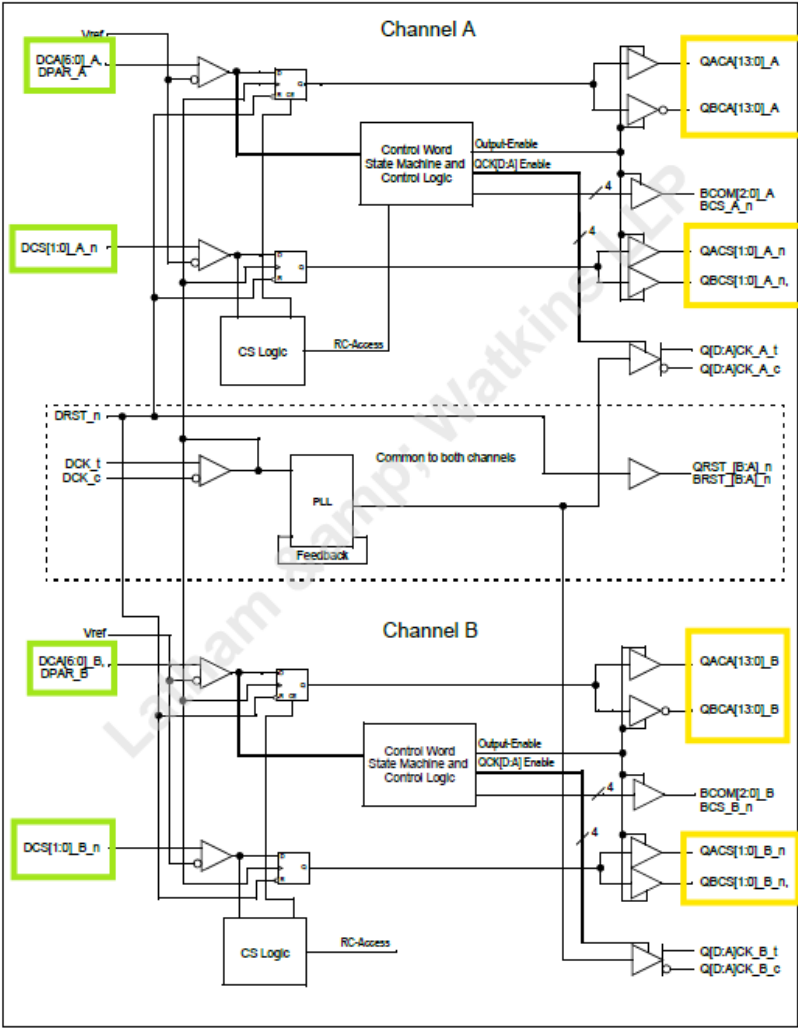
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	3.3.3 QCA Bus Operation Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this: • If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked. • During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is signaled to a rank. • When DRAM interface Block MR Mode is enabled <code>RW01[1] = 0</code>. • When CA Pass-through mode is enabled <code>RW00[2] = 1</code>. Ex. CC-32 (JEDEC 82-512) at 9.  NOTE Timing Diagram for 0 nCK command latency adder. Figure 9 — DDR CA Multi-Command Timing Diagram Example Ex. CC-32 (JEDEC 82-512) at 17 (timing diagram showing the generation of module C/A signals (QCA[13:0]) in response to the input C/A signals (DCA[6:0]).

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	3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.

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	 The diagram illustrates a dual-channel RCD circuit, labeled Channel A and Channel B. Each channel processes input signals DCA[6:0] and DPAR to generate output signals QACA[13:0] and QBCA[13:0]. The circuit includes a common PLL block and CS Logic. Input signals DCA[6:0] and DPAR are highlighted in green boxes. Output signals QACA[13:0] and QBCA[13:0] are highlighted in yellow boxes. The diagram also shows intermediate signals like BCOM[2:0] and BCS, and control signals like Q[D:A]CK_A_t and Q[D:A]CK_A_c. A common PLL block is shared between both channels, receiving DCK_t and DCK_c inputs and outputting QDST[B:A]_n and BRST[B:A]_n signals. Ex. CC-30 (JEDEC 82-513) at 7 (showing the RCD circuitry generates module C/A signals shown in yellow in response to the input C/A signals shown in green).

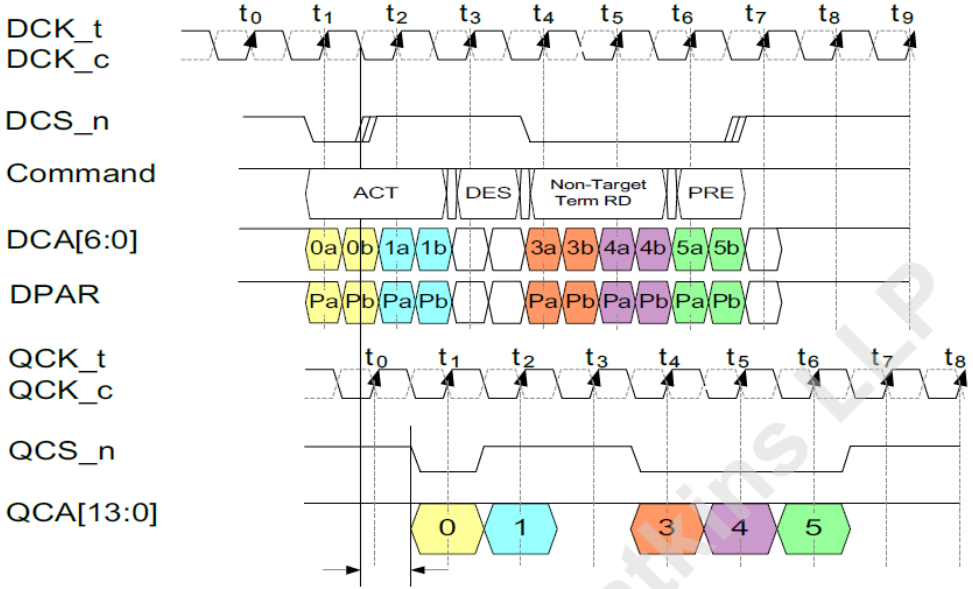
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It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.		BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.	Output	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.	Control bus				Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
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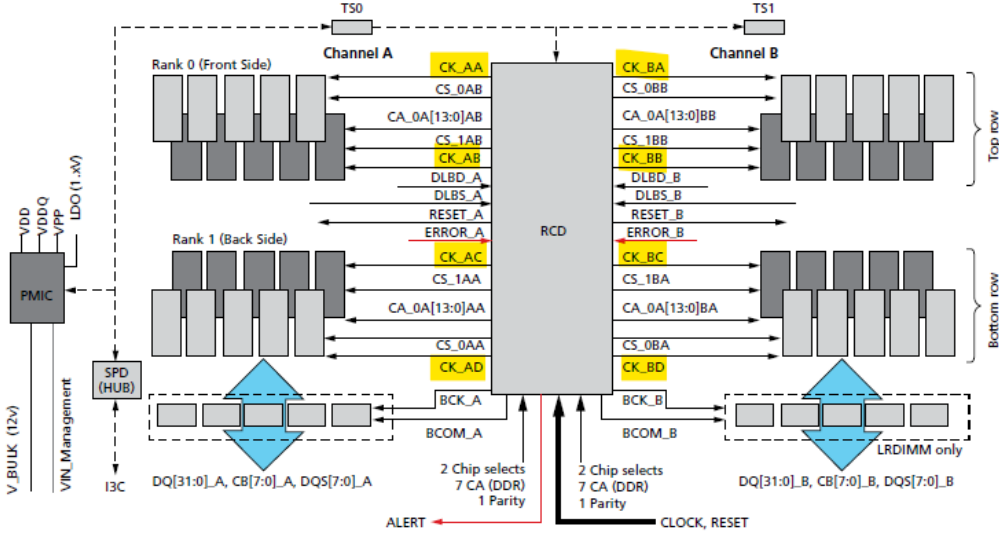
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	3.3.3 QCA Bus Operation Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this: • If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked. • During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank. • When DRAM interface Block MR Mode is enabled <code>RW01[1] = 0</code>. • When CA Pass-through mode is enabled <code>RW00[2] = 1</code>. Ex. CC-30 (JEDEC 82-513) at 9.

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	3.4.1 Double Data Rate DCA (cont'd)  NOTE Timing Diagram for 0 nCK command latency adder. Figure 9 — DDR CA Multi-Command Timing Diagram Example Ex. CC-30 (JEDEC 82-513) at 17 (timing diagram showing the generation of module C/A signals (QCA[13:0]) in response to the input C/A signals (DCA[6:0]).
[1.c.3] generate a plurality of local clocks corresponding, respectively, to the plurality of groups	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to generate a plurality of local clocks corresponding, respectively, to the plurality of groups of the memory devices. For example, the RCD circuitry is configurable to generate local clock outputs corresponding to the groups of memory devices associated with Channels A and B in different ranks. <i>See, e.g.,</i>

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of the memory devices,	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

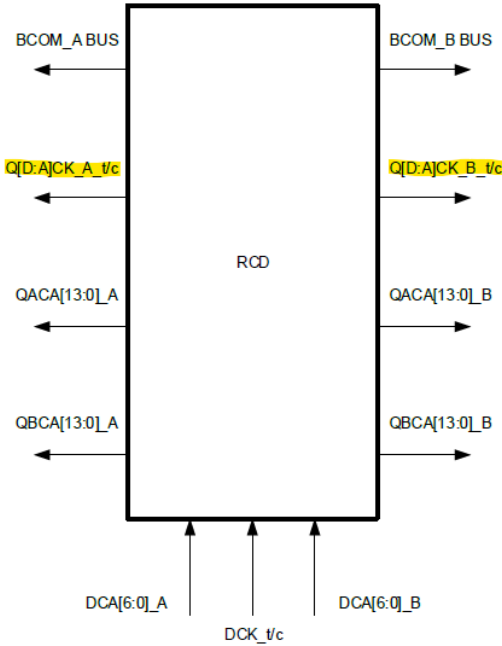
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	Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing generation of local clocks and reception of the system clock). To reduce the electrical load on the host memory controller's command, address, and control bus, Micron's RDIMMs and LRDIMMs utilize a DDR5 registering clock driver (RCD). The RCD presents a single load to the controller while redriving signals to the DDR5 SDRAM devices, which helps enable higher densities and increase signal integrity. The RCD also provides a low-jitter, low-skew PLL that redistributes a differential clock pair to multiple differential pairs of clock outputs. Ex. CC-03 (DDR5 RDIMM Datasheet) at 12 (stating the RCD provides multiple clock outputs).

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	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

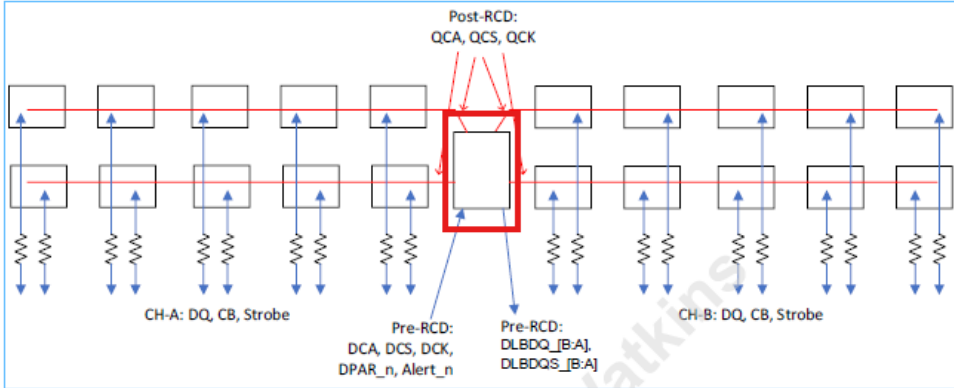
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Claim Language	'407 Accused Products ⁴
	Block Diagram  Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. As described in the JEDEC 82-512 and 82-513 standards that the '407 Accused Products comply with, the RCD has a “common clock input and PLL, but produces separate clock outputs to the DRAM channels.” <i>See, e.g.,</i>

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Claim Language	'407 Accused Products ⁴
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6. 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.

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Claim Language	'407 Accused Products ⁴
	6.1 Signal Groups (cont'd)  Figure 2 — (Example) RDIMM 2Rx4 Topology Ex. CC-31 (JESD305B.01 (April 2026)) at 13 (illustrating RCD, in red, receiving “Pre-RCD” clock (“DCK”) and generating “Post-RCD” control and address signals (“QCK”).

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

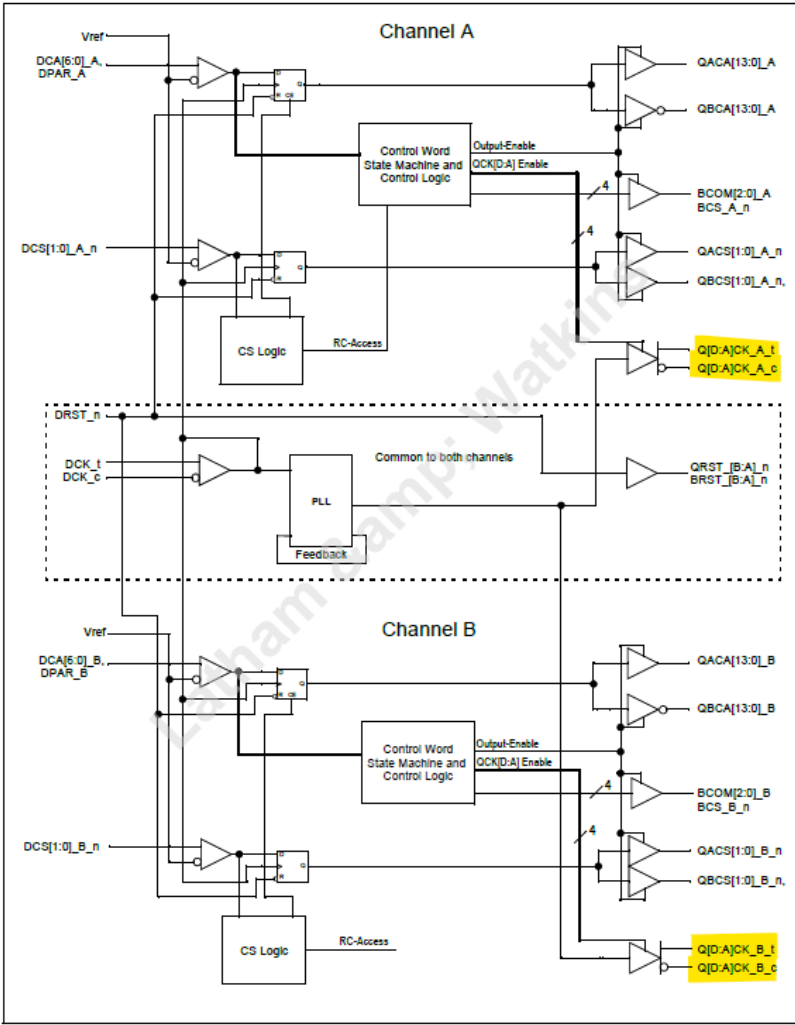
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Claim Language	'407 Accused Products ⁴																																																																																																																																																																						
	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="11">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td>CB[7:0]</td></tr><tr><th colspan="11">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td>[7:0]</td><td></td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td>CB[7:0]</td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive corresponding local clocks, e.g., QACK, QBCK, QCK, QDCK, generated by the RCD). Furthermore, the '407 Accused Products comply with the JEDEC 82-512 and 82-513 standards, which further demonstrates that the RCD includes local clocks corresponding, respectively, to the plurality of groups of the memory devices. <table><tr><td>Clock outputs</td><td>Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c</td><td>POD differential</td><td>cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.</td></tr><tr><td>nonback</td><td>OT RD</td><td>POD</td><td>nonback outputs to Host</td></tr></table> Ex. CC-32 (JEDEC 82-512) at 4.	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCK	QACS0	QACA	0	Bottom	Lower	QCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top				[15:8]				[31:24]			Bottom		[7:0]				[23:16]				CB[7:0]	1Rx8 Sub Channel 1 (Channel B)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top		[7:0]				[23:16]					Bottom				[15:8]				[31:24]		CB[7:0]	Clock outputs	Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.	nonback	OT RD	POD	nonback outputs to Host
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Bottom		[7:0]				[23:16]				CB[7:0]																																																																																																																																																													
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Clock outputs	Q[D:A]CK [B:A]_t Q[D:A]CK [B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.																																																																																																																																																																				
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Claim Language	'407 Accused Products ⁴
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.

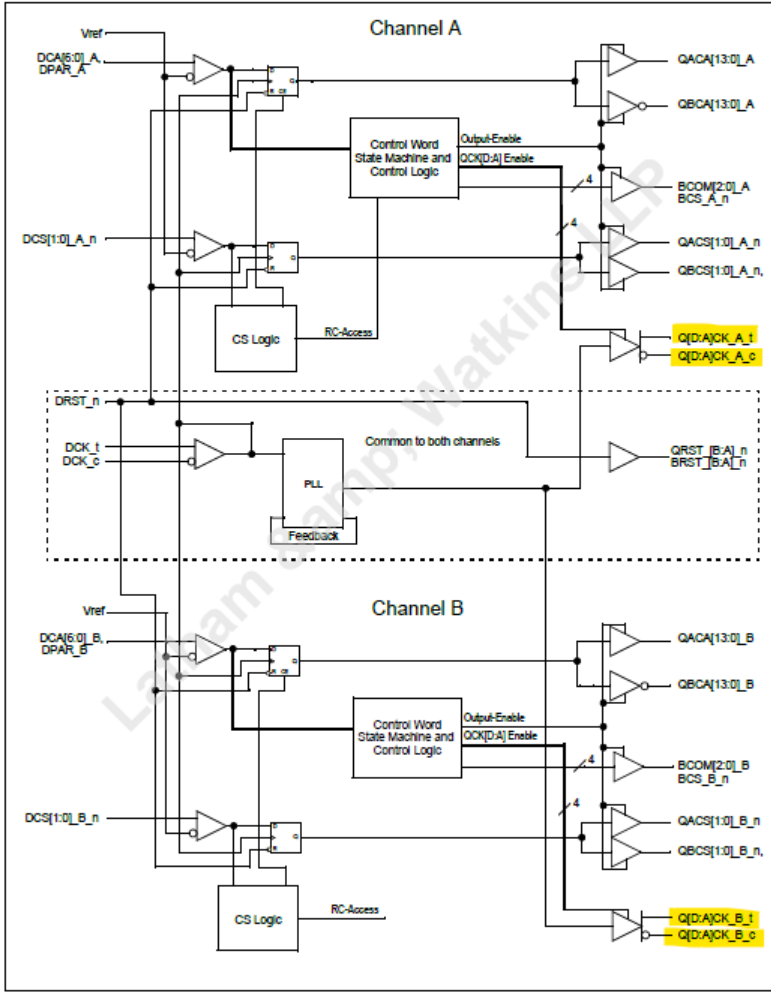
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Claim Language	'407 Accused Products ⁴
	3.2 Logic Diagram Positive Logic  Ex. CC-32 (JEDEC 82-512) at 7.

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Claim Language	'407 Accused Products ⁴				
	<table><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c</td><td>POD differential</td><td>Clock outputs to the DRAMs. Four copies per channel.</td></tr></table> Ex. CC-30 (JEDEC 82-513) at 4. 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.		

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Claim Language	'407 Accused Products ⁴
	3.2 Logic Diagram Positive Logic  Ex. CC-30 (JEDEC 82-513) at 7.

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Claim Language	'407 Accused Products ⁴
[1.c.3.i] the plurality of local clocks having respective phase relationships with the system clock,	In the '407 Accused Products, the plurality of local clocks have respective phase relationships with the system clock. For example: Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1.

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	The RCD of the '407 Accused Products support “output delay control” where the local clocks have respective phase relationships with the system clock. <i>See, e.g.,</i> 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disable independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-32 (JEDEC 82-512) at 12. Table 23 — Control Words Duplicated for Frequency Context Support <table> <tr> <th>Register Address</th><th>Description</th></tr> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCCK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </table>	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCCK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
Register Address	Description																																								
RW0A	QCK Signals Driver Characteristics Control Word																																								
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RW13	QBCK Output Delay Control Word																																								
RW14	QCCK Output Delay Control Word																																								
RW15	QDCK Output Delay Control Word																																								
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RW1B, RW1C	QxCA Output Delay Control Word																																								
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RW40 .. RW47	Internal VrefCA Control Word																																								
RW48, RW49	Internal VrefCS Control Word																																								
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U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																																																																																																										
	Ex. CC-32 (JEDEC 82-512) at 68 (showing local clocks QACK, QBCK, QCK, and QDCK that have respective phase relationships with the system clock). 8.13.2 RW12 - QACK Output Delay Control Word Table 118 — RW12: QACK Output Delay Control Word <table><tr><th colspan="8">Setting</th><th rowspan="2">Definition</th><th rowspan="2">Encoding</th></tr><tr><th>OP 7</th><th>OP 6</th><th>OP 5</th><th>OP 4</th><th>OP 3</th><th>OP 2</th><th>OP 1</th><th>OP 0</th></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td rowspan="4">Output Delay Control for QACK_t/ QACK_c Output Signals^{1,2}</td><td>Delay Outputs by +(0/64) * t_{CK} (Same as Default)</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Delay Outputs by +(1/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Delay Outputs by +(2/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td colspan="6">...</td><td>...</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td rowspan="4">Reserved</td><td>Delay Outputs by +(61/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Delay Outputs by +(62/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Delay Outputs by +(63/64) * t_{CK}</td></tr><tr><td>x</td><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td></tr><tr><td>x</td><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td><td>Reserved</td></tr><tr><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Output Delay Feature Enable for QACK_t/QACK_c²</td><td>(Default) Feature Disabled</td></tr><tr><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td></td><td>Feature Enabled³</td></tr></table> NOTE 1 These control bits do not have any effect unless RW12[7] = 1. NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset. NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{DDP} for the delay to become stable on the outputs. Ex. CC-32 (JEDEC 82-512) at 146; <i>see also id.</i> at 147-148 (showing that local clocks QACK, QBCK, QCK, and QDCK have respective phase relationships with the system clock). 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 t_{CK}. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{DDP} after the Output Delay Control Word has been written to a signal group before the output becomes stable. 3.3.8 Per-group Output Delay Control (cont'd) The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111).	Setting								Definition	Encoding	OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0	x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)	x	x	0	0	0	0	0	1	Delay Outputs by +(1/64) * t _{CK}	x	x	0	0	0	0	1	0	Delay Outputs by +(2/64) * t _{CK}	x	x	...						...	x	x	1	1	1	1	0	1	Reserved	Delay Outputs by +(61/64) * t _{CK}	x	x	1	1	1	1	1	0	Delay Outputs by +(62/64) * t _{CK}	x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	x	0	x	x	x	x	x	x	Reserved	x	1	x	x	x	x	x	x	Reserved	Reserved	0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled	1	x	x	x	x	x	x	x		Feature Enabled ³
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OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0																																																																																																																				
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)																																																																																																																		
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}																																																																																																																		
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}																																																																																																																		
x	x	...							...																																																																																																																		
x	x	1	1	1	1	0	1	Reserved	Delay Outputs by +(61/64) * t _{CK}																																																																																																																		
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}																																																																																																																		
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x	0	x	x	x	x	x	x		Reserved																																																																																																																		
x	1	x	x	x	x	x	x	Reserved	Reserved																																																																																																																		
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled																																																																																																																		
1	x	x	x	x	x	x	x		Feature Enabled ³																																																																																																																		

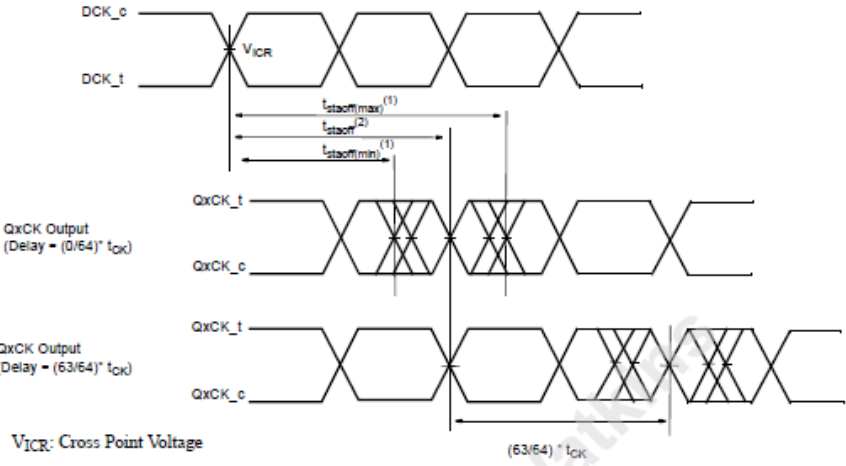
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																										
	Ex. CC-30 (JEDEC 82-513) at 11-12. Table 23 — Control Words Duplicated for Frequency Context Support <table> <tr> <th>Register Address</th><th>Description</th></tr> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCKK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW33[7:6]</td><td>DFE Tap 5 and Tap 6 Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </table> Ex. CC-30 (JEDEC 82-513) at 70 (showing local clocks QACK, QBCK, QCKK, and QDCK that have respective phase relationships with the system clock).	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCKK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW33[7:6]	DFE Tap 5 and Tap 6 Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
Register Address	Description																																										
RW0A	QCK Signals Driver Characteristics Control Word																																										
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RW11[3:0]	Command Latency Adder Configuration Control Word																																										
RW12	QACK Output Delay Control Word																																										
RW13	QBCK Output Delay Control Word																																										
RW14	QCKK Output Delay Control Word																																										
RW15	QDCK Output Delay Control Word																																										
RW17 .. RW1A	QxCS_n Output Delay Control Word																																										
RW1B, RW1C	QxCA Output Delay Control Word																																										
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	8.13.2 RW12 - QACK Output Delay Control Word Table 119 — RW12: QACK Output Delay Control Word <table><tr><th colspan="8">Setting</th><th rowspan="2">Definition</th><th rowspan="2">Encoding</th></tr><tr><th>OP 7</th><th>OP 6</th><th>OP 5</th><th>OP 4</th><th>OP 3</th><th>OP 2</th><th>OP 1</th><th>OP 0</th></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td rowspan="7">Output Delay Control for QACK_t/ QACK_c Output Signals^{1,2}</td><td>Delay Outputs by +(0/64) * t_{CK} (Same as Default)</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Delay Outputs by +(1/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Delay Outputs by +(2/64)* t_{CK}</td></tr><tr><td>x</td><td>x</td><td colspan="7">...</td><td>...</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Delay Outputs by +(61/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Delay Outputs by +(62/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Delay Outputs by +(63/64) * t_{CK}</td></tr><tr><td>x</td><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td rowspan="2">Reserved</td><td>Reserved</td></tr><tr><td>x</td><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td></tr><tr><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td rowspan="2">Output Delay Feature Enable for QACK_t/QACK_c²</td><td>(Default) Feature Disabled</td></tr><tr><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Feature Enabled³</td></tr></table> NOTE 1 These control bits do not have any effect unless RW12[7] = 1. NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset. NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs. Ex. CC-30 (JEDEC 82-513) at 150; <i>see also id.</i> at 150-156 (showing that local clocks QACK, QBCK, QCK, and QDCK have respective phase relationships with the system clock). For example, local clocks (QxCK) shown below have respective phase relationships with respect to the system clock (DCK).	Setting								Definition	Encoding	OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0	x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)	x	x	0	0	0	0	0	1	Delay Outputs by +(1/64) * t _{CK}	x	x	0	0	0	0	1	0	Delay Outputs by +(2/64)* t _{CK}	x	x	...							...	x	x	1	1	1	1	0	1	Delay Outputs by +(61/64) * t _{CK}	x	x	1	1	1	1	1	0	Delay Outputs by +(62/64) * t _{CK}	x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	x	0	x	x	x	x	x	x	Reserved	Reserved	x	1	x	x	x	x	x	x	Reserved	0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled	1	x	x	x	x	x	x	x	Feature Enabled ³
Setting								Definition	Encoding																																																																																																																	
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0																																																																																																																			
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)																																																																																																																	
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x	x	0	0	0	0	1	0		Delay Outputs by +(2/64)* t _{CK}																																																																																																																	
x	x	...							...																																																																																																																	
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}																																																																																																																	
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}																																																																																																																	
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}																																																																																																																	
x	0	x	x	x	x	x	x	Reserved	Reserved																																																																																																																	
x	1	x	x	x	x	x	x		Reserved																																																																																																																	
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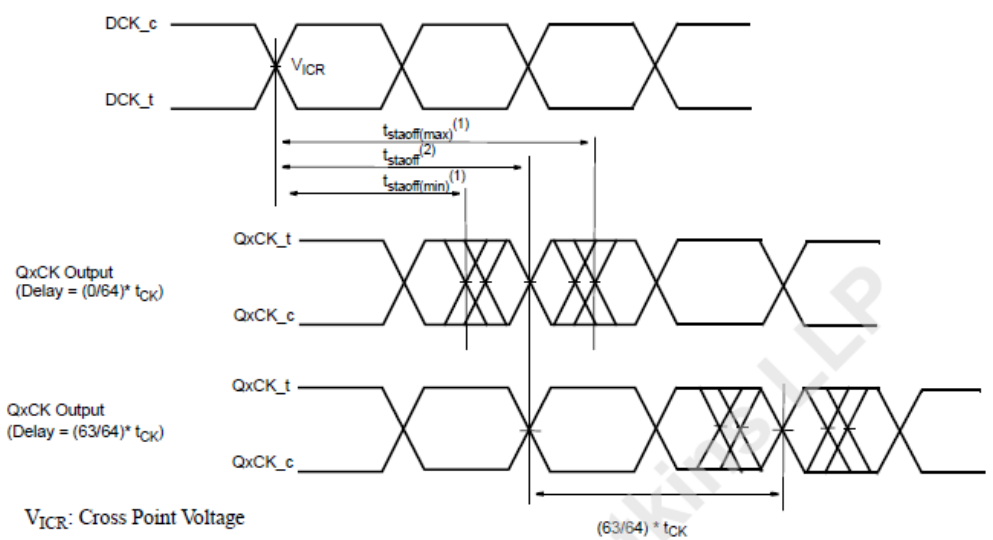
U.S. Patent No. 12,675,407 Infringement Claim Chart

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	14.7 Output Clock Driver Characteristics (cont'd)  Figure 112 — Definition for t_{staoff} and t_{dynoff} in SDR Mode Ex. CC-32 (JEDEC 82-512) at 242.

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x	0	x	x	x	x	x	x	Reserved	Reserved																																																																																																																		
x	1	x	x	x	x	x	x	Reserved	Reserved																																																																																																																		
0	x	x	x	x	x	x	x	Output Delay Feature Enable for	(Default) Feature Disabled																																																																																																																		
1	x	x	x	x	x	x	x	QACK_t/QACK_c ²	Feature Enabled ³																																																																																																																		
	Ex. CC-32 (JEDEC 82-512) at 146; <i>see also id.</i> at 147-148 (showing tables 119, 120, and 121).																																																																																																																										

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	14.7 Output Clock Driver Characteristics (cont'd)  V_{ICR}: Cross Point Voltage 1. $t_{staoft(max/min)}$ = propagation delay specification limits for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles) over process, voltage and temperature. 2. t_{staoft} = measured propagation delay for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles). 3. t_{dynoff} = maximum propagation delay variation over voltage and temperature within t_{staoft} window. This includes all sources of jitter and drift (e.g. thermal noise, supply noise, voltage/temperature drift, SSC tracking, SSO, etc.) except reference clock noise. Figure 112 — Definition for t_{staoft} and t_{dynoff} in SDR Mode Ex. CC-30 (JEDEC 82-513) at 253.

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Claim Language	'407 Accused Products ⁴																																																																																																																																																						
	8.13.2 RW12 - QACK Output Delay Control Word Table 119 — RW12: QACK Output Delay Control Word <table><tr><th colspan="8">Setting</th><th rowspan="2">Definition</th><th rowspan="2">Encoding</th></tr><tr><th>OP 7</th><th>OP 6</th><th>OP 5</th><th>OP 4</th><th>OP 3</th><th>OP 2</th><th>OP 1</th><th>OP 0</th></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td rowspan="7">Output Delay Control for QACK_t/ QACK_c Output Signals^{1,2}</td><td>Delay Outputs by +(0/64) * t_{CK} (Same as Default)</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Delay Outputs by +(1/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Delay Outputs by +(2/64)* t_{CK}</td></tr><tr><td>x</td><td>x</td><td colspan="6">...</td><td>...</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Delay Outputs by +(61/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Delay Outputs by +(62/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Delay Outputs by +(63/64) * t_{CK}</td></tr><tr><td>x</td><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td rowspan="2">Reserved</td><td>Reserved</td></tr><tr><td>x</td><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td></tr><tr><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td rowspan="2">Output Delay Feature Enable for QACK_t/QACK_c²</td><td>(Default) Feature Disabled</td></tr><tr><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Feature Enabled³</td></tr><tr><td colspan="10">NOTE 1 These control bits do not have any effect unless RW12[7] = 1.</td></tr><tr><td colspan="10">NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.</td></tr><tr><td colspan="10">NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.</td></tr></table> Ex. CC-30 (JEDEC 82-513) at 150; <i>see also id.</i> at 150-156 (showing tables 120, 121, and 122).	Setting								Definition	Encoding	OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0	x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)	x	x	0	0	0	0	0	1	Delay Outputs by +(1/64) * t _{CK}	x	x	0	0	0	0	1	0	Delay Outputs by +(2/64)* t _{CK}	x	x	...						...	x	x	1	1	1	1	0	1	Delay Outputs by +(61/64) * t _{CK}	x	x	1	1	1	1	1	0	Delay Outputs by +(62/64) * t _{CK}	x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	x	0	x	x	x	x	x	x	Reserved	Reserved	x	1	x	x	x	x	x	x	Reserved	0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled	1	x	x	x	x	x	x	x	Feature Enabled ³	NOTE 1 These control bits do not have any effect unless RW12[7] = 1.										NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.										NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t _{ODU} for the delay to become stable on the outputs.									
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x	0	x	x	x	x	x	x	Reserved	Reserved																																																																																																																																														
x	1	x	x	x	x	x	x		Reserved																																																																																																																																														
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[1.c.3.ii] the respective phase relationships being programmable independently of each other; and	In the '407 Accused Products, the respective phase relationships are programmable independently of each other. For example, the delay control of the local clocks (QxCK) with respect to the system clock (DCK) can be independently programmed. <i>See, e.g.,</i>																																																																																																																																																						

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Claim Language	'407 Accused Products ⁴																																								
	3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-32 (JEDEC 82-512) at 12 (showing “output delay” for output signal groups). Table 23 — Control Words Duplicated for Frequency Context Support <table border="1"> <thead> <tr> <th>Register Address</th><th>Description</th></tr> </thead> <tbody> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCs_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCKK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCs_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </tbody> </table>	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCs_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCKK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCs_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
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	Ex. CC-32 (JEDEC 82-512) at 68 (showing delays in QACK, QBCK, QCKK, and QDCK can be set independently). 8.13.2 RW12 - QACK Output Delay Control Word Table 118 — RW12: QACK Output Delay Control Word <table><tr><th colspan="8">Setting</th><th rowspan="2">Definition</th><th rowspan="2">Encoding</th></tr><tr><th>OP 7</th><th>OP 6</th><th>OP 5</th><th>OP 4</th><th>OP 3</th><th>OP 2</th><th>OP 1</th><th>OP 0</th></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td rowspan="7">Output Delay Control for QACK_ t/ QACK_ c Output Signals^{1,2}</td><td>Delay Outputs by +(0/64) * t_{CK} (Same as Default)</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Delay Outputs by +(1/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Delay Outputs by +(2/64)* t_{CK}</td></tr><tr><td>x</td><td>x</td><td colspan="6">...</td><td>...</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Delay Outputs by +(61/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Delay Outputs by +(62/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Delay Outputs by +(63/64) * t_{CK}</td></tr><tr><td>x</td><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td><td>Reserved</td></tr><tr><td>x</td><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td><td>Reserved</td></tr><tr><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Output Delay Feature Enable for QACK_ t/QACK_ c²</td><td>(Default) Feature Disabled</td></tr><tr><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td></td><td>Feature Enabled³</td></tr></table> NOTE 1 These control bits do not have any effect unless RW12[7] = 1. NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset. NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs. Ex. CC-32 (JEDEC 82-512) at 146; <i>see also id.</i> at 147-148 (showing how to program the delays independently, in tables 119, 120, and 121). 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 t_{CK}. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. 3.3.8 Per-group Output Delay Control (cont'd) The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111).	Setting								Definition	Encoding	OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0	x	x	0	0	0	0	0	0	Output Delay Control for QACK_ t/ QACK_ c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)	x	x	0	0	0	0	0	1	Delay Outputs by +(1/64) * t _{CK}	x	x	0	0	0	0	1	0	Delay Outputs by +(2/64)* t _{CK}	x	x	...						...	x	x	1	1	1	1	0	1	Delay Outputs by +(61/64) * t _{CK}	x	x	1	1	1	1	1	0	Delay Outputs by +(62/64) * t _{CK}	x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	x	0	x	x	x	x	x	x	Reserved	Reserved	x	1	x	x	x	x	x	x	Reserved	Reserved	0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_ t/QACK_ c ²	(Default) Feature Disabled	1	x	x	x	x	x	x	x		Feature Enabled ³
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RW40 .. RW47	Internal VrefCA Control Word																																										
RW48, RW49	Internal VrefCS Control Word																																										
RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words																																										
PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver																																										
PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words																																										

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language

'407 Accused Products⁴

8.13.2 RW12 - QACK Output Delay Control Word

Table 119 — RW12: QACK Output Delay Control Word

Setting								Definition	Encoding
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0		
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64)* t _{CK}
x	x	...							...
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}
x	0	x	x	x	x	x	x	Reserved	Reserved
x	1	x	x	x	x	x	x	Reserved	Reserved
0	x	x	x	x	x	x	x	Output Delay Feature Enable for	(Default) Feature Disabled
1	x	x	x	x	x	x	x	QACK_t/QACK_c ²	Feature Enabled ³

NOTE 1

These control bits do not have any effect unless RW12[7] = 1.

NOTE 2

RW12[7:0] will be sticky, cleared by power cycle not Reset.

NOTE 3

When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.

Ex. CC-30 (JEDEC 82-513) at 150; *see also id.* at 150-156 (showing how to program the delays independently, in tables 120, 121, and 122).

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																																																									
	Table 232 — Output Clock Driver Characteristics at Application Mode (Frequency Band 1) (cont'd) <table><tr><th rowspan="2">Symbol</th><th rowspan="2">Parameter</th><th rowspan="2">Conditions</th><th colspan="2">DDR5-3200</th><th colspan="2">DDR5-3600</th><th colspan="2">DDR5-4000/ 4400/4800</th><th colspan="2">DDR5-5200/ 5600</th><th rowspan="2">Unit</th><th rowspan="2">Note</th></tr><tr><th>Min</th><th>Max</th><th>Min</th><th>Max</th><th>Min</th><th>Max</th><th>Min</th><th>Max</th></tr><tr><td>tSTAB_DBDATA</td><td>LRDIMM stabilization time for commands requiring use of data path</td><td>DCK_t/DCK_c stable</td><td>-</td><td>4.14</td><td>-</td><td>4.14</td><td>-</td><td>4.14</td><td>-</td><td>4.14</td><td>μs</td><td>4,5</td></tr><tr><td>tCKsk</td><td>Fractional Clock Output Skew</td><td></td><td>-</td><td>20</td><td>-</td><td>20</td><td>-</td><td>20</td><td>-</td><td>20</td><td>ps</td><td>6</td></tr><tr><td>tstaoff</td><td>Clock delay through the register between the input clock and output clock</td><td></td><td colspan="8">Min:(tPDM(min) + 1/2 tCK); Max: (tPDM(max) + 1/2 tCK)</td><td>ps</td><td>7,8</td></tr><tr><td>tdynoff</td><td>Maximum re-driven dynamic clock offset</td><td></td><td>-</td><td>40</td><td>-</td><td>35</td><td>-</td><td>35</td><td>-</td><td>35</td><td>ps</td><td>9,10</td></tr></table> NOTE 1 At Down-bin Data Rate (1980 MT/s ≤ f ≤ 2100 MT/s), the max value of tSTAB01 is 10μs. NOTE 2 This parameter applies to control word writes to RW05/RW06 and Bit OP0 in RW00. Since application of RW05/RW06 is postponed in the frequency change sequence defined in Section 5.4.2, a tSTAB01 waiting time is always sufficient after exit from Self Refresh with Clock Stop (i.e., tSTAB02 is not required). See timing diagrams in Figure 35 on page 48 and Figure 36 on page 49. NOTE 3 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does not require the use of data path (e.g., SRX, NOP, etc.). NOTE 4 Applies only to Self Refresh Exit without frequency change. NOTE 5 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and requires the use of data path (e.g., WR, RD). NOTE 6 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCK_A_t/QCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCK_B_t/QCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only. NOTE 7 RW05/RW06 must be set to correct speed and the device must run at one of the speed nodes defined in Table 199 on page 191. See Figure 111 and Figure 112. NOTE 8 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock tPD. NOTE 9 VDD measurement DC bandwidth is limited to 20 MHz. NOTE 10 See Figure 111 and Figure 112.	Symbol	Parameter	Conditions	DDR5-3200		DDR5-3600		DDR5-4000/ 4400/4800		DDR5-5200/ 5600		Unit	Note	Min	Max	Min	Max	Min	Max	Min	Max	tSTAB_DBDATA	LRDIMM stabilization time for commands requiring use of data path	DCK_t/DCK_c stable	-	4.14	-	4.14	-	4.14	-	4.14	μs	4,5	tCKsk	Fractional Clock Output Skew		-	20	-	20	-	20	-	20	ps	6	tstaoff	Clock delay through the register between the input clock and output clock		Min:(tPDM(min) + 1/2 tCK); Max: (tPDM(max) + 1/2 tCK)								ps	7,8	tdynoff	Maximum re-driven dynamic clock offset		-	40	-	35	-	35	-	35	ps	9,10
Symbol	Parameter				Conditions	DDR5-3200		DDR5-3600		DDR5-4000/ 4400/4800		DDR5-5200/ 5600			Unit	Note																																																										
		Min	Max	Min		Max	Min	Max	Min	Max																																																																
tSTAB_DBDATA	LRDIMM stabilization time for commands requiring use of data path	DCK_t/DCK_c stable	-	4.14	-	4.14	-	4.14	-	4.14	μs	4,5																																																														
tCKsk	Fractional Clock Output Skew		-	20	-	20	-	20	-	20	ps	6																																																														
tstaoff	Clock delay through the register between the input clock and output clock		Min:(tPDM(min) + 1/2 tCK); Max: (tPDM(max) + 1/2 tCK)								ps	7,8																																																														
tdynoff	Maximum re-driven dynamic clock offset		-	40	-	35	-	35	-	35	ps	9,10																																																														
	Ex. CC-32 (JEDEC 82-512) at 240 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”). <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 250.																																																																									

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language

'407 Accused Products⁴

14.7 Output Clock Driver Characteristics (cont'd)

Table 233 — Output Clock Driver Characteristics at Test Mode (Frequency Band 2)

Symbol	Parameter	Conditions	Test Frequency (140 to 990 MHz)		Unit	Note
			Min	Max		
t _{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	50	μs	
t _{STAB02}	RCD Stabilization time from control word write to PLL lock	DCK_t/DCK_c stable	-	50	μs	1
t _{CKsk}	Fractional Clock Output Skew		-	100	ps	2
t _{stao}	Clock delay through the register between the input clock and output clock		Min: (t _{PD} (min) + 1/2 tCK); Max: (t _{PD} (max) + 1/2 tCK)		ps	3,4
t _{dyno}	Maximum re-driven dynamic clock offset		-	500	ps	5,6

NOTE 1 This parameter applies to control word writes to [RW05](#) and Bit OP0 in [RW00](#).

NOTE 2 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCCK_A_t/QCCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCCK_B_t/QCCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only.

NOTE 3 [RW05](#) must not be set to enable PLL bypass mode and the device must run within the Test Frequency range defined in Table 199 on page 191. See Figure 111 and Figure 112.

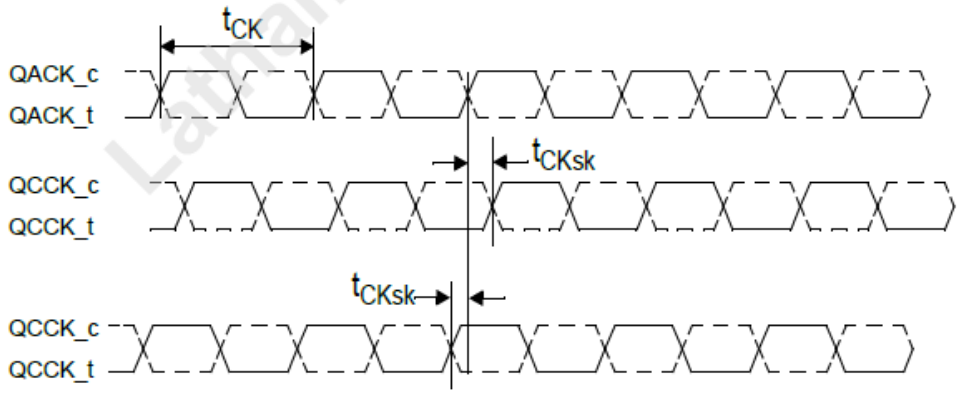
NOTE 4 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t_{PD}.

NOTE 5 V_{DD} measurement DC bandwidth is limited to 20 MHz.

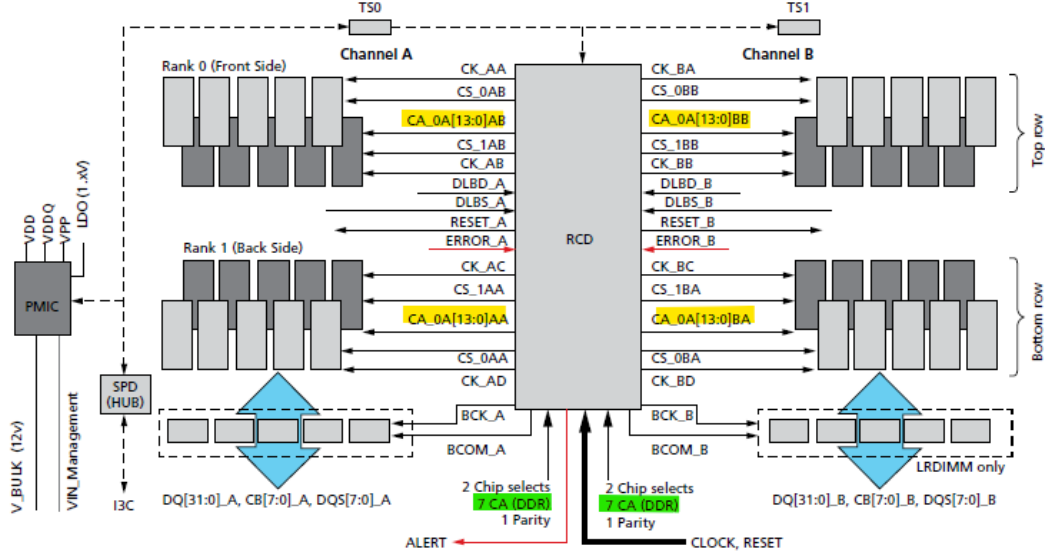
NOTE 6 See Figure 111 and Figure 112.

Ex. CC-32 (JEDEC 82-512) at 241 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”). See also Ex. CC-30 (JEDEC 82-513) at 251.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 The diagram shows four digital signals over time. The top two signals are QACK_c (dashed line) and QACK_t (solid line). The bottom two signals are QCCK_c (dashed line) and QCCK_t (solid line). A horizontal double-headed arrow labeled t_{CK} spans one full period of the QACK signals. A vertical line is drawn through the middle of the diagram. Two horizontal double-headed arrows labeled t_{CKsk} indicate the skew between the rising edges of the QCCK signals relative to the vertical line. Figure 113 — Clock Output (QxCK) Skew Ex. CC-32 (JEDEC 82-512) at 242. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 254.
[1.c.4] output the module C/A signals to the memory devices; and	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to output the module C/A signals to the memory devices. For example, the module C/A signals (yellow) are output to DRAM memory devices as shown below. <i>See, e.g.,</i>

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

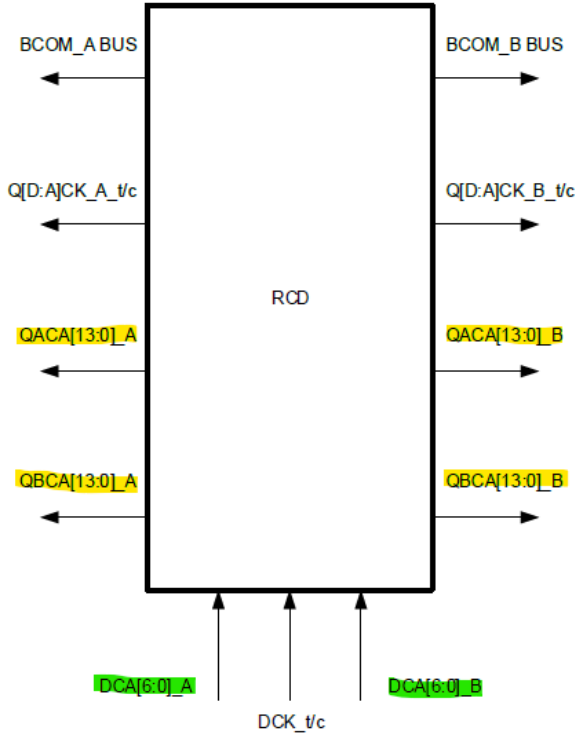
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	The diagram illustrates the electrical interface between a Host Side and a DRAM Side. The Host Side signals are connected to the DRAM Side pins. The DRAM Side signals are connected to the Host Side pins. The ZQ pin is connected to Vss. The QACA and QBQA signals are highlighted in yellow. Ex. CC-07 (DDR5 RDIMM Addendum) at 8.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

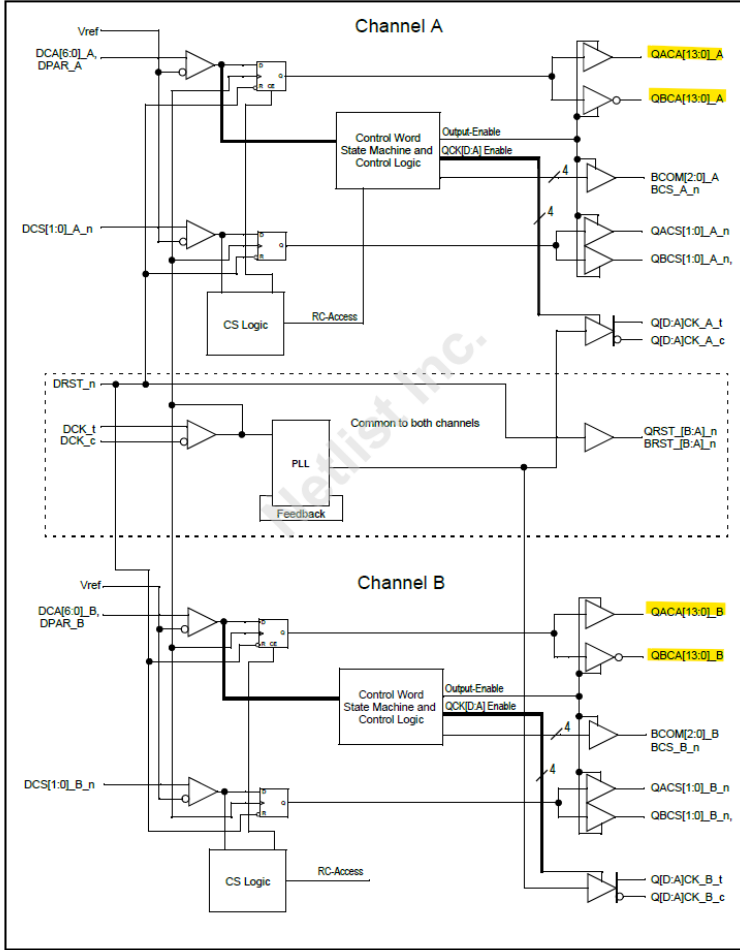
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Block Diagram  Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 The diagram illustrates a dual-channel device with two identical processing paths, Channel A and Channel B. Each channel includes a differential input stage (DCA/DPA), a control word state machine and control logic block, and a CS Logic block. The control logic is connected to a common PLL block. The outputs of each channel include a 13-bit data bus (QACA/QBCA), a 2-bit bus (BCOM/BCS), and a 1-bit bus (QACS/QBCS). The device also features a common clock input (DCK_t/DCK_c) and a reset input (DRST_n). The outputs are labeled as Q(D:A)CK_A_t, Q(D:A)CK_A_c, Q(D:A)CK_B_t, and Q(D:A)CK_B_c. Ex. CC-32 (JEDEC 82-512) at 7.

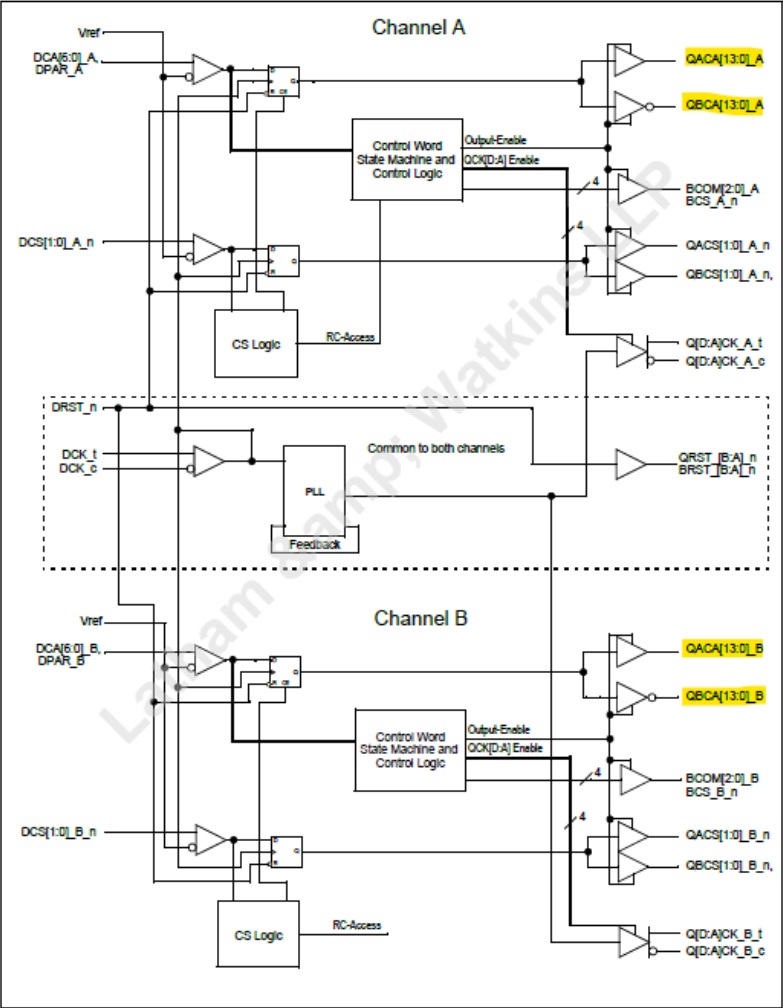
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴			
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U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-32 (JEDEC 82-512) at 15. 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 The diagram illustrates a dual-channel ADC system. Channel A and Channel B are shown, each with its own input multiplexer, CS Logic, and output multiplexer. A shared PLL block is connected to both channels. The system includes various control signals and data outputs. Channel A components and signals: - Inputs: V_{ref}, $DCA[6:0]_A$, $DPAR_A$, $DCS[1:0]_{A,n}$ - Control: Control Word State Machine and Control Logic, CS Logic, RC-Access - Outputs: $QACA[13:0]_A$, $QBCA[13:0]_A$, $B[COM][2:0]_A$, $BCS_{A,n}$, $QACS[1:0]_{A,n}$, $QBCS[1:0]_{A,n}$, $Q(DA)CK_{A,t}$, $Q(DA)CK_{A,c}$ Channel B components and signals: - Inputs: V_{ref}, $DCA[6:0]_B$, $DPAR_B$, $DCS[1:0]_{B,n}$ - Control: Control Word State Machine and Control Logic, CS Logic, RC-Access - Outputs: $QACB[13:0]_B$, $QBCB[13:0]_B$, $B[COM][2:0]_B$, $BCS_{B,n}$, $QACS[1:0]_{B,n}$, $QBCS[1:0]_{B,n}$, $Q(DA)CK_{B,t}$, $Q(DA)CK_{B,c}$ Shared components and signals: - Inputs: $DRST_n$, DCK_t, DCK_c - Block: PLL (Common to both channels) - Outputs: $QRST_{[B:A]_n}$, $BRST_{[B:A]_n}$ Ex. CC-30 (JEDEC 82-513) at 7.

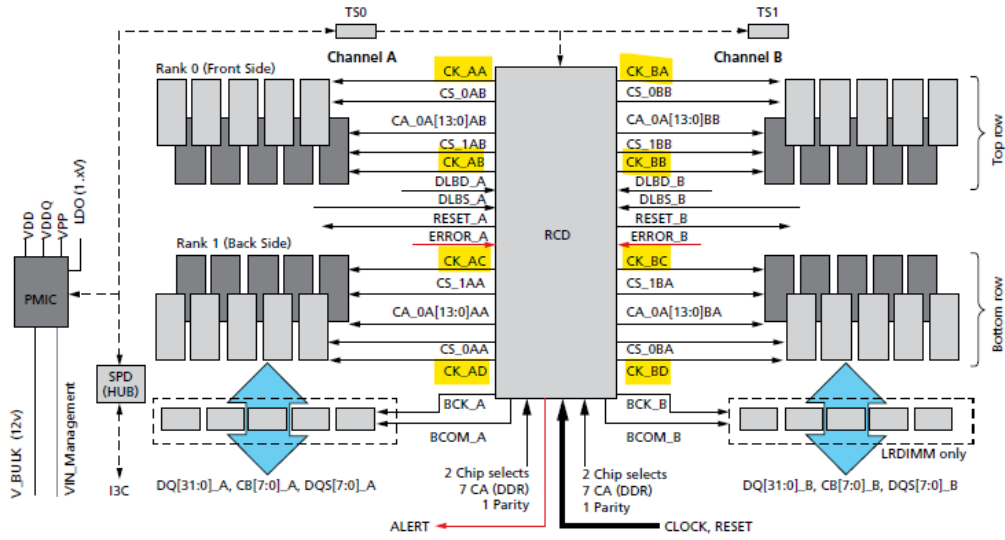
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴			
	Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
		BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
		BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
		BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
Ex. CC-30 (JEDEC 82-513) at 4.				
3.3.3 QCA Bus Operation				
Under normal operation, the RCD passes commands through from the D inputs to the Q outputs when a command is present on the bus, and drives HIGH on the QCA outputs when no command is being sent. The RCD passes the DCA bus through when either DCS input is active but there are exceptions to this:				
• If the register detects a parity error and parity checking is enabled, the DCA bus is still passed through to DRAM and QCS is blocked. • During the 2nd UI of a 2 UI command, the DCA bus must be passed through. There will only be a DCS signal active for the 2nd UI if a non-target termination is being signaled to a rank. • When DRAM interface Block MR Mode is enabled RW01[1] = 0. • When CA Pass-through mode is enabled RW00[2] = 1.				
Ex. CC-30 (JEDEC 82-513) at 9.				

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																								
	3.4.1 Double Data Rate DCA The DDR5 RCD will support DDR (Double Data Rate) on the host-interface DCA signals. The host-interface CA bus is 7 bits, plus parity. The DRAM-interface CA bus remains SDR (Single Data Rate) and is 14 bits and without parity. The RCD will expand the incoming host-interface 7-bit DCA bus to 14 bits on the DRAM interface. Table 8 shows the mapping of the incoming DCA bits to the output QCA. The Chip Selects remain SDR on both the host-interface and DRAM interface. The two channel DDR5RCD has six CA ports, two DCA input ports and four QCA output ports as shown in Figure 6. Table 8 — DDR CA Mapping DCA to QCA <table><tr><th>UI</th><th>DCA0</th><th>DCA1</th><th>DCA2</th><th>DCA3</th><th>DCA4</th><th>DCA5</th><th>DCA6</th></tr><tr><td>0</td><td>QCA0</td><td>QCA1</td><td>QCA2</td><td>QCA3</td><td>QCA4</td><td>QCA5</td><td>QCA6</td></tr><tr><td>1</td><td>QCA7</td><td>QCA8</td><td>QCA9</td><td>QCA10</td><td>QCA11</td><td>QCA12</td><td>QCA13</td></tr></table> Figure 6 — RCD DCA and QCA Ports Ex. CC-30 (JEDEC 82-513) at 15.	UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6	0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6	1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13
UI	DCA0	DCA1	DCA2	DCA3	DCA4	DCA5	DCA6																		
0	QCA0	QCA1	QCA2	QCA3	QCA4	QCA5	QCA6																		
1	QCA7	QCA8	QCA9	QCA10	QCA11	QCA12	QCA13																		
[1.c.5] output the plurality of local clocks to the memory devices,	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to output the plurality of local clocks to the memory devices. For example: The DDR5 RCD circuitry is configurable to output local clocks (QACK, QBCK, QCCK, and QDCK) to the memory devices.																								

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	The diagram illustrates the electrical interface between a Host Side and a DRAM Side. On the Host Side, various control and data signals are shown, including DCS[1:0]A_n, DCS[1:0]B_n, DCA_[6:0]A, DCA_[6:0]B, ALERT_n, LBDQ, RSP_A_n, LBDQS, RSP_B_n, PAR_A, PAR_B, HSCL, HSDA, CK_c, CK_t, and RESET_n. On the DRAM Side, corresponding signals are shown, including QACS[1:0][B:A]_n, QBCS[1:0][B:A]_n, QACA[13:0][B:A], QBCA[13:0][B:A], DERROR_IN_[B:A]_n, DLBD_[B:A], DLBS_[B:A], Q[D:A]CK[A:B]_c, Q[D:A]CK[A:B]_t, and QRST[B:A]. The central block represents the RDIMM, with pins labeled R, E, G, I, S, T, E, R, &, P, S, C, and ZQ. The ZQ pin is connected to Vss. Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing generation of local clocks). To reduce the electrical load on the host memory controller's command, address, and control bus, Micron's RDIMMs and LRDIMMs utilize a DDR5 registering clock driver (RCD). The RCD presents a single load to the controller while redriving signals to the DDR5 SDRAM devices, which helps enable higher densities and increase signal integrity. The RCD also provides a low-jitter, low-skew PLL that redistributes a differential clock pair to multiple differential pairs of clock outputs. Ex. CC-03 (DDR5 RDIMM Datasheet) at 12 (stating the RCD provides multiple clock outputs).

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Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

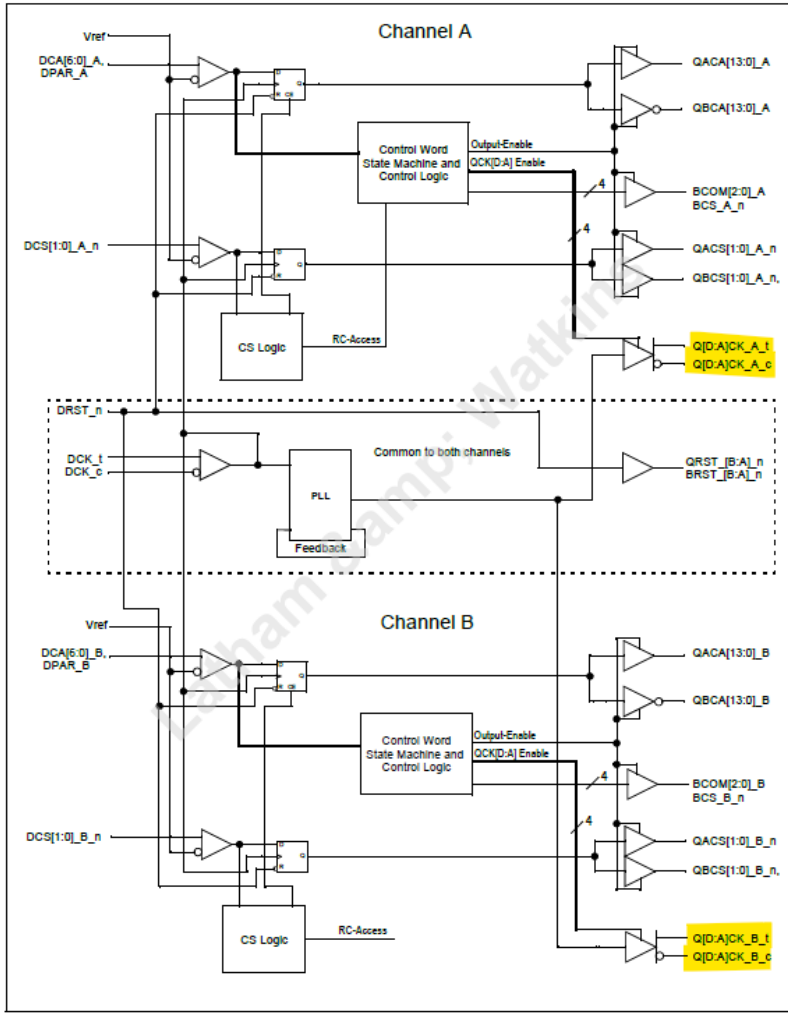
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴								
	Block Diagram The diagram shows a central rectangular block labeled 'RCD'. It has several input and output lines. On the left side, from top to bottom, there are: an output line labeled 'BCOM_A BUS' with an arrow pointing left; an input line labeled 'Q[D:A]CK_A_t/c' with an arrow pointing right; an input line labeled 'QACA[13:0]_A' with an arrow pointing right; and an input line labeled 'QBCA[13:0]_A' with an arrow pointing right. On the right side, from top to bottom, there are: an output line labeled 'BCOM_B BUS' with an arrow pointing right; an input line labeled 'Q[D:A]CK_B_t/c' with an arrow pointing left; an input line labeled 'QACA[13:0]_B' with an arrow pointing left; and an input line labeled 'QBCA[13:0]_B' with an arrow pointing left. At the bottom, there are three input lines: 'DCA[6:0]_A' on the left, 'DCK_t/c' in the center, and 'DCA[6:0]_B' on the right, all with arrows pointing up into the block. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. <table><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c</td><td>POD differential</td><td>cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.</td></tr><tr><td>Feedback</td><td>OT RN</td><td>POD</td><td>Feedback outputs to Host</td></tr></table> Ex. CC-32 (JEDEC 82-512) at 4 (showing the RCD includes clock outputs for the memory devices).	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.	Feedback	OT RN	POD	Feedback outputs to Host
Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.						
Feedback	OT RN	POD	Feedback outputs to Host						

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Claim Language	'407 Accused Products ⁴
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.

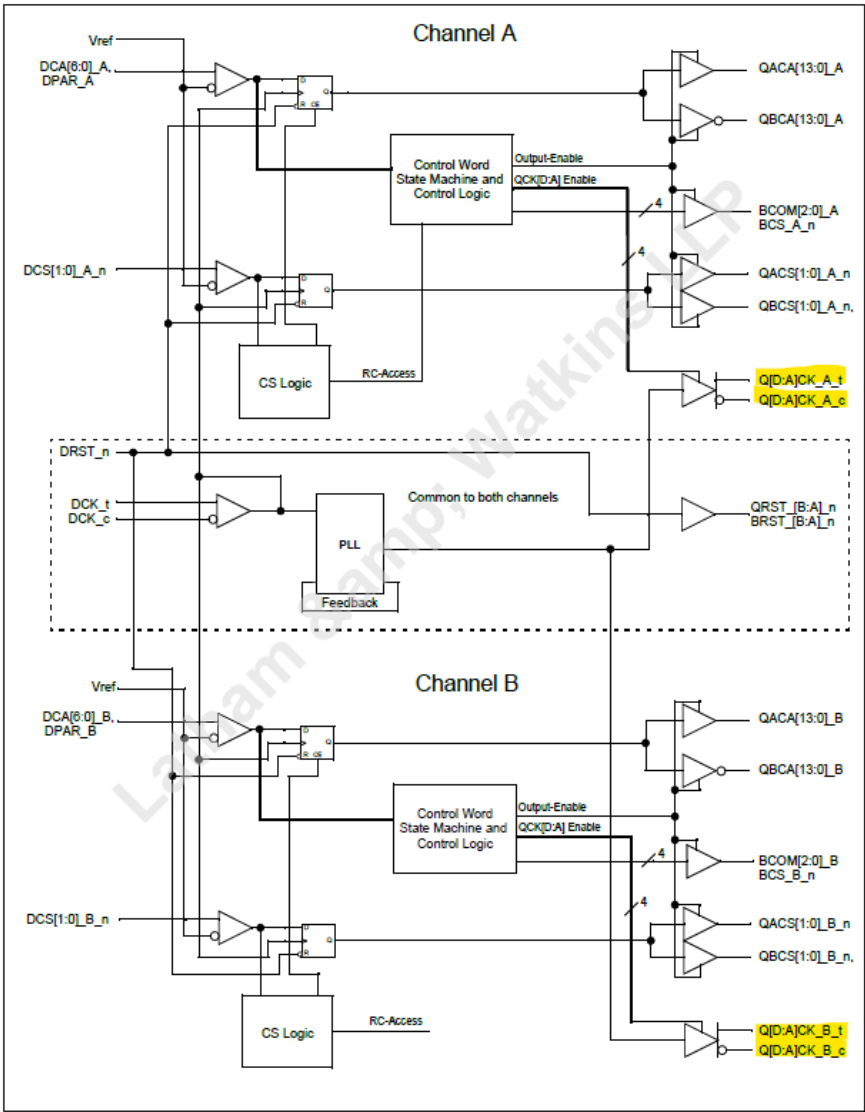
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	3.2 Logic Diagram Positive Logic  Ex. CC-32 (JEDEC 82-512) at 7.

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Claim Language	'407 Accused Products ⁴			
	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
	Ex. CC-30 (JEDEC 82-513) at 4 (showing the RCD includes clock outputs for the memory devices). 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.			

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 The diagram illustrates a dual-channel ADC system with two channels, Channel A and Channel B, sharing a common PLL and control logic. Channel A inputs include Vref, DCA[6:0]_A, DPAR_A, DCS[1:0]_A_n, and DCA[6:0]_B. Channel B inputs include Vref, DCA[6:0]_B, DPAR_B, DCS[1:0]_B_n, and DCA[6:0]_A. The shared PLL block receives DRST_n, DCK_t, and DCK_c, and outputs GRST_[B:A]_n and BRST_[B:A]_n. The control logic block receives RC-Access and outputs QCK[D:A] Enable. The outputs for Channel A are QACA[13:0]_A, QBCA[13:0]_A, BCOM[2:0]_A, BCS_A_n, QACS[1:0]_A_n, QBCS[1:0]_A_n, Q[D:A]JCK_A_t, and Q[D:A]JCK_A_c. The outputs for Channel B are QACA[13:0]_B, QBCA[13:0]_B, BCOM[2:0]_B, BCS_B_n, QACS[1:0]_B_n, QBCS[1:0]_B_n, Q[D:A]JCK_B_t, and Q[D:A]JCK_B_c. A watermark 'Latham & Watkins LLP' is visible across the diagram. Ex. CC-30 (JEDEC 82-513) at 7.

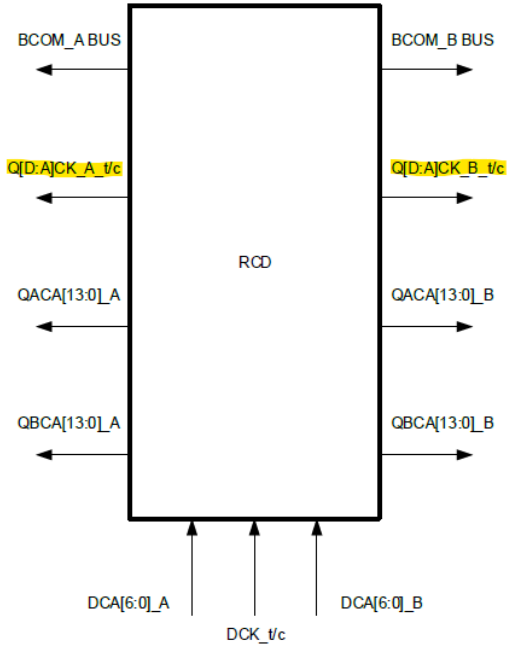
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
[1.c.5.i] wherein a respective local clock having a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices,	In the '407 Accused Products, a respective local clock has a respective programmable phase relationship with the system clock is output to a corresponding group of the memory devices and not to any other group of the memory devices. For example: The local clocks generated by the RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 1.c.3.i and 1.c.3.ii above) are output to a corresponding group of the memory devices in a rank and not to any other group of the memory devices. <i>See, e.g.,</i>
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing the RCD circuitry outputs each local clock to a specific group of devices).

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Block Diagram  The diagram shows a central rectangular block labeled 'RCD'. It has several input and output lines. On the left side, from top to bottom, there are: an output line labeled 'BCOM_A BUS' with an arrow pointing left; an input line labeled 'Q[D:A]CK_A_1/c' with an arrow pointing right; an input line labeled 'QACA[13:0]_A' with an arrow pointing right; an input line labeled 'QBCA[13:0]_A' with an arrow pointing right. On the right side, from top to bottom, there are: an output line labeled 'BCOM_B BUS' with an arrow pointing right; an input line labeled 'Q[D:A]CK_B_1/c' with an arrow pointing left; an output line labeled 'QACA[13:0]_B' with an arrow pointing left; an output line labeled 'QBCA[13:0]_B' with an arrow pointing left. At the bottom, there are three input lines: 'DCA[6:0]_A' on the left, 'DCK_1/c' in the center, and 'DCA[6:0]_B' on the right, all with arrows pointing up into the RCD block. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. The JEDEC DDR5 RDIMM standard further demonstrates the local clocks generated by the Accused Products' RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 1.c.3, 1.c.3.i, and 1.c.3.ii above) are output to a corresponding group of the memory devices and not to any other group of the memory devices.

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

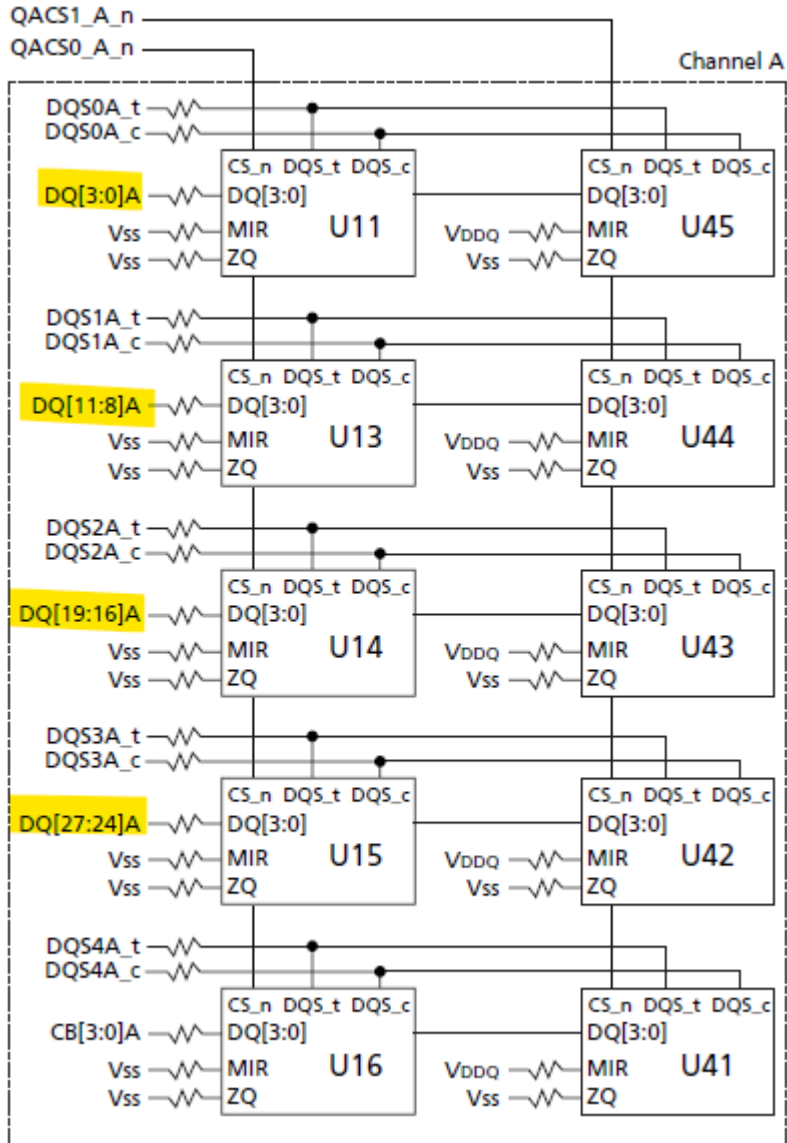
2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

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Claim Language	'407 Accused Products ⁴
	6.9.1 DIMM Connectivity Wiring (cont'd) Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing local clocks, e.g., QACK, QBCK, QCCK, QDCK, being output to certain groups of memory devices but not others).
[1.c.5.ii] wherein the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data signal lines in accordance with the respective local clock.	In the '407 Accused Products, the corresponding group of the memory devices is configurable to perform memory read or write operations by communicating data signals with the memory controller via a corresponding subset of data signal lines in accordance with the respective local clock. For example, a group of memory devices associated with channel A perform memory read and write operations by communicating data signals with the memory controller via a subset of data signal lines (DQ[]A). <i>See, e.g.:</i>

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Claim Language	'407 Accused Products ⁴
	 The diagram illustrates the internal structure of Channel A, which is a multi-channel memory interface. It consists of six pairs of memory blocks, labeled U11 through U16 and U45 through U41. Each block is connected to a common data bus (DQ[3:0]A, DQ[11:8]A, DQ[19:16]A, DQ[27:24]A, and CB[3:0]A) and has its own control signals (CS_n, DQS_t, DQS_c) and power pins (Vss, VDDQ). The blocks are arranged in a 3x2 grid, with U11, U13, U14, U15, U16 on the left and U45, U44, U43, U42, U41 on the right. The data buses are connected to the DQ[3:0] pins of each block. The control signals are connected to the CS_n, DQS_t, and DQS_c pins. The power pins are connected to Vss and VDDQ. The diagram also shows the connection of the data buses to the external signals QACS1_A_n and QACS0_A_n.

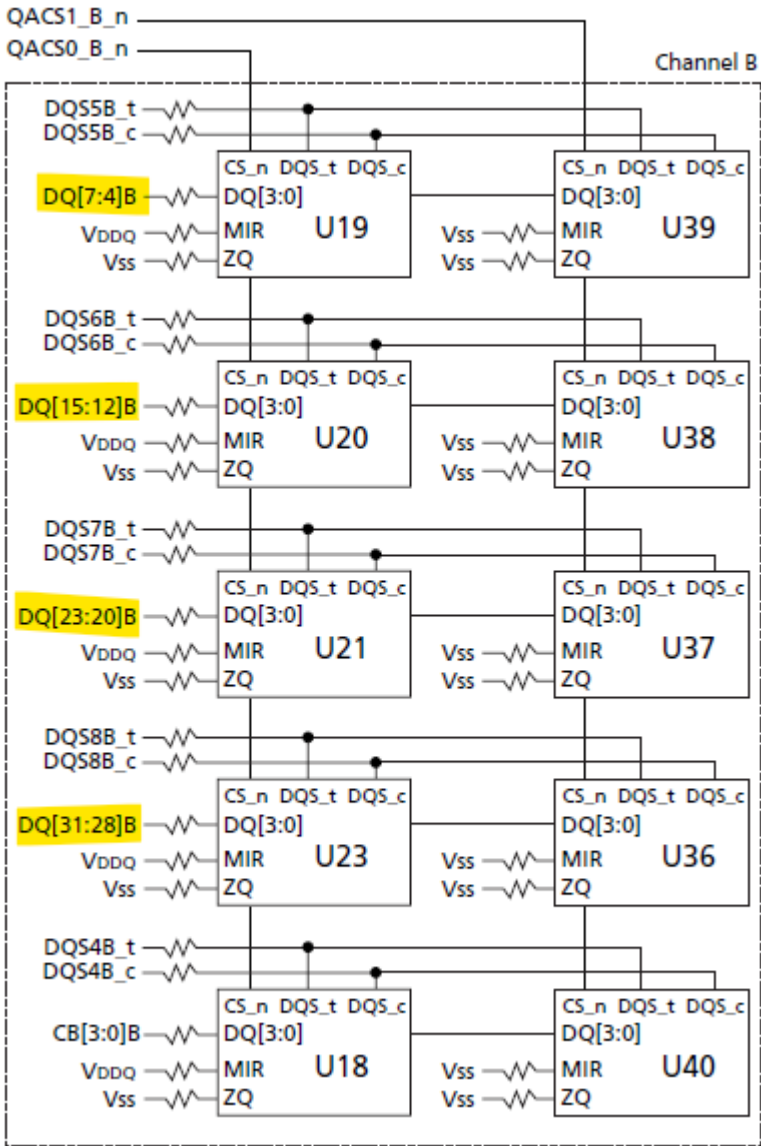
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	The diagram illustrates the internal structure of Channel A, which consists of five pairs of memory blocks (U1/U35, U2/U34, U3/U33, U4/U32, U5/U31). Each block is connected to a data bus (DQ[7:4]A, DQ[15:12]A, DQ[23:20]A, DQ[31:28]A, and CB[7:4]A) and has control signals (CS_n, DQS_t, DQS_c) and power connections (Vss, VDDQ). The blocks are arranged in a vertical stack, with each pair of blocks sharing a common data bus and control signals. The diagram also shows the connection of the data buses to the external signals QBCS1_A_n and QBCS0_A_n.

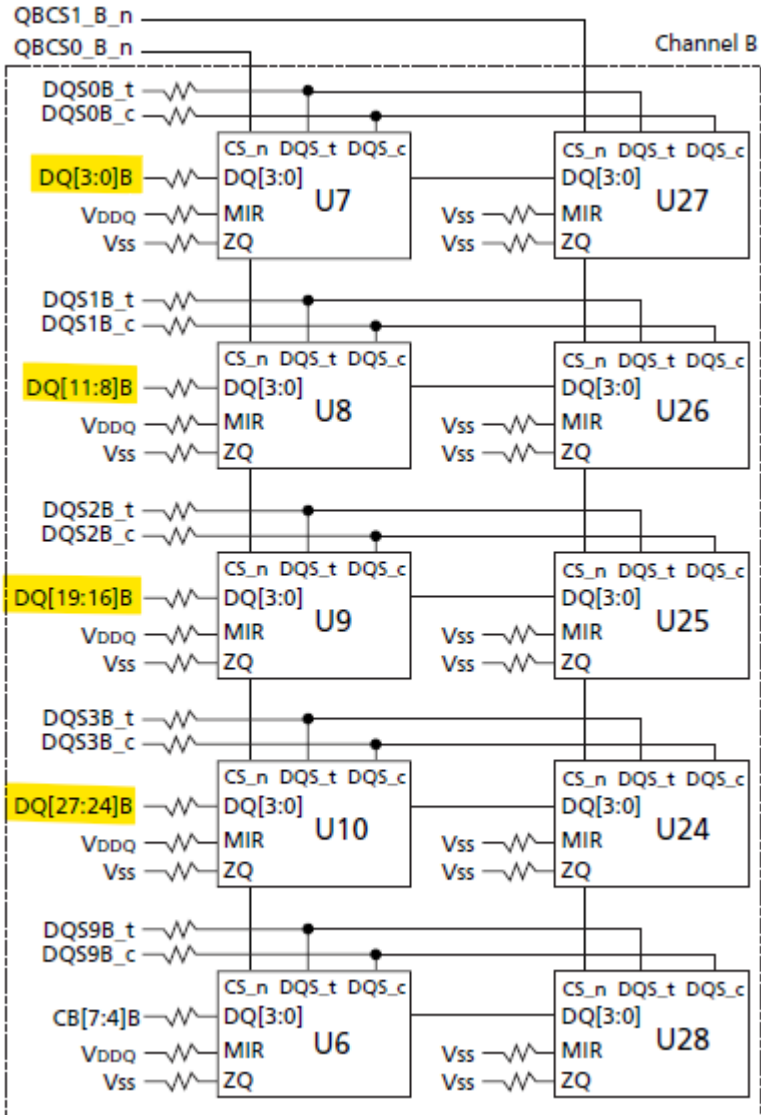
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (also showing similar for data signals DQ[]B). Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. Furthermore, a group of memory devices associated with channel B perform memory read and write operations by communicating data signals with the memory controller via a subset of data signal lines (DQ[]B). <i>See, e.g.,</i>

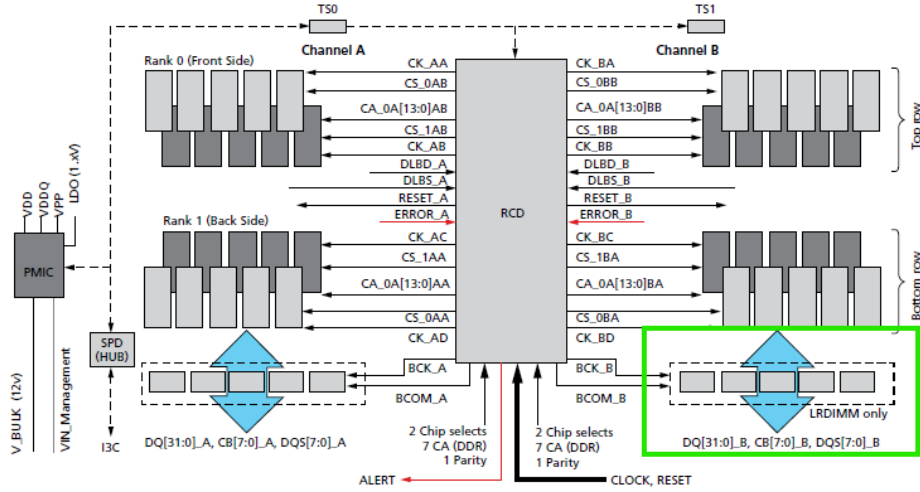
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Claim Language	'407 Accused Products ⁴
	 The diagram illustrates the internal structure of Channel B, which contains ten memory blocks arranged in two columns. The left column includes blocks U19, U20, U21, U23, and U18, while the right column includes blocks U39, U38, U37, U36, and U40. Each block is connected to a common data bus (DQ) and has its own address bus (CS_n) and power connections (VDDQ, Vss, VDD, Vss). The data bus connections are labeled with DQ[t:c] and DQ[3:0]. The address bus connections are labeled with CS_n, DQS_t, and DQS_c. The power connections are labeled with VDDQ, Vss, VDD, and Vss. The data bus segments DQ[7:4]B, DQ[15:12]B, DQ[23:20]B, DQ[31:28]B, and CB[3:0]B are highlighted in yellow. The diagram also shows connections to external signals QACS1_B_n and QACS0_B_n.

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Claim Language	'407 Accused Products ⁴
	 The diagram illustrates the internal structure of Channel B, which is a memory array. It consists of six rows of memory blocks, each row containing two blocks (labeled U7, U27, U8, U26, U9, U25, U10, U24, U6, U28). Each block is connected to a common data bus (DQ[3:0]B, DQ[11:8]B, DQ[19:16]B, DQ[27:24]B, and CB[7:4]B) and a common clock bus (DQS_t, DQS_c). The blocks are also connected to a common chip select bus (CS_n) and a common mirror bus (MIR). The blocks are connected to a common ZQ bus (ZQ). The blocks are connected to a common VDDQ bus (VDDQ) and a common Vss bus (Vss).

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Claim Language	'407 Accused Products ⁴
	Ex. CC-07 (DDR5 RDIMM Addendum) at 8. Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. Furthermore, data is read or written in memory devices in accordance with the respective local clock. The JEDEC DDR5 RDIMM standard further demonstrates groups of memory devices perform memory read and write operations by communicating data signals with the memory controller via corresponding subsets of data signal lines (e.g., DQS[]).

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'407 Accused Products⁴

6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

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6.9.1 DIMM Connectivity Wiring (cont'd)

1Rx8 (less than 8000 Mbps)

DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QACK	QACS0	QACA	0

1Rx8 (8000 Mbps and Faster)

DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QCCK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS0	QACA	0

1Rx8 Sub Channel 0 (Channel A)

Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top			[15:8]				[31:24]			
Bottom		[7:0]			[23:16]				CB[7:0]	

1Rx8 Sub Channel 1 (Channel B)

Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]			[23:16]					
Bottom			[15:8]				[31:24]			CB[7:0]

Figure 12 — DIMM Connectivity Wiring

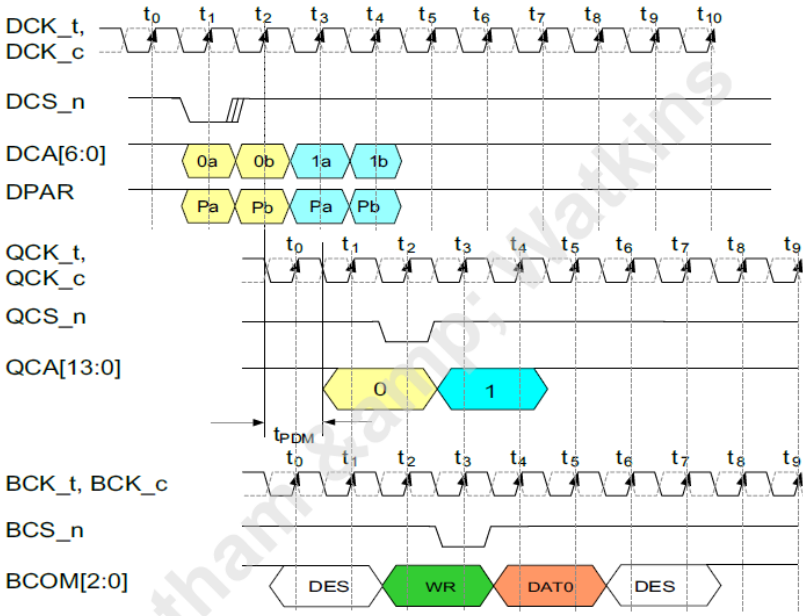
Ex. CC-88 (JEDEC 305B) at 31-32 (showing memory devices receiving data from the memory controller via subsets of data signal lines, e.g., [3:0], [11:8], [15:12]).

The JEDEC DDR5 RCD definition further demonstrates the chip select signal (QCS) and command and address signal (QCA) received by each DDR memory device is synchronized with the local clock (QCK):

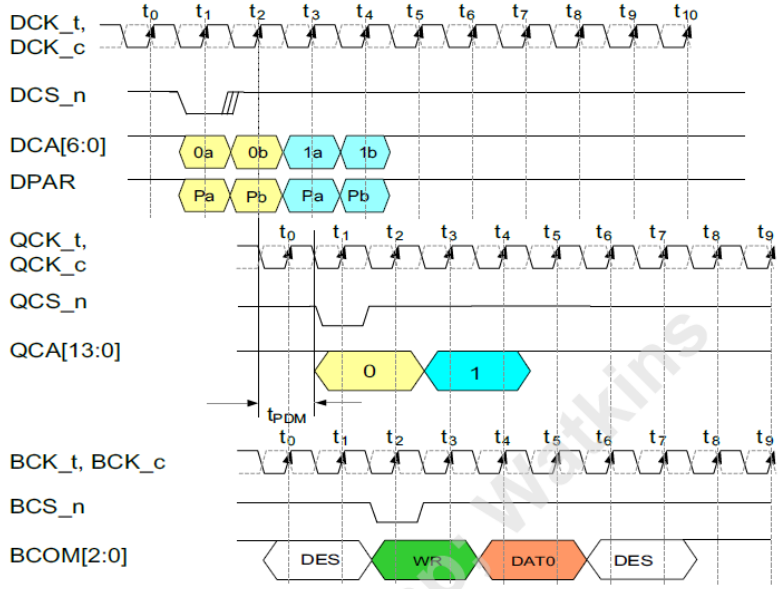
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Claim Language	'407 Accused Products ⁴
	3.4.1 Double Data Rate DCA (cont'd) NOTE Timing Diagram for 0 nCK command latency adder. Figure 7 — One UI DRAM Command Timing Diagram NOTE Timing Diagram for 0 nCK command latency adder. Figure 8 — Two UI DRAM Command Timing Diagram Ex. CC-32 (JEDEC 82-512) at 16. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 16.

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	The JEDEC DDR5 RCD definition explains the Accused Products are configurable to perform READ and WRITE operations synchronized with the local clock QCK. <i>See, e.g.,</i>:  Figure 11 — SDR1 Mode Example of DRAM WR Command with RW00[1] = 0

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	 Figure 12 — SDR2 Mode Example of DRAM WR Command with RW00[1] = 1 Ex. CC-32 (JEDEC 82-512) at 18-19. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 18-19. Further, the timing diagrams for read and write operations show that DQ signals appear after a set latency in terms of clock cycles after the module C/A signals are received. As another example, DQ is output/received in relation to DQS, and DQS is aligned with clock signals in predetermined manner. So the DQ output/input is in accordance with local clock signals.

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	4.7 Read Operation The Read Operation causes the DRAM to retrieve and output data stored in its array. The Read Operation is initiated by the Read Command during which the beginning column address and bank/group address for the data to be retrieved from the array is provided. The data is driven by the DRAM on its DQ pins RL (CL) cycles after the Read Command along with the proper waveform on the DQS inputs. Read Latency (RL or CL) is defined from the Read command to data and is not affected by the Read DQS offset timing (MR40 OP[2:0]). 4.7.1 READ Burst Operation During a READ or WRITE command, DDR5 shall support BC8, BL16, BL32 (optional) and BL32 OTF (optional) during the READ or WRITE. MR0[1:0] is used to select burst operation mode. NOTES: 1. BL=16, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble 2. DES commands are shown for ease of illustration; other commands may be valid at these times. 3. In this example, Read DQS Offset Timing is set to 0 Clocks. Figure 35 — READ Burst Operation (BL16) Ex. CC-05 (JESD79-5) at 135.

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	4.8.2 Write Burst Operation The following write timing diagrams are to help understand the meaning of each write parameter; the diagrams are just examples. The details of each parameter are defined separately. In these write timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown center-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate.

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controller and a system bus, the system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines, the memory module comprising:	The '407 Accused Products include a “memory module.” <i>See, e.g.,</i> Ex. 51 to Complaint (Micron RDIMM Importation) at 5.

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	Ex. 51 to Complaint (Micron RDIMM Importation) at 8.

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Claim Language	'407 Accused Products ⁴
	Ex. 51 to Complaint (Micron MRDIMM Importation) at 5.  Ex. 51 to Complaint (Micron MRDIMM Importation) at 12.

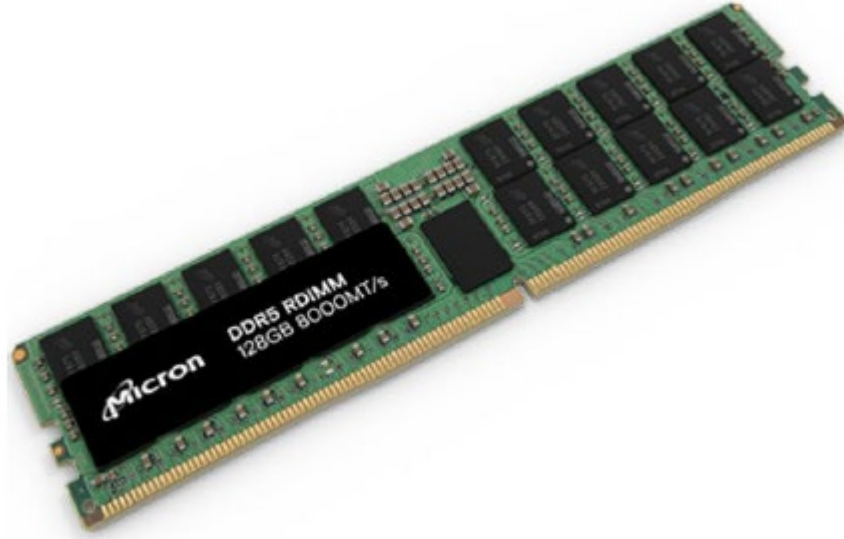
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Ex. 55 to Complaint (HPE RDIMM Importation at 6). Ex. 55 to Complaint (Lenovo RDIMM Importation) at 4.

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Claim Language	'407 Accused Products ⁴
	Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6. Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4.

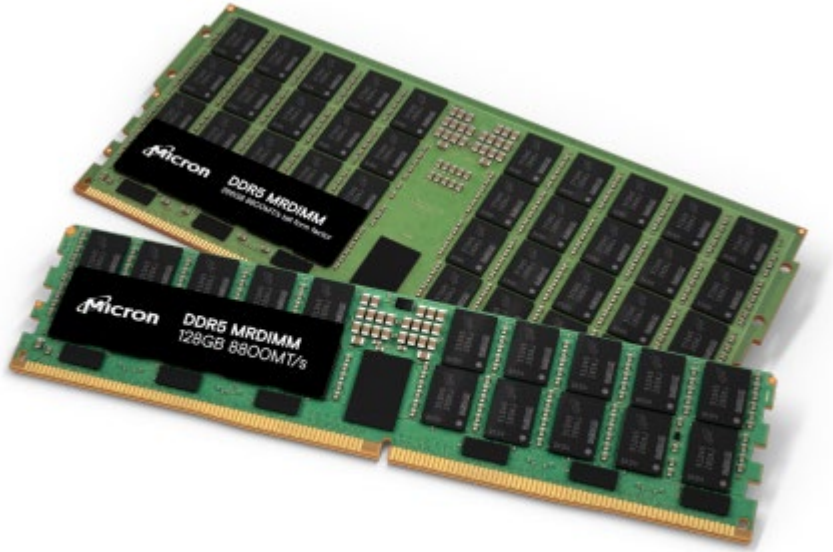
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Claim Language	'407 Accused Products ⁴
	 Ex. CC-01 (DDR5 Brochure) at 1.

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Claim Language	'407 Accused Products ⁴
	Figure 3: 288-Pin DDR5 RDIMM Notes: 1. All dimensions are in millimeters; MAX/MIN or typical (TYP) where noted. 2. Tolerance on all dimensions ± 0.15 unless otherwise specified. 3. The dimensional diagram is for reference only. Ex. CC-03 (DDR5 RDIMM Datasheet) at 24.

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Claim Language	'407 Accused Products ⁴
	 Figure 1: MRDIMM low-profile form factor (foreground) and TFF (background) Ex. CC-02 (MRDIMM Product Brief) at 1. The memory module of the '407 Accused Products are further operable in a computer system having a memory controller and a system bus. For example, the '407 Accused Products are operable in “server platforms,” and are configured to exchange information with a “host memory controller” over a “command, address and control bus” and a “memory bus.” <i>See, e.g.,</i> Micron is first to enable the market with a monolithic 32Gb-based DDR5 128GB DRAM module that delivers the fastest speed and lowest latency for enabling generations of server platforms to come. Ex. CC-01 (DDR5 Brochure) at 1.

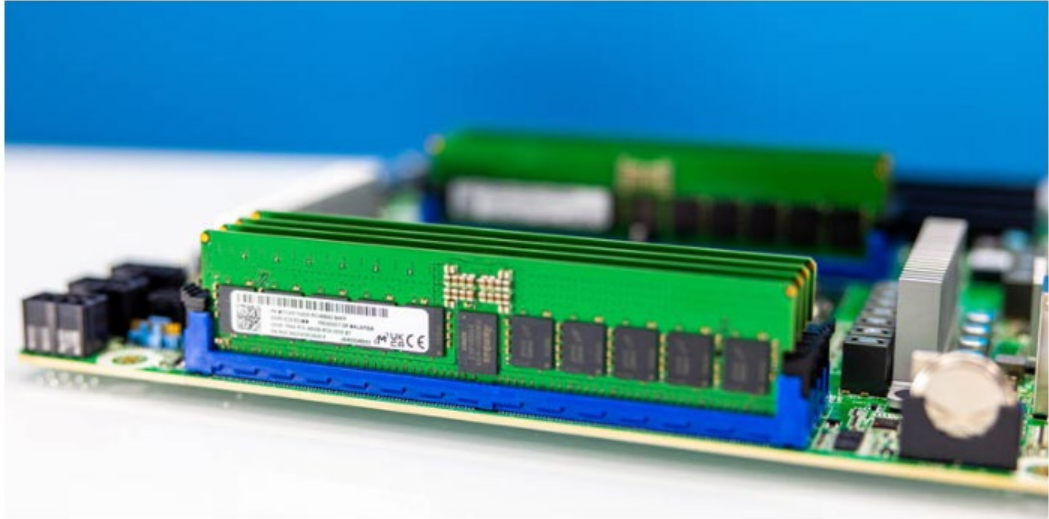
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Claim Language	'407 Accused Products ⁴
	Product Description This specification defines the electrical and mechanical requirements for 287-pin and 288-pin, 1.1V (V_{DD}) double data rate, synchronous DRAM, registered dual in-line memory modules (DDR5 SDRAM RDIMMs). These DDR5 RDIMMs are intended for use as main memory when installed in servers. Some specifications are part number-specific; refer to the module data sheet addendum of the specific Micron part number (MPN) for the complete specification. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. To reduce the electrical load on the host memory controller's command, address, and control bus, Micron's RDIMMs and LRDIMMs utilize a DDR5 registering clock driver (RCD). The RCD presents a single load to the controller while redriving signals to the DDR5 SDRAM devices, which helps enable higher densities and increase signal integrity. The RCD also provides a low-jitter, low-skew PLL that redistributes a differential clock pair to multiple differential pairs of clock outputs. Ex. CC-03 (DDR5 RDIMM Datasheet) at 12; <i>see also id.</i> at 12-13, 21 (discussing operations from the host). Micron memory modules are designed to optimize signal integrity through carefully designed terminations, controlled board impedances, routing topologies, trace length matching, and decoupling. However, good signal integrity starts at the system level. Micron encourages designers to simulate the signal characteristics of the system's memory bus to ensure adequate signal integrity of the entire memory system. Ex. CC-03 (DDR5 RDIMM Datasheet) at 17 (showing Accused Devices communicate over signals with a system memory bus).

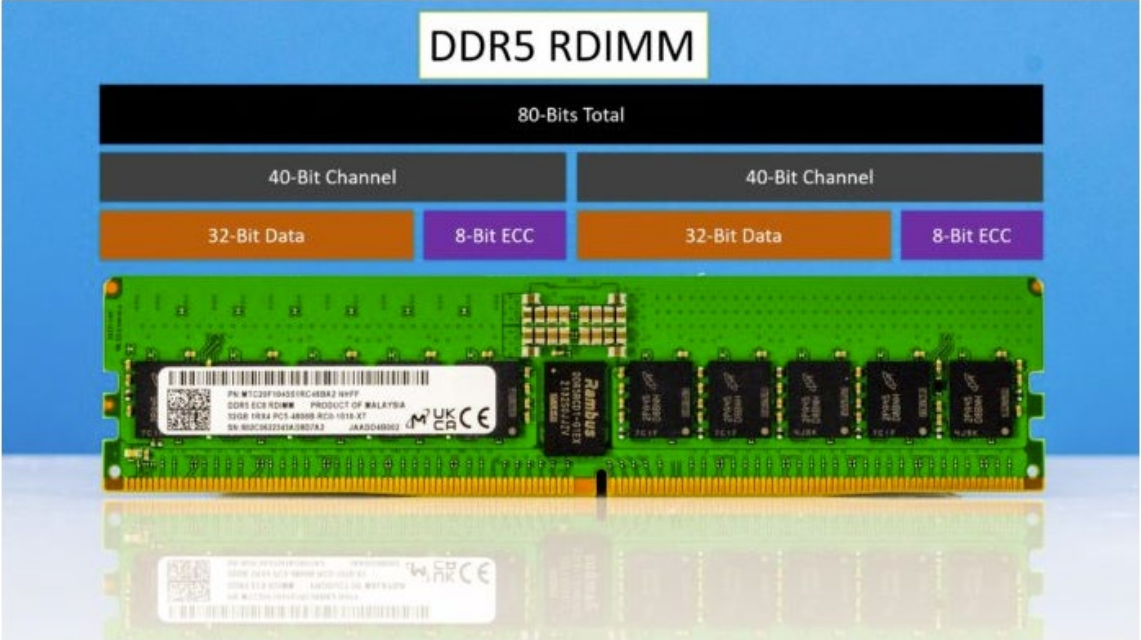
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Meeting Next-Generation CPU Requirements At the system-level, despite only modest clock rate improvements, the transition to multicore CPU architectures has enabled continuous year-over-year compute performance gains. Signal integrity, power delivery, layout complexity, and other system-level challenges are limiting advances in CPU core frequency. Simultaneously, CPU core counts are continuously increasing, limiting the available memory bandwidth-per-core. New memory architectures beyond DDR4 SDRAM are needed to meet the next-generation bandwidth-per-core requirements. The chart below illustrates the bandwidth trend in real systems and shows how bandwidth-per-core has been relatively constant and is even beginning to trend down (as illustrated in the lower right corner of the chart). Ex. CC-08 (DDR5 White Paper) at 2. Micron 1β enables generations of server platforms to come Industry's fastest speed, lowest latency DDR5 based on a 32Gb monolithic die, Micron 1β will be available in 5600MT/s, 6400MT/s, 7200MT/s, and 8000MT/s, enabling future generations of server platforms. Micron's 1β 16Gb-based modules are available in 16GB, 32GB, and 64GB module densities for general compute applications not requiring larger capacities. Ex. CC-01 (DDR5 Brochure) at 2.

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	 <i>Micron DDR5 RDIMM In ASRock AMD EPYC Genoa Motherboard 2</i> Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 1.

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	 Micron DDR5 RDIMM Two 40 Bit Channels Ex. CC-06 Patrick Kennedy, <i>Why DDR5 is Absolutely Necessary in Modern Servers</i>, Serve the Home (Feb. 10, 2023) at 2. Furthermore, the Accused Products are operable with a system bus including one or more clock signal lines, control/address (C/A) signal lines and data/strobe signal lines. <i>See, e.g.,</i>

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Claim Language	'407 Accused Products ⁴												
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (connection to system bus in red); <i>see also id.</i> at 6-7 (showing pin assignments for clock signal lines, control/address (C/A) signal lines and data/strobe signal lines). <table><thead><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr></thead><tbody><tr><td>CK_t, CK_c</td><td>Input</td><td>V_{DD}</td><td>Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>DDR Input</td><td>V_{DD}</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.</td></tr></tbody></table>	Symbol	Type	I/O Level	Description	CK_t, CK_c	Input	V _{DD}	Clock: CK_t and CK_c are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.
Symbol	Type	I/O Level	Description										
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	Ex. CC-03 (DDR5 RDIMM Datasheet) at 8. <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.</td></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">V_{DD}</td><td rowspan="2">Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td></tr></table> Ex. CC-03 (DDR5 RDIMM datasheet) at 9. The JEDEC DDR5 RDIMM standard connector pinout and signal description further indicates the system bus of the computer system includes one or more clock signal lines, control/address (C/A) signal lines, and data signal lines. <i>See, e.g.,</i> Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr><tr><td>PAR_A PAR_B</td><td>Parity input</td><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td>Data Strobes (positive line of differential pair)</td></tr><tr><td>CK_t</td><td>Clock (true/positive)</td><td>DQS[9:0]_A_c DQS[9:0]_B_c</td><td>Data Strobes (negative line of differential pair)</td></tr><tr><td>CK_c</td><td>Clock (complement/negative)</td><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Not valid for x4 operation. Enabled via Mode Register.</td></tr></table>	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	V _{DD}	ECC Check Bits Input/Output: Bidirectional data bus. On x72 ECC DIMMs, only 4 check bits per sub channel are present.	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	V _{DD}	Data Strobe: Output with read data, input with write data. Edge-aligned with read data, centered in write data. The data strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM only supports differential data strobe. It does not support single-ended strobe.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input/ Output	V _{DD}	Termination Data Strobe: Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs. Not used on LRDIMMs. On DDR5, TDQS_t is a shared function with DM_n. Refer to the DRAM component data sheet for shared pin functionality.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	PAR_A PAR_B	Parity input	DQS[9:0]_A_t DQS[9:0]_B_t	Data Strobes (positive line of differential pair)	CK_t	Clock (true/positive)	DQS[9:0]_A_c DQS[9:0]_B_c	Data Strobes (negative line of differential pair)	CK_c	Clock (complement/negative)	TDQS[9:5]_A_t TDQS[9:5]_B_t	Not valid for x4 operation. Enabled via Mode Register.
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Claim Language	'407 Accused Products ⁴																				
	Ex. CC-88 (JEDEC 305B) at 3 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”), control/address (C/A) signal lines (e.g., “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)). Table 4 — Input/Output Functional Description <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>CK_t, CK_c.</td><td>Input</td><td>VDD</td><td>Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.</td></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Input</td><td>VDD</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating clock signal lines (e.g., “CK_t” and “CK_c”) and control/address (C/A) signal lines (e.g., “CA[]_A” and “CA[]_B”)). <table><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr><tr><td>CB[7:0]_A CB[7:0]_B</td><td>Input/ Output</td><td>VDD</td><td>ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (illustrating data signal lines (e.g., “DQ[]_A” and “DQ[]_B”)).	Symbol	Type	I/O Levels	Function	CK_t, CK_c.	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.	CA[6:0]_A CA[6:0]_B	Input	VDD	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during Mode Register Set commands.	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.	CB[7:0]_A CB[7:0]_B	Input/ Output	VDD	ECC Checkbits Input/ Output: Bi-directional data bus - for 8-bit ECC (EC8) all 8 bits are used - for 4-bit ECC (EC4) [3:0] bits are used; [7:4] bits are floating
Symbol	Type	I/O Levels	Function																		
CK_t, CK_c.	Input	VDD	Clock: CK_t and CK_c are differential clock inputs. All CA and CS_n input signals are sampled on the crossing of the positive edge of CK_t and negative edge of CK_c.																		
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DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 A4=High. Refer to vendor specific data sheets to determine which DQ is used.																		
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	Table 4 — Input/Output Functional Description (cont'd) <table><tr><th>Symbol</th><th>Type</th><th>I/O Levels</th><th>Function</th></tr><tr><td>DQS[9:0]_A_t DQS[9:0]_B_t</td><td rowspan="2">Input/ Output</td><td rowspan="2">VDD</td><td rowspan="2">Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.</td></tr><tr><td>DQS[9:0]_A_c DQS[9:0]_B_c</td></tr><tr><td>TDQS[9:5]_A_t TDQS[9:5]_B_t</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr><tr><td>TDQS[9:5]_A_c TDQS[9:5]_B_c</td><td>Input</td><td>VDD</td><td>Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 5.	Symbol	Type	I/O Levels	Function	DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.	DQS[9:0]_A_c DQS[9:0]_B_c	TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.	TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.
Symbol	Type	I/O Levels	Function															
DQS[9:0]_A_t DQS[9:0]_B_t	Input/ Output	VDD	Data Strobe: output with read data, input with write data. Edge-aligned with read data, centered in write data. The Data Strobe DQS_t is paired with differential signals DQS_c, respectively, to provide differential pair signaling to the system during reads and writes. DDR5 SDRAM supports differential Data Strobe only and does not support single-ended.															
DQS[9:0]_A_c DQS[9:0]_B_c																		
TDQS[9:5]_A_t TDQS[9:5]_B_t	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.															
TDQS[9:5]_A_c TDQS[9:5]_B_c	Input	VDD	Dummy load for matching the loading for mixed populations of x8 based RDIMMs and x4 based RDIMMs.															

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	Front Side Pin Label	Pin	Pin	Back side Pin Label	Front Side Pin Label	Pin	Pin	Back side Pin Label		
	VIN_BULK	1	145	VIN_BULK	PAR_A	74	218	CK_c		
	RFU	2	146	VIN_BULK	VSS	75	219	VSS		
	VIN_MGMT	3	147	PCAMP			Key			
	HSCL	4	148	HSA						
	HSDA	5	149	RFU	CA0_B	76	220	RFU / No Pin Present		
	VSS	6	150	RFU	VSS	77	221	CA1_B		
	DQ0_A	7	151	VSS	CA2_B	78	222	VSS		
	VSS	8	152	DQ2_A	VSS	79	223	CA3_B		
	DQ1_A	9	153	VSS	CA4_B	80	224	VSS		
	VSS	10	154	DQ3_A	VSS	81	225	CA5_B		
	DQS0_A_t	11	155	VSS	CA6_B	82	226	VSS		
	DQS0_A_c	12	156	DQS5_A_c, TDQS5_A_c	VSS	83	227	PAR_B		
	VSS	13	157	DQS5_A_t, TDQS5_A_t	CS0_B_n	84	228	VSS		
	DQ4_A	14	158	VSS	VSS	85	229	CS1_B_n		
	VSS	15	159	DQ6_A	DLBDQ	86	230	VSS		
	DQ5_A	16	160	VSS	DLBDQS	87	231	RFU		
	VSS	17	161	DQ7_A	VSS	88	232	RFU		
	DQ8_A	18	162	VSS	CB4_B	89	233	VSS		
	VSS	19	163	DQ10_A	VSS	90	234	CB6_B		
	DQ9_A	20	164	VSS	CB5_B	91	235	VSS		
	VSS	21	165	DQ11_A	VSS	92	236	CB7_B		
	DQS1_A_t	22	166	VSS	DQS9_B_t, TDQS9_B_t	93	237	VSS		
	DQS1_A_c	23	167	DQS6_A_c, TDQS6_A_c	DQS9_B_c, TDQS9_B_c	94	238	DQS4_B_c		
	VSS	24	168	DQS6_A_t, TDQS6_A_t	VSS	95	239	DQS4_B_t		
	DQ12_A	25	169	VSS	CB0_B	96	240	VSS		
	VSS	26	170	DQ14_A	VSS	97	241	CB2_B		
	DQ13_A	27	171	VSS	CB1_B	98	242	VSS		
	VSS	28	172	DQ15_A	VSS	99	243	CB3_B		
	DQ16_A	29	173	VSS	DQ0_B	100	244	VSS		
	VSS	30	174	DQ18_A	VSS	101	245	DQ2_B		
	DQ17_A	31	175	VSS	DQ1_B	102	246	VSS		
	VSS	32	176	DQ19_A	VSS	103	247	DQ3_B		
	DQS2_A_t	33	177	VSS	DQS0_B_t	104	248	VSS		
	DQS2_A_c	34	178	DQS7_A_c, TDQS7_A_c	DQS0_B_c	105	249	DQS5_B_c, TDQS5_B_c		
	VSS	35	179	DQS7_A_t, TDQS7_A_t	VSS	106	250	DQS5_B_t, TDQS5_B_t		
	DQ20_A	36	180	VSS	DQ4_B	107	251	VSS		
	VSS	37	181	DQ22_A	VSS	108	252	DQ6_B		
	DQ21_A	38	182	VSS	DQ5_B	109	253	VSS		

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Claim Language	'407 Accused Products ⁴
	Ex. CC-88 (JEDEC 305B) at 6 (illustrating clock signal lines (<i>e.g.</i> , “CK_t” and “CK_c”), control/address (C/A) signal lines (<i>e.g.</i> , “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (<i>e.g.</i> , “DQ[]_A” and “DQ[]_B”)).

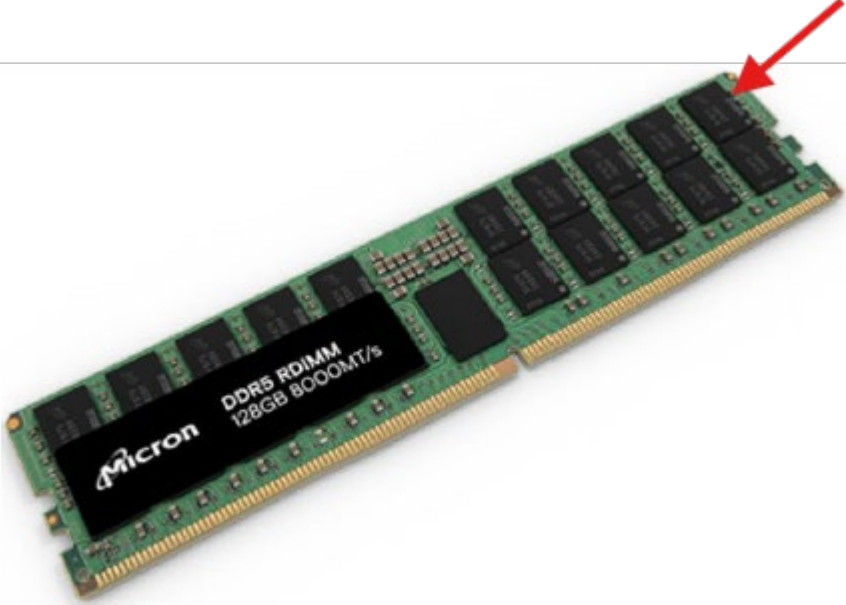
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Claim Language	'407 Accused Products ⁴											
	Front Side		Pin	Pin	Back side		Front Side		Pin	Pin	Back side	
	Pin Label				Pin Label		Pin Label				Pin Label	
	VSS	39	183	DQ23_A			VSS	110	254	DQ7_B		
	DQ24_A	40	184	VSS			DQ8_B	111	255	VSS		
	VSS	41	185	DQ26_A			VSS	112	256	DQ10_B		
	DQ25_A	42	186	VSS			DQ9_B	113	257	VSS		
	VSS	43	187	DQ27_A			VSS	114	258	DQ11_B		
	DQS3_A_t	44	188	VSS			DQS1_B_t	115	259	VSS		
	DQS3_A_c	45	189	DQS8_A_c, TDQS8_A_c			DQS1_B_c	116	260	DQS6_B_c, TDQS6_B_c		
	VSS	46	190	DQS8_A_t, TDQS8_A_t			VSS	117	261	DQS6_B_t, TDQS6_B_t		
	DQ28_A	47	191	VSS			DQ12_B	118	262	VSS		
	VSS	48	192	DQ30_A			VSS	119	263	DQ14_B		
	DQ29_A	49	193	VSS			DQ13_B	120	264	VSS		
	VSS	50	194	DQ31_A			VSS	121	265	DQ15_B		
	CB0_A	51	195	VSS			DQ16_B	122	266	VSS		
	VSS	52	196	CB2_A			VSS	123	267	DQ18_B		
	CB1_A	53	197	VSS			DQ17_B	124	268	VSS		
	VSS	54	198	CB3_A			VSS	125	269	DQ19_B		
	DQS4_A_t	55	199	VSS			DQS2_B_t	126	270	VSS		
	DQS4 A c	56	200	DQS9 A c, TDQS9 A c			DQS2 B c	127	271	DQS7 B c, TDQS7 B c		
	VSS	57	201	DQS9_A_t, TDQS9_A_t			VSS	128	272	DQS7_B_t, TDQS7_B_t		
	CB4_A	58	202	VSS			DQ20_B	129	273	VSS		
	VSS	59	203	CB6_A			VSS	130	274	DQ22_B		
	CB5_A	60	204	VSS			DQ21_B	131	275	VSS		
	VSS	61	205	CB7_A			VSS	132	276	DQ23_B		
	ALERT_n	62	206	VSS			DQ24_B	133	277	VSS		
	VSS	63	207	RESET_n			VSS	134	278	DQ26_B		
	CS0_A_n	64	208	VSS			DQ25_B	135	279	VSS		
	VSS	65	209	CS1_A_n			VSS	136	280	DQ27_B		
	CA0_A	66	210	VSS			DQS3_B_t	137	281	VSS		
	VSS	67	211	CA1_A			DQS3_B_c	138	282	DQS8_B_c, TDQS8_B_c		
	CA2_A	68	212	VSS			VSS	139	283	DQS8_B_t, TDQS8_B_t		
	VSS	69	213	CA3 A			DQ28 B	140	284	VSS		
	CA4_A	70	214	VSS			VSS	141	285	DQ30_B		
	VSS	71	215	CA5_A			DQ29_B	142	286	VSS		
	CA6_A	72	216	VSS			VSS	143	287	DQ31_B		
	VSS	73	217	CK_t			RFU	144	288	VSS		
NOTE 1 Individual Annex standards for RDIMM define pin use where multiple functions are possible.												
NOTE 2 Pin 220 for 8000 Mbps and faster RDIMM raw cards do not have the edge pin present.												

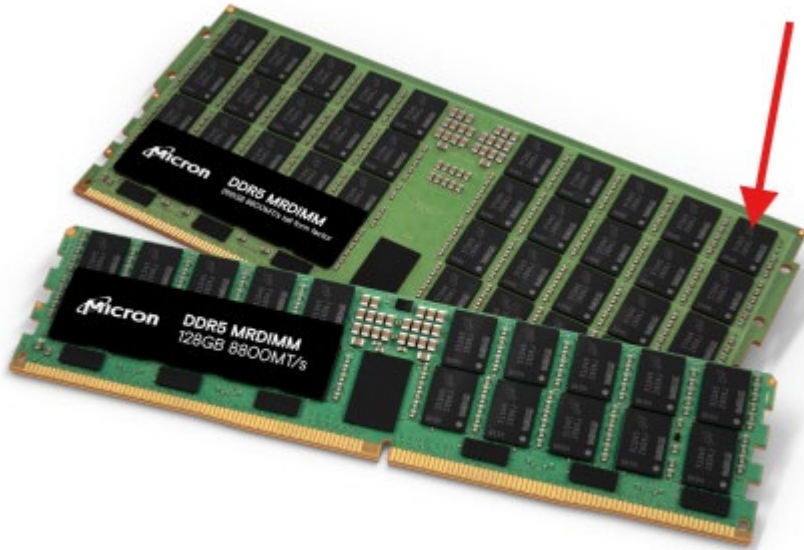
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Ex. CC-88 (JEDEC 305B) at 7 (illustrating clock signal lines (<i>e.g.</i> , “CK_t” and “CK_c”), control/address (C/A) signal lines (<i>e.g.</i> , “CA[]_A,” “CA[]_B,” “CS[]_A_n,” and “CS[]_B_n”), and data signal lines (<i>e.g.</i> , “DQ[]_A” and “DQ[]_B”)).
[6.a] a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus;	The '407 Accused Products comprise a printed circuit board (PCB) having connectors configured to provide electrical connections between the memory module and the system bus. For example: <i>See supra</i> limitation [1.a].
[6.b] memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices;	The '407 Accused Products comprise memory devices mounted on the PCB, the memory devices including first memory devices and second memory devices. For example, The Accused Products have memory devices mounted on the PCB. <i>See, e.g.</i>,:

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Claim Language	'407 Accused Products ⁴
	 Ex. CC-01 (DDR5 Brochure) at 1 (showing an example of a memory device mounted on the printed circuit board).

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Claim Language	'407 Accused Products ⁴
	 Figure 1: MRDIMM low-profile form factor (foreground) and TFF (background) Ex. CC-02 (MRDIMM Product Brief) at 1.  Ex. 51 to Complaint (Micron RDIMM Importation) at 8.

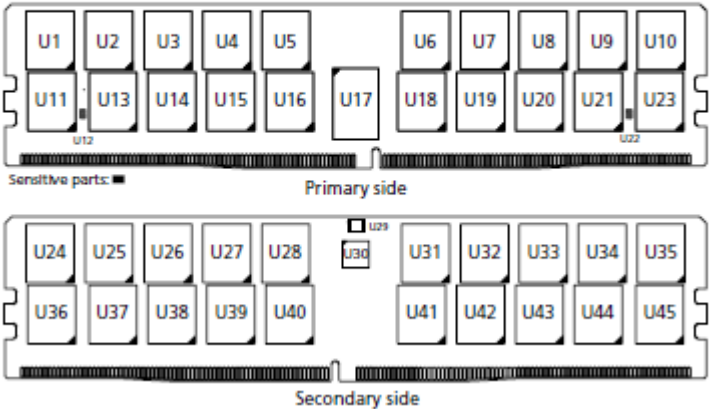
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Claim Language	'407 Accused Products ⁴
	Ex. 51 to Complaint (Micron MRDIMM Importation) at 12. Ex. 51 to Complaint (HPE RDIMM Importation at 6).

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Claim Language	'407 Accused Products ⁴
	Ex. 57 to Complaint (Lenovo RDIMM Importation) at 6. Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 4.

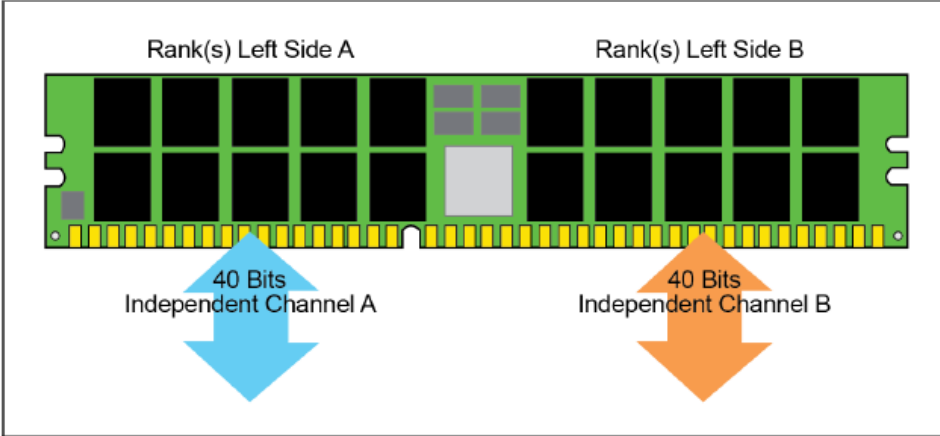
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	General Description High-speed DDR5 SDRAM modules use DDR5 SDRAM devices with four or eight internal memory bank groups. DDR5 SDRAM modules utilizing 4- and 8-bit-wide DDR5 SDRAM devices have eight internal bank groups consisting of four memory banks each, providing a total of 32 banks. 16-bit-wide DDR5 SDRAM devices have four internal bank groups consisting of four memory banks each, providing a total of sixteen banks. DDR5 SDRAM modules benefit from DDR5 SDRAM's use of an 16n-prefetch architecture with an interface designed to transfer two data words per clock cycle at the I/O pins. A single READ or WRITE operation for the DDR5 SDRAM effectively consists of a single 16n-bit-wide, eight-clock data transfer at the internal DRAM core and sixteen corresponding n-bit-wide, one-half-clock-cycle data transfers at the I/O pins. Ex. CC-03 (DDR5 RDIMM Datasheet) at 5. Figure 1: 288-Pin DDR5 RDIMM (R/C-A0)  Ex. CC-07 (DDR5 RDIMM Addendum) at 1 (showing the memory devices); <i>see also id.</i> at 4-6 (“Table-to-Module DQ Map” showing pin mapping for the memory devices).

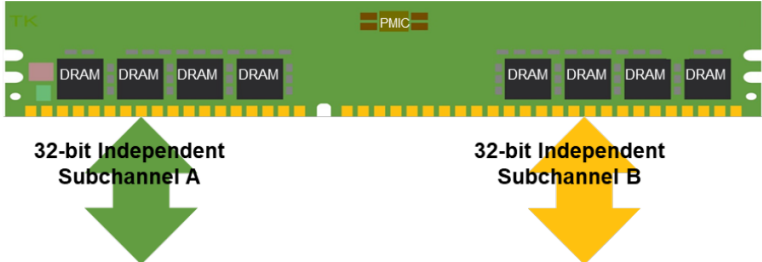
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Claim Language	'407 Accused Products ⁴
	The diagram illustrates a DDR5 DIMM with two ranks, Rank(s) Left Side A and Rank(s) Left Side B. Each rank is connected to an independent channel, Channel A and Channel B, via 40-bit data buses. Channel A is represented by a blue double-headed arrow and Channel B by an orange double-headed arrow. Figure 4: DDR5 DIMM Ex. CC-08 (DDR5 White Paper) at 4. Furthermore, the memory devices include first memory devices and second memory devices. For example, the Accused Products include first memory devices that communicate data with the memory controller via independent channel A (or subsets of data signal lines in channel A), and second memory devices that communicate with the memory controller via independent channel B (or subsets of data signal lines in channel B). <i>See, e.g.,:</i>

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Claim Language	'407 Accused Products ⁴
	In addition to higher data rates and improvements to the I/O circuitry, DDR5 introduces other new protocol features unrelated to data rate that are integral to increasing bandwidth and performance. For example, DDR5 DIMMs feature two 40-bit (32 bits plus ECC) independent channels. When combined with a new default burst length of 16 (BL16) in the DDR5 component, this allows a single burst to access 64B of data (the typical CPU cache line size) using only one of the independent channels, or only half of the DIMM. Providing this ability to interleave accesses from these two independent channels enables tremendous improvements to concurrency, essentially turning an 8-channel system as we know it today into a 16-channel system.  Figure 4: DDR5 DIMM Ex. CC-08 (DDR5 White Paper) at 4.

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Claim Language	'407 Accused Products ⁴
	Module Layout and Data Access A key difference of a DDR5 module as compared to a DDR4 module is the presence of subchannels (Figure 1). A standard DDR5 module has two independent subchannels. Each subchannel has up to two physical package ranks. Each DRAM package can be configured in an initiator/target topology to enable additional logical ranks for increased density.  <i>Figure 1: DDR5 1Rx8 UDIMM Module Illustrating Two Independent Subchannels</i> By enabling the new DDR5 subchannel module layout, the default burst length (BL) needed to be increased from 8 to 16; this achieves the same data payload as a DDR4 module per transaction. The doubling of the BL implies a halving of data inputs/outputs (I/Os) required to fulfill the same amount of data for a given system access size. The independent subchannels increase concurrency and support better scheduling from the memory controller. As an example, each subchannel allows for 32 data I/Os with a BL of 16, resulting in 64-byte payloads. A READ operation from the combined subchannels results in an output of 128 bytes. Ex. CC-11 Micron DDR5: Client Module Features, Micron at 4 (“DDR5 Features White Paper”) at 1.

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	Features • DDR5 functionality and operations supported as defined in the component data sheet • 287/288-pin RDIMM • Supports ECC error detection and correction • On-DIMM SPD EEPROM with Hub function and integrated temperature sensor (TS) • Two on-DIMM discrete TS5 temperature sensors • On-DIMM power management integrated circuit (PMIC) • Sideband access with I3C-basic/I2C support • Two independent I/O sub channels for increased bandwidth • DDR command/address bus to RCD • Gold edge contacts • Halogen-free • Fly-by topology • Terminated clock, control and command/address bus Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

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	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing multiple channels with corresponding subset of data lines). <table><tr><th>Parameter</th><th>Options</th><th>Notes</th></tr><tr><td>DIMM organization</td><td>x80 ECC (EC8), x72 ECC (EC4)</td><td>Two independent 40-bit or 36-bit I/O sub-channels</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 2. Rank Addressing Two chip select inputs per sub-channel (CS0_n and CS1_n) on Micron's modules are used to select a specific rank of DRAM. Each sub-channel has two chip select inputs (CS0x_n and CS1x_n). Ex. CC-03 (DDR5 RDIMM Datasheet) at 12.	Parameter	Options	Notes	DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels
Parameter	Options	Notes					
DIMM organization	x80 ECC (EC8), x72 ECC (EC4)	Two independent 40-bit or 36-bit I/O sub-channels					

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																
	<table><tr><td>CA[6:0]_A CA[6:0]_B</td><td>DDR Input</td><td>V_{DD}</td><td>Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.</td></tr><tr><td>CS[1:0]_A CS[1:0]_B</td><td>Input</td><td>V_{DD}</td><td>Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V_{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V_{DD}.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 8 (showing channels A and B have separate CA Bus and CS commands for first memory devices and second memory devices). <table><tr><th>Symbol</th><th>Type</th><th>I/O Level</th><th>Description</th></tr><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>V_{DD}</td><td>Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V_{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.</td></tr></table> Ex. CC-03 (DDR5 RDIMM Datasheet) at 9 (showing channels A and B have separate dedicated data lines for first memory devices and second memory devices). The JEDEC DDR5 RDIMM standard further demonstrates the Accused Products’ edge connectors allow each respective group of the memory devices to be configured to communicate data with the memory controller via a corresponding subset of the data signal lines. <i>See, e.g.,</i>	CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.	CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .	Symbol	Type	I/O Level	Description	DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.
CA[6:0]_A CA[6:0]_B	DDR Input	V _{DD}	Command/Address Inputs: CA signals provide the command and address inputs according to the Command Truth Table. Note: Since some commands are multi-cycle, the pins may not be interchanged between devices on the same bus. The address inputs also provide the op-code during MODE REGISTER SET commands.														
CS[1:0]_A CS[1:0]_B	Input	V _{DD}	Chip Select: All commands are masked when CS_n is registered HIGH. CS_n provides for external rank selection on systems with multiple ranks. CS_n is considered part of the command code. CS_n is also used to enter and exit the parts from power down mode and self refresh mode. While not in self refresh mode the CS_n input buffer operates with the same ODT and V _{REF} parameters as configured by the CA_ODT strap setting or mode register. When in self refresh, the CS_n is a CMOS rail-to-rail signal with DC HIGH and LOW at 80% and 20% of V _{DD} .														
Symbol	Type	I/O Level	Description														
DQ[31:0]_A DQ[31:0]_B	Input/ Output	V _{DD}	Data Input/Output: Bidirectional data bus. If CRC is enabled via the mode register, then CRC code is added at the end of data burst. Any DQ from DQ0–DQ3 may indicate the internal V _{REF} level during test via mode register setting MR4 A4 = HIGH. Refer to the vendor-specific data sheets to determine which DQ is used.														

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	Table 3 — Pin Definitions <table><tr><th>Pin Name</th><th>Description</th><th>Pin Name</th><th>Description</th></tr><tr><td>CA[6:0]_A CA[6:0]_B</td><td>Command and Address (CA) Bus</td><td>DQ[31:0]_A DQ[31:0]_B</td><td>DIMM memory Data bus channel A and B</td></tr><tr><td>CS[1:0]_A_n CS[1:0]_B_n</td><td>Control (Chip Select) (CS_n)</td><td>CB[7:0]_A CB[7:0]_B</td><td>DIMM ECC Checkbits (CB) channel A and B</td></tr></table> Ex. CC-88 (JEDEC 305B) at 3 (showing each channel has separate CA Bus and CS commands). <table><tr><td>DQ[31:0]_A DQ[31:0]_B</td><td>Input/ Output</td><td>VDD</td><td>Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 R4=High. Refer to vendor specific data sheets to determine which DQ is used.</td></tr></table> Ex. CC-88 (JEDEC 305B) at 4 (showing each channel has separate data lines).	Pin Name	Description	Pin Name	Description	CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B	CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B	DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 R4=High. Refer to vendor specific data sheets to determine which DQ is used.
Pin Name	Description	Pin Name	Description														
CA[6:0]_A CA[6:0]_B	Command and Address (CA) Bus	DQ[31:0]_A DQ[31:0]_B	DIMM memory Data bus channel A and B														
CS[1:0]_A_n CS[1:0]_B_n	Control (Chip Select) (CS_n)	CB[7:0]_A CB[7:0]_B	DIMM ECC Checkbits (CB) channel A and B														
DQ[31:0]_A DQ[31:0]_B	Input/ Output	VDD	Data Input/ Output: Bi-directional data bus. If CRC is enabled via Mode register, then CRC code is added at the end of Data Burst. Any DQ from DQ0-DQ3 may indicate the internal Vref level during test via Mode Register Setting MR4 R4=High. Refer to vendor specific data sheets to determine which DQ is used.														

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

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	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><td>DIMM Side</td><td>Row</td><td>Clock</td><td>CS</td><td>CA</td><td>Rank</td></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><td>DIMM Side</td><td>Row</td><td>Clock</td><td>CS</td><td>CA</td><td>Rank</td></tr><tr><td>Top</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="11">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><td>Side</td><td>DQS0</td><td>DQS5</td><td>DQS1</td><td>DQS6</td><td>DQS2</td><td>DQS7</td><td>DQS3</td><td>DQS8</td><td>DQS4</td><td>DQS9</td></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td colspan="2"></td><td colspan="2">[15:8]</td><td colspan="2"></td><td colspan="2">[31:24]</td><td colspan="2"></td></tr><tr><td>Bottom</td><td colspan="2">[7:0]</td><td colspan="2"></td><td colspan="2">[23:16]</td><td colspan="2"></td><td colspan="2">CB[7:0]</td></tr><tr><th colspan="11">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><td>Side</td><td>DQS0</td><td>DQS5</td><td>DQS1</td><td>DQS6</td><td>DQS2</td><td>DQS7</td><td>DQS3</td><td>DQS8</td><td>DQS4</td><td>DQS9</td></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td colspan="2">[7:0]</td><td colspan="2"></td><td colspan="2">[23:16]</td><td colspan="2"></td><td colspan="2">[31:24]</td></tr><tr><td>Bottom</td><td colspan="2"></td><td colspan="2">[15:8]</td><td colspan="2"></td><td colspan="2">[31:24]</td><td colspan="2">CB[7:0]</td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive CA, CS, and data signals via different subsets of CA, CS, and data signal lines).	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCCK	QACS0	QACA	0	Bottom	Lower	QCCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top			[15:8]				[31:24]				Bottom	[7:0]				[23:16]				CB[7:0]		1Rx8 Sub Channel 1 (Channel B)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top	[7:0]				[23:16]				[31:24]		Bottom			[15:8]				[31:24]		CB[7:0]	
1Rx8 (less than 8000 Mbps)																																																																																																																																																															
DIMM Side	Row	Clock	CS	CA	Rank																																																																																																																																																										
Top	Lower	QACK	QACS0	QACA	0																																																																																																																																																										
Bottom	Lower	QACK	QACS0	QACA	0																																																																																																																																																										
1Rx8 (8000 Mbps and Faster)																																																																																																																																																															
DIMM Side	Row	Clock	CS	CA	Rank																																																																																																																																																										
Top	Lower	QCCK	QACS0	QACA	0																																																																																																																																																										
Bottom	Lower	QCCK	QACS0	QACA	0																																																																																																																																																										
1Rx8 Sub Channel 0 (Channel A)																																																																																																																																																															
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																					
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																					
Top			[15:8]				[31:24]																																																																																																																																																								
Bottom	[7:0]				[23:16]				CB[7:0]																																																																																																																																																						
1Rx8 Sub Channel 1 (Channel B)																																																																																																																																																															
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																					
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																					
Top	[7:0]				[23:16]				[31:24]																																																																																																																																																						
Bottom			[15:8]				[31:24]		CB[7:0]																																																																																																																																																						
[6.c] circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable to:	The '407 Accused Products comprise circuitry on the PCB and coupled to the memory devices, wherein the circuitry is configurable. For example, the '407 Accused Products that are RDIMM devices include RCD (Registering Clock Driver) circuitry on the PCB. See, e.g.,																																																																																																																																																														

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	 Ex. 51 to Complaint (Micron RDIMM Importation) at 11.

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Claim Language	'407 Accused Products ⁴
	 Ex. 55 to Complaint (HPE RDIMM Importation) at 10.

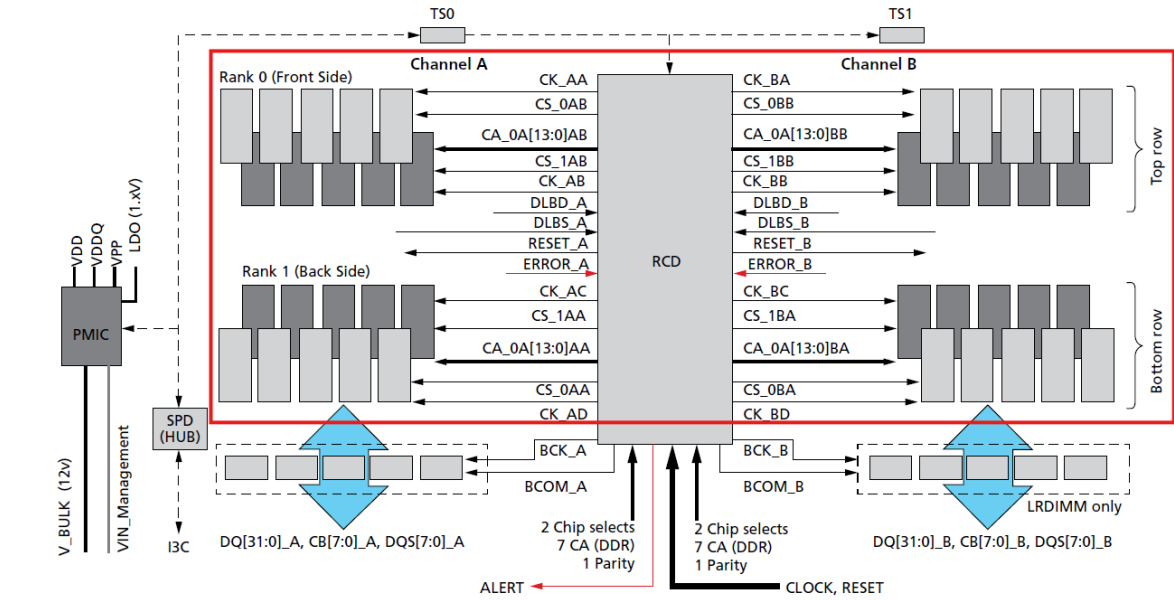
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	 Ex. 57 to Complaint (Lenovo RDIMM Importation) at 10. 

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	Ex. 59 to Complaint (SuperMicro RDIMM Importation) at 10. For example, the '407 Accused Products that are MRDIMM devices include MRCD (Multiplexed Registering Clock Driver) circuitry on the PCB. <i>See, e.g.,</i>  Ex. 53 to Complaint (Micron MRDIMM Importation at 11). Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. The '407 Accused Products' circuitry is coupled to the memory devices. For example, the '407 Accused Products RCD circuitry and MRCD circuitry are coupled to the memory devices:

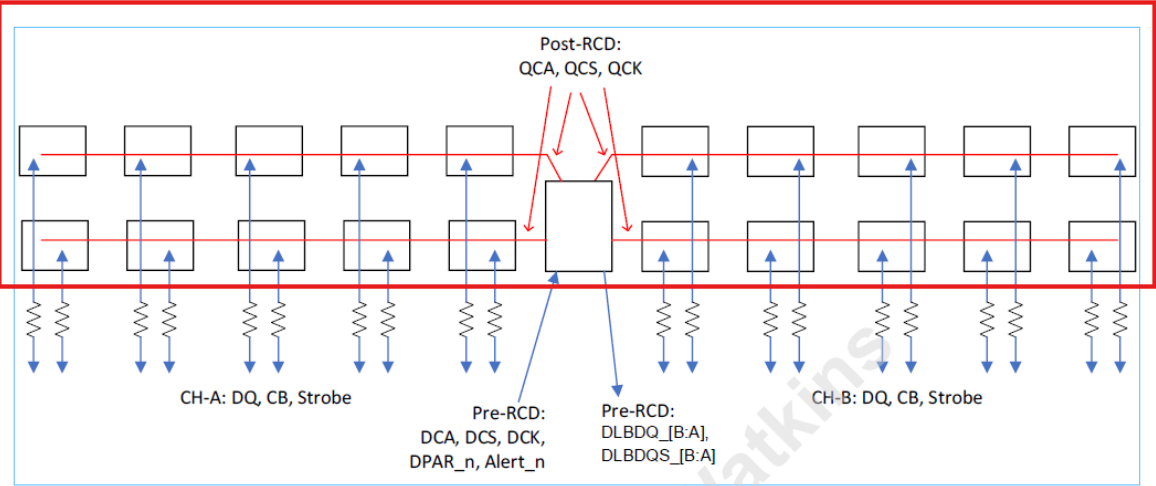
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	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

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	Host Side DRAM Side Ex. CC-07 (DDR5 RDIMM Addendum) at 8.

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	6.1 Signal Groups (cont'd)  Figure 2 — (Example) RDIMM 2Rx4 Topology Ex. CC-31 (JESD305B.01 (April 2026)) at 13.

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Claim Language	'407 Accused Products ⁴																																											
	<table><tr><td rowspan="4">Data buffer control and communication outputs</td><td>BCS_[B:A]_n</td><td>POD</td><td>the register. There is a separate signal per channel.</td></tr><tr><td>BCOM[2:0]_[B:A]</td><td>POD</td><td>Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.</td></tr><tr><td>BRST_[B:A]_n</td><td>CMOS</td><td>Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.</td></tr><tr><td>BCK_[B:A]_t BCK_[B:A]_c</td><td>POD differential</td><td>Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.</td></tr><tr><td>Output Control bus</td><td>Q[B:A]CS[1:0]_[B:A]_n</td><td>POD</td><td>Differential clock output pair to the data buffer. Separate per DRAM channel.</td></tr><tr><td>Output Address and Command bus</td><td>Q[B:A]CA[13:0]_[B:A]</td><td>POD</td><td>Chip Select signals to the DRAMs. 2 copies of each signal.</td></tr><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c</td><td>POD differential</td><td>Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).</td></tr><tr><td>Loopback</td><td>QLBD QLBS</td><td>POD</td><td>Clock outputs to the DRAMs. Four copies per channel.</td></tr><tr><td>Reset output</td><td>QRST_[B:A]_n</td><td>CMOS</td><td>Loopback outputs to Host QLBD - loopback Data QLBS - loopback Strobe</td></tr><tr><td>Error out</td><td>ALERT_n</td><td>POD</td><td>Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD QRST_n to reset the DDR5 SDRAM on all DIMM topologies. The QRST outputs are asserted at power up. RCD requires MRW to independently de-assert to allow staggering of sub-channels.</td></tr><tr><td></td><td></td><td></td><td>When LOW, this output indicates that a parity error was identified associated with the CA inputs when parity checking is enabled or that the DERROR_IN_n input was asserted, regardless of whether parity checking is enabled or not. One signal for the two channels.</td></tr></table>	Data buffer control and communication outputs	BCS_[B:A]_n	POD	the register. There is a separate signal per channel.	BCOM[2:0]_[B:A]	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.	BRST_[B:A]_n	CMOS	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Differential clock output pair to the data buffer. Separate per DRAM channel.	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Chip Select signals to the DRAMs. 2 copies of each signal.	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).	Loopback	QLBD QLBS	POD	Clock outputs to the DRAMs. Four copies per channel.	Reset output	QRST_[B:A]_n	CMOS	Loopback outputs to Host QLBD - loopback Data QLBS - loopback Strobe	Error out	ALERT_n	POD	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD QRST_n to reset the DDR5 SDRAM on all DIMM topologies. The QRST outputs are asserted at power up. RCD requires MRW to independently de-assert to allow staggering of sub-channels.				When LOW, this output indicates that a parity error was identified associated with the CA inputs when parity checking is enabled or that the DERROR_IN_n input was asserted, regardless of whether parity checking is enabled or not. One signal for the two channels.		
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3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels.																																												
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Claim Language	'407 Accused Products ⁴			
	Data buffer control and communication outputs	BCS_[B:A]_n	POD	the register. There is a separate signal per channel. Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
		BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
		BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
		BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
	Loopback	QLBD QLBS	POD	Loopback outputs to Host QLBD - loopback Data QLBS - loopback Strobe
	Reset output	QRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD QRST_n to reset the DDR5 SDRAM on all DIMM topologies. The QRST outputs are asserted at power up. RCD requires MRW to independently de-assert to allow staggering of sub-channels.
	Error out	ALERT_n	POD	When LOW, this output indicates that a parity error was identified associated with the CA inputs when parity checking is enabled or that the DERROR_IN_n input was asserted, regardless of whether parity checking is enabled or not. One signal for the two channels.
Ex. CC-30 (JEDEC 82-513) at 4.				
3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels.				

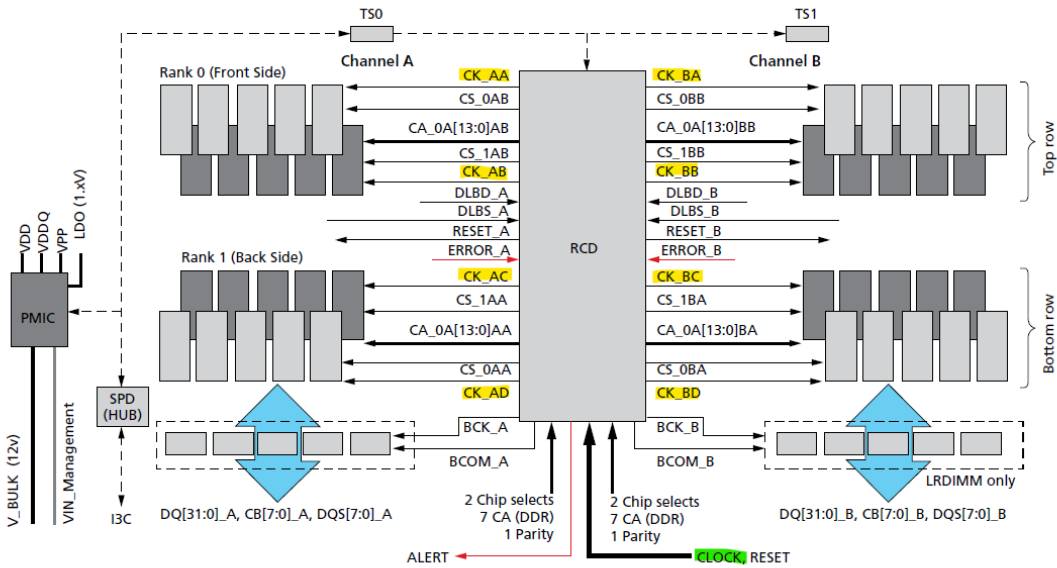
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Ex. CC-30 (JEDEC 82-513) at 6. Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. The circuitry on the PCB is configurable as discussed below. <i>See</i> limitations [6.c.1]-[6.c.5].

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[6.c.1] receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines;	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the C/A signal lines. For example: <i>See supra</i> limitation [1.c.1].
[6.c.2] generate module C/A signals in response to the input C/A signals;	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to generate module C/A signals in response to the input C/A signals. For example: <i>See supra</i> limitation [1.c.2].
[6.c.3] generate local clocks from the system clock,	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to generate local clocks from the system clock. For example, the RCD circuitry is configurable to generate local clock outputs (yellow) from the system clock input (green). <i>See, e.g.,</i>

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	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. Registering Clock Driver Operation Registered DDR5 SDRAM modules use a registering clock driver device consisting of a register and a phase-lock loop (PLL). The device complies with the JEDEC DDR5 RCD specification. Refer to the JESD82-511 (DDR5RCD01) and JESD82-512 (DDR5RCD02) specifications for more detail. To reduce the electrical load on the host memory controller's command, address, and control bus, Micron's RDIMMs and LRDIMMs utilize a DDR5 registering clock driver (RCD). The RCD presents a single load to the controller while redriving signals to the DDR5 SDRAM devices, which helps enable higher densities and increase signal integrity. The RCD also provides a low-jitter, low-skew PLL that redistributes a differential clock pair to multiple differential pairs of clock outputs. Ex. CC-03 (DDR5 RDIMM Datasheet) at 12.

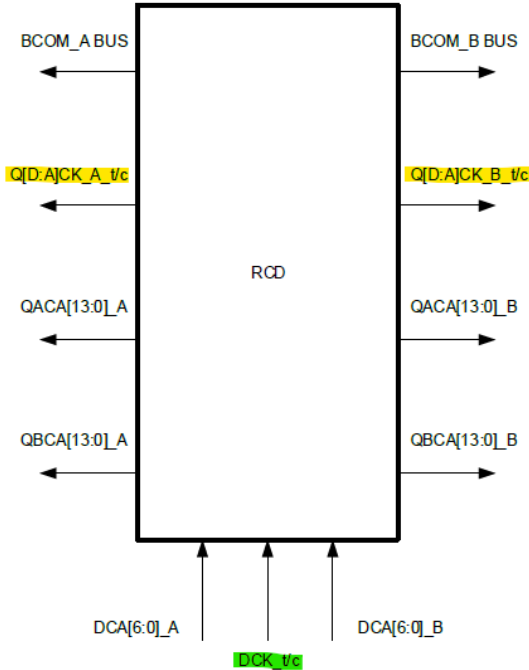
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Claim Language	'407 Accused Products ⁴
	<u>Host Side</u> DCS[1:0]A_n DCS[1:0]B_n DCA_[6:0]A DCA_[6:0]B ALERT_n LBDQ, RSP_A_n LBDQS, RSP_B_n PAR_A PAR_B HSCL HSDA CK_c CK_t RESET_n R E G I S T E R & P S C ZQ <u>DRAM Side</u> QACS[1:0][B:A]_n: QBCS[1:0][B:A]_n: QACA[13:0][B:A] QBCA[13:0][B:A] DERROR_IN_[B:A]_n DLBD_[B:A] DLBS_[B:A] QRST[B:A] V_{SS} Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing generation of local clocks and reception of the system clock).

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Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

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Claim Language	'407 Accused Products ⁴
	Block Diagram  The diagram shows a central rectangular block labeled 'RCD'. It has four bidirectional connections on its left side: 'BCOM_A BUS' at the top, 'Q[D:A]CK_A_1/c' in yellow below it, 'QACA[13:0]_A' below that, and 'QBCA[13:0]_A' at the bottom. On its right side, it has four bidirectional connections: 'BCOM_B BUS' at the top, 'Q[D:A]CK_B_1/c' in yellow below it, 'QACA[13:0]_B' below that, and 'QBCA[13:0]_B' at the bottom. At the bottom of the RCD block, three arrows point upwards into it, labeled from left to right as 'DCA[6:0]_A', 'DCK_1/c' in green, and 'DCA[6:0]_B'. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. As described in the JEDEC 82-512 and 82-513 standards that the '407 Accused Products comply with, the RCD has a “common clock input and PLL, but produces separate clock outputs to the DRAM channels.” <i>See, e.g.,</i>

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Claim Language	'407 Accused Products ⁴
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6. 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.

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	6.1 Signal Groups (cont'd) Figure 2 — (Example) RDIMM 2Rx4 Topology Ex. CC-31 (JESD305B.01 (April 2026)) at 13 (illustrating RCD, in red, receiving “Pre-RCD” clock (“DCK”) and generating “Post-RCD” control and address signals (“QCK”).

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

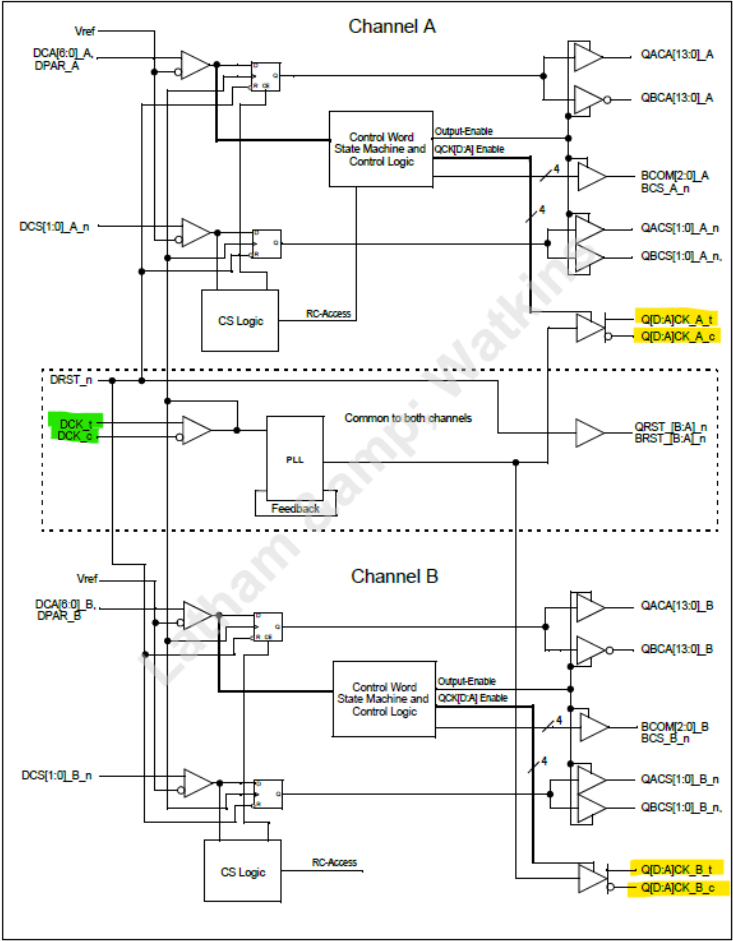
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																																																																																																																																														
	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCKK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCKK</td><td>QACS0</td><td>QACA</td><td>0</td></tr></table> <table><tr><th colspan="11">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td>[7:0]</td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td>CB[7:0]</td><td></td></tr></table> <table><tr><th colspan="11">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td></td><td>[7:0]</td><td></td><td></td><td>[23:16]</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Bottom</td><td></td><td></td><td>[15:8]</td><td></td><td></td><td></td><td>[31:24]</td><td></td><td>CB[7:0]</td><td></td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing groups of memory devices on various sides and in various rows on the module configured to receive corresponding local clocks, e.g., QACK, QBCK, QCKK, QDCK, generated by the RCD). Furthermore, the '407 Accused Products comply with the JEDEC 82-512 and 82-513 standards, which further demonstrates that the RCD generates local clocks from the system clock.	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCKK	QACS0	QACA	0	Bottom	Lower	QCKK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top			[15:8]				[31:24]				Bottom		[7:0]			[23:16]				CB[7:0]		1Rx8 Sub Channel 1 (Channel B)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top		[7:0]			[23:16]						Bottom			[15:8]				[31:24]		CB[7:0]	
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	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.
	Loopback	DLBD_[B:A] DLBS_[B:A]	POD V _{REF} based	Loopback Inputs to RCD DLBD - loopback Data DLBS - loopback Strobe
	Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.
	Error input	ERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.
	Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.
		BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.
		BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.
		BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.
	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.
	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).
	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
Ex. CC-32 (JEDEC 82-512) at 4. 3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.				

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Claim Language	'407 Accused Products ⁴
	3.2 Logic Diagram Positive Logic  Ex. CC-32 (JEDEC 82-512) at 7.

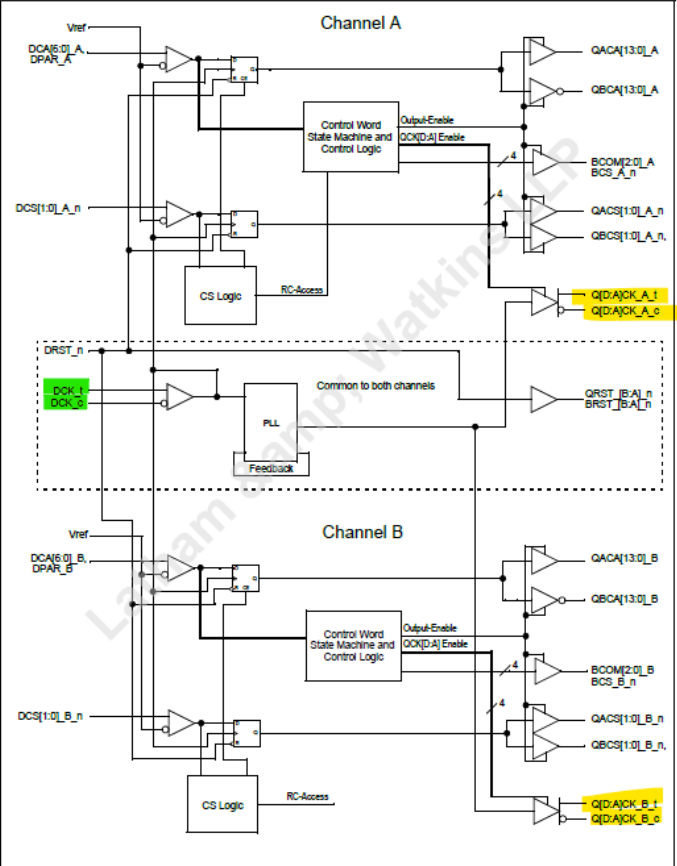
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13.1 Input Clock Architecture The RCD is provided a clock by the entity that is controlling it, i.e., memory controller, or any test equipment. This clock is used by the RCD to generate all the IO specific timing, and is the same clock used by the controlling entity as the one and only deterministic source of all the response timings to and from the RCD. This allows the memory controller to have a deterministic control of every event in the RCD. Spread Spectrum Clocking (SSC) Capability Ex. CC-32 (JEDEC 82-512) at 213.																																																
	<table><tr><td>Clock inputs</td><td>DCK_t/DCK_c</td><td>POD differential</td><td>Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.</td></tr><tr><td>Loopback</td><td>DLBD_[B:A]</td><td>POD V_{REF} based</td><td>Loopback Inputs to RCD</td></tr><tr><td></td><td>DLBS_[B:A]</td><td></td><td>DLBD - loopback Data DLBS - loopback Strobe</td></tr><tr><td>Reset input</td><td>DRST_n</td><td>CMOS input</td><td>Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.</td></tr><tr><td>Error input</td><td>DERROR_IN_[B:A]_n</td><td>Low voltage swing POD input</td><td>DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.</td></tr><tr><td rowspan="4">Data buffer control and communication outputs</td><td>BCS_[B:A]_n</td><td>POD</td><td>Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.</td></tr><tr><td>BCOM[2:0]_[B:A]</td><td>POD</td><td>Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.</td></tr><tr><td>BRST_[B:A]_n</td><td>CMOS</td><td>Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.</td></tr><tr><td>BCK_[B:A]_t BCK_[B:A]_c</td><td>POD differential</td><td>Differential clock output pair to the data buffer. Separate per DRAM channel.</td></tr><tr><td>Output Control bus</td><td>Q[B:A]CS[1:0]_[B:A]_n</td><td>POD</td><td>Chip Select signals to the DRAMs. 2 copies of each signal.</td></tr><tr><td>Output Address and Command bus</td><td>Q[B:A]CA[13:0]_[B:A]</td><td>POD</td><td>Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).</td></tr><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c</td><td>POD differential</td><td>Clock outputs to the DRAMs. Four copies per channel.</td></tr></table>	Clock inputs	DCK_t/DCK_c	POD differential	Differential system clock input pair to the PLL. The clock is common to both RCD channels, CH_A and CH_B.	Loopback	DLBD_[B:A]	POD V _{REF} based	Loopback Inputs to RCD		DLBS_[B:A]		DLBD - loopback Data DLBS - loopback Strobe	Reset input	DRST_n	CMOS input	Active LOW asynchronous reset input. When LOW, it causes a reset of the internal latches and disables the outputs, thereby forcing the outputs to float.	Error input	DERROR_IN_[B:A]_n	Low voltage swing POD input	DRAM CRC ALERT_n output is connected to this input pin, which in turn is buffered and redriven to the ALERT_n output of the register. There is a separate signal per channel.	Data buffer control and communication outputs	BCS_[B:A]_n	POD	Data buffer chip selects. Indicate a valid command on the BCOM bus, and used to signal non-target termination. Separate per DRAM channel.	BCOM[2:0]_[B:A]	POD	Data buffer command. The start of each command is signaled by an active chip select. Separate per DRAM channel.	BRST_[B:A]_n	CMOS	Re-driven or CMD based reset. This is an asynchronous output. It is the responsibility of the RCD BRST to reset the DDR5 Data Buffer on all DIMM topologies.	BCK_[B:A]_t BCK_[B:A]_c	POD differential	Differential clock output pair to the data buffer. Separate per DRAM channel.	Output Control bus	Q[B:A]CS[1:0]_[B:A]_n	POD	Chip Select signals to the DRAMs. 2 copies of each signal.	Output Address and Command bus	Q[B:A]CA[13:0]_[B:A]	POD	Command/Address bus outputs to the DRAMs, valid after the specified clock count and immediately following a rising edge of the clock. Two copies of each signal. When Output Inversion is enabled in RW00[5], Copy B output signals get inverted during all commands other than Deselect. Both copies drive High during idle cycles (i.e., Deselect).	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.		
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Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.																																													
Ex. CC-30 (JEDEC 82-513) at 4.																																																

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Claim Language	'407 Accused Products ⁴
	3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.

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Claim Language	'407 Accused Products ⁴
	3.2 Logic Diagram Positive Logic  Ex. CC-30 (JEDEC 82-513) at 7.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	13.1 Input Clock Architecture The RCD is provided a clock by the entity that is controlling it, i.e., memory controller, or any test equipment. This clock is used by the RCD to generate all the IO specific timing, and is the same clock used by the controlling entity as the one and only deterministic source of all the response timings to and from the RCD. This allows the memory controller to have a deterministic control of every event in the RCD. Spread Spectrum Clocking (SSC) Capability Ex. CC-30 (JEDEC 82-513) at 222.
[6.c.3.i] the local clocks including a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock,	In the '407 Accused Products, the local clocks include a first local clock having a first phase shift from the system clock and a second local clock having a second phase shift from the system clock. For example:

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Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. The RCD of the '407 Accused Products support “output delay control” where a first local clock has a first phase shift from the system clock and a second local clock has a second phase shift from the system clock. <i>See, e.g.,</i>

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	3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disabled independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-32 (JEDEC 82-512) at 12. Table 23 — Control Words Duplicated for Frequency Context Support <table border="1"> <thead> <tr> <th>Register Address</th><th>Description</th></tr> </thead> <tbody> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCCCK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </tbody> </table> Ex. CC-32 (JEDEC 82-512) at 68 (showing local clocks QACK, QBCK, QCCCK, and QDCK that each have a phase shift from the system clock).	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCCCK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
Register Address	Description																																								
RW0A	QCK Signals Driver Characteristics Control Word																																								
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U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																																																																																																										
	8.13.2 RW12 - QACK Output Delay Control Word Table 118 — RW12: QACK Output Delay Control Word <table><tr><th colspan="8">Setting</th><th rowspan="2">Definition</th><th rowspan="2">Encoding</th></tr><tr><th>OP 7</th><th>OP 6</th><th>OP 5</th><th>OP 4</th><th>OP 3</th><th>OP 2</th><th>OP 1</th><th>OP 0</th></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td rowspan="7">Output Delay Control for QACK_ t/QACK_c Output Signals^{1,2}</td><td>Delay Outputs by +(0/64) * t_{CK} (Same as Default)</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Delay Outputs by +(1/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Delay Outputs by +(2/64)* t_{CK}</td></tr><tr><td>x</td><td>x</td><td></td><td></td><td></td><td></td><td></td><td></td><td>...</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Delay Outputs by +(61/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Delay Outputs by +(62/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Delay Outputs by +(63/64) * t_{CK}</td></tr><tr><td>x</td><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td><td>Reserved</td></tr><tr><td>x</td><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td><td>Reserved</td></tr><tr><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Output Delay Feature Enable for QACK_ t/QACK_c²</td><td>(Default) Feature Disabled</td></tr><tr><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td></td><td>Feature Enabled³</td></tr></table> NOTE 1 These control bits do not have any effect unless RW12[7] = 1. NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset. NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.	Setting								Definition	Encoding	OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0	x	x	0	0	0	0	0	0	Output Delay Control for QACK_ t/QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)	x	x	0	0	0	0	0	1	Delay Outputs by +(1/64) * t _{CK}	x	x	0	0	0	0	1	0	Delay Outputs by +(2/64)* t _{CK}	x	x							...	x	x	1	1	1	1	0	1	Delay Outputs by +(61/64) * t _{CK}	x	x	1	1	1	1	1	0	Delay Outputs by +(62/64) * t _{CK}	x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	x	0	x	x	x	x	x	x	Reserved	Reserved	x	1	x	x	x	x	x	x	Reserved	Reserved	0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_ t/QACK_c ²	(Default) Feature Disabled	1	x	x	x	x	x	x	x		Feature Enabled ³
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x	0	x	x	x	x	x	x	Reserved	Reserved																																																																																																																		
x	1	x	x	x	x	x	x	Reserved	Reserved																																																																																																																		
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	Ex. CC-32 (JEDEC 82-512) at 146; <i>see also id.</i> at 147-148 (showing that local clocks QACK, QBCK, QCCK, and QDCK that each have a phase shift from the system clock). 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 tCK. Each signal group can be enabled or disable independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. 3.3.8 Per-group Output Delay Control (cont'd) The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111). Ex. CC-30 (JEDEC 82-513) at 11-12.																																																																																																																										

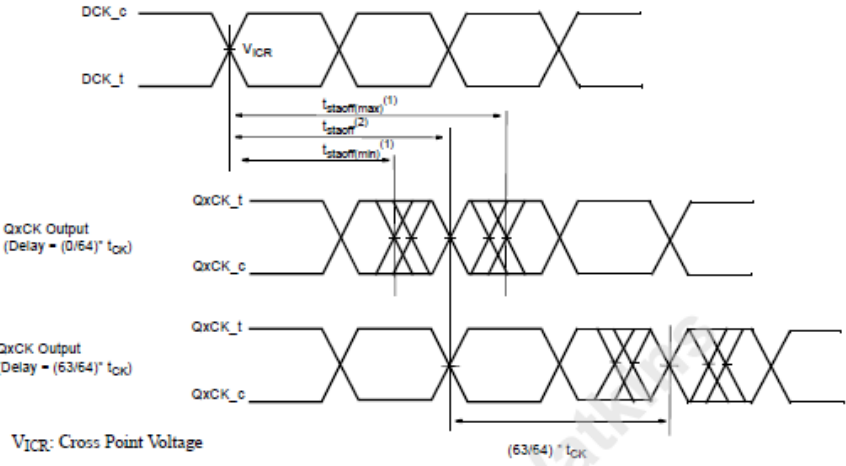
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																										
	Table 23 — Control Words Duplicated for Frequency Context Support <table> <tr> <th>Register Address</th><th>Description</th></tr> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCKK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW33[7:6]</td><td>DFE Tap 5 and Tap 6 Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </table> Ex. CC-30 (JEDEC 82-513) at 70 (showing local clocks QACK, QBCK, QCKK, and QDCK that each have a phase shift from the system clock).	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCKK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW33[7:6]	DFE Tap 5 and Tap 6 Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
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RW13	QBCK Output Delay Control Word																																										
RW14	QCKK Output Delay Control Word																																										
RW15	QDCK Output Delay Control Word																																										
RW17 .. RW1A	QxCS_n Output Delay Control Word																																										
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OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0																																																																																																																		
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x	0	x	x	x	x	x	x	Reserved	Reserved																																																																																																																
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	Ex. CC-30 (JEDEC 82-513) at 150; <i>see also id.</i> at 150-156 (showing that local clocks QACK, QBCK, QCCCK, and QDCK that each have a phase shift from the system clock). For example, local clocks (QxCK) shown below each have a phase shift from the system clock (DCK).																																																																																																																								

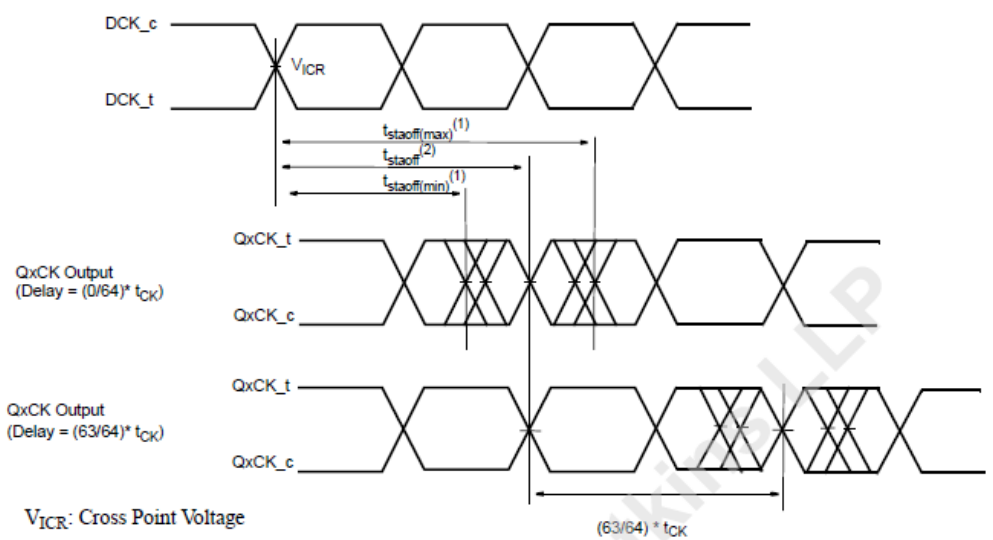
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	14.7 Output Clock Driver Characteristics (cont'd)  Figure 112 — Definition for t_{staoff} and t_{dynoff} in SDR Mode Ex. CC-32 (JEDEC 82-512) at 242.

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x	0	x	x	x	x	x	x		Reserved																																																																																																															
x	1	x	x	x	x	x	x	Reserved																																																																																																																
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	Ex. CC-32 (JEDEC 82-512) at 146; <i>see also id.</i> at 147-148 (showing tables 119, 120, and 121).																																																																																																																							

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	14.7 Output Clock Driver Characteristics (cont'd)  V_{ICR}: Cross Point Voltage 1. $t_{staoFF(max/min)}$ = propagation delay specification limits for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles) over process, voltage and temperature. 2. t_{staoFF} = measured propagation delay for clock signal (rising DCK input clock edge to rising QxCK output clock edge for corresponding data cycles). 3. t_{dynoff} = maximum propagation delay variation over voltage and temperature within t_{staoFF} window. This includes all sources of jitter and drift (e.g. thermal noise, supply noise, voltage/temperature drift, SSC tracking, SSO, etc.) except reference clock noise. Figure 112 — Definition for t_{staoFF} and t_{dynoff} in SDR Mode Ex. CC-30 (JEDEC 82-513) at 253.

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Claim Language	'407 Accused Products ⁴																																																																																																																																																								
	8.13.2 RW12 - QACK Output Delay Control Word Table 119 — RW12: QACK Output Delay Control Word <table><tr><th colspan="8">Setting</th><th rowspan="2">Definition</th><th rowspan="2">Encoding</th></tr><tr><th>OP 7</th><th>OP 6</th><th>OP 5</th><th>OP 4</th><th>OP 3</th><th>OP 2</th><th>OP 1</th><th>OP 0</th></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td rowspan="7">Output Delay Control for QACK_t/ QACK_c Output Signals^{1,2}</td><td>Delay Outputs by +(0/64) * t_{CK} (Same as Default)</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Delay Outputs by +(1/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Delay Outputs by +(2/64)* t_{CK}</td></tr><tr><td>x</td><td>x</td><td colspan="6">...</td><td>...</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Delay Outputs by +(61/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Delay Outputs by +(62/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Delay Outputs by +(63/64) * t_{CK}</td></tr><tr><td>x</td><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td><td>Reserved</td></tr><tr><td>x</td><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td><td>Reserved</td></tr><tr><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Output Delay Feature Enable for</td><td>(Default) Feature Disabled</td></tr><tr><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>QACK_t/QACK_c²</td><td>Feature Enabled³</td></tr><tr><td colspan="10">NOTE 1 These control bits do not have any effect unless RW12[7] = 1.</td></tr><tr><td colspan="10">NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.</td></tr><tr><td colspan="10">NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.</td></tr></table> Ex. CC-30 (JEDEC 82-513) at 150; <i>see also id.</i> at 150-156 (showing tables 120, 121, and 122).	Setting								Definition	Encoding	OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0	x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)	x	x	0	0	0	0	0	1	Delay Outputs by +(1/64) * t _{CK}	x	x	0	0	0	0	1	0	Delay Outputs by +(2/64)* t _{CK}	x	x	...						...	x	x	1	1	1	1	0	1	Delay Outputs by +(61/64) * t _{CK}	x	x	1	1	1	1	1	0	Delay Outputs by +(62/64) * t _{CK}	x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	x	0	x	x	x	x	x	x	Reserved	Reserved	x	1	x	x	x	x	x	x	Reserved	Reserved	0	x	x	x	x	x	x	x	Output Delay Feature Enable for	(Default) Feature Disabled	1	x	x	x	x	x	x	x	QACK_t/QACK_c ²	Feature Enabled ³	NOTE 1 These control bits do not have any effect unless RW12[7] = 1.										NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset.										NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t _{ODU} for the delay to become stable on the outputs.									
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x	0	x	x	x	x	x	x	Reserved	Reserved																																																																																																																																																
x	1	x	x	x	x	x	x	Reserved	Reserved																																																																																																																																																
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[6.c.3.ii] the first phase shift and the second phase shift being programmable independently of each other; and	In the '407 Accused Products, the first phase shift and the second phase shift are programmable independently of each other. For example: For example, the delay control of the local clocks (QxCK) with respect to the system clock (DCK) can be independently programmed. <i>See, e.g.,</i> 3.3.8 Per-group Output Delay Control The RCD supports output delay control for output signal groups defined and located in RW12 to RW1E. The output delay settings are incremental CK fractions of 1/64 t_{CK}. Each signal group can be enabled or disable independently through RCD Control Words. The Host must wait t_{ODU} after the Output Delay Control Word has been written to a signal group before the output becomes stable. The output delay feature controlled by RW12 to RW1E is not supported when the RCD is configured in Test Frequency mode (Band 2, RW05[7] = 1) or in PLL bypass mode (RW05[3:0] = 1111).																																																																																																																																																								

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	Ex. CC-32 (JEDEC 82-512) at 12 (showing “output delay” for output signal groups). Table 23 — Control Words Duplicated for Frequency Context Support <table border="1"> <thead> <tr> <th>Register Address</th><th>Description</th></tr> </thead> <tbody> <tr> <td>RW0A</td><td>QCK Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0C</td><td>QxCA and QxCS_n Signals Driver Characteristics Control Word</td></tr> <tr> <td>RW0D</td><td>Data Buffer Interface Driver Characteristics Control Word</td></tr> <tr> <td>RW10</td><td>IBT Control Word</td></tr> <tr> <td>RW11[3:0]</td><td>Command Latency Adder Configuration Control Word</td></tr> <tr> <td>RW12</td><td>QACK Output Delay Control Word</td></tr> <tr> <td>RW13</td><td>QBCK Output Delay Control Word</td></tr> <tr> <td>RW14</td><td>QCKK Output Delay Control Word</td></tr> <tr> <td>RW15</td><td>QDCK Output Delay Control Word</td></tr> <tr> <td>RW17 .. RW1A</td><td>QxCS_n Output Delay Control Word</td></tr> <tr> <td>RW1B, RW1C</td><td>QxCA Output Delay Control Word</td></tr> <tr> <td>RW1D</td><td>BCS_n and BCOM[2:0] Output Delay Control Word</td></tr> <tr> <td>RW1E</td><td>BCK Output Delay Control Word</td></tr> <tr> <td>RW31[7:4, 0]</td><td>DFE Configuration Control Word</td></tr> <tr> <td>RW40 .. RW47</td><td>Internal VrefCA Control Word</td></tr> <tr> <td>RW48, RW49</td><td>Internal VrefCS Control Word</td></tr> <tr> <td>RW50[0], RW51, RW52[3:0], RW53[3:0], RW54</td><td>CTLE Control Words</td></tr> <tr> <td>PG[00, 01]</td><td>DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver</td></tr> <tr> <td>PG[5]RW[7B:60]</td><td>Per-bit QCA Output Delay Control Words</td></tr> </tbody> </table> Ex. CC-32 (JEDEC 82-512) at 68 (showing delays in QACK, QBCK, QCKK, and QDCK can be set independently).	Register Address	Description	RW0A	QCK Signals Driver Characteristics Control Word	RW0C	QxCA and QxCS_n Signals Driver Characteristics Control Word	RW0D	Data Buffer Interface Driver Characteristics Control Word	RW10	IBT Control Word	RW11[3:0]	Command Latency Adder Configuration Control Word	RW12	QACK Output Delay Control Word	RW13	QBCK Output Delay Control Word	RW14	QCKK Output Delay Control Word	RW15	QDCK Output Delay Control Word	RW17 .. RW1A	QxCS_n Output Delay Control Word	RW1B, RW1C	QxCA Output Delay Control Word	RW1D	BCS_n and BCOM[2:0] Output Delay Control Word	RW1E	BCK Output Delay Control Word	RW31[7:4, 0]	DFE Configuration Control Word	RW40 .. RW47	Internal VrefCA Control Word	RW48, RW49	Internal VrefCS Control Word	RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words	PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4 Coefficients per Input Receiver	PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words
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RW48, RW49	Internal VrefCS Control Word																																										
RW50[0], RW51, RW52[3:0], RW53[3:0], RW54	CTLE Control Words																																										
PG[00, 01]	DFE Gain and Tap 1, Tap 2, Tap 3, Tap 4, Tap 5, Tap 6 Coefficients per Input Receiver																																										
PG[5]RW[7B:60]	Per-bit QCA Output Delay Control Words																																										

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																																																																																																									
	8.13.2 RW12 - QACK Output Delay Control Word Table 119 — RW12: QACK Output Delay Control Word <table><tr><th colspan="8">Setting</th><th rowspan="2">Definition</th><th rowspan="2">Encoding</th></tr><tr><th>OP 7</th><th>OP 6</th><th>OP 5</th><th>OP 4</th><th>OP 3</th><th>OP 2</th><th>OP 1</th><th>OP 0</th></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td rowspan="7">Output Delay Control for QACK_t/ QACK_c Output Signals^{1,2}</td><td>Delay Outputs by +(0/64) * t_{CK} (Same as Default)</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>Delay Outputs by +(1/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>Delay Outputs by +(2/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td colspan="7">...</td><td>...</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>Delay Outputs by +(61/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>Delay Outputs by +(62/64) * t_{CK}</td></tr><tr><td>x</td><td>x</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>Delay Outputs by +(63/64) * t_{CK}</td></tr><tr><td>x</td><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td rowspan="2">Reserved</td><td>Reserved</td></tr><tr><td>x</td><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Reserved</td></tr><tr><td>0</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td rowspan="2">Output Delay Feature Enable for QACK_t/QACK_c²</td><td>(Default) Feature Disabled</td></tr><tr><td>1</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>x</td><td>Feature Enabled³</td></tr></table> NOTE 1 These control bits do not have any effect unless RW12[7] = 1. NOTE 2 RW12[7:0] will be sticky, cleared by power cycle not Reset. NOTE 3 When feature is enabled the delay settings in RW12[5:0] require a time of t_{ODU} for the delay to become stable on the outputs.	Setting								Definition	Encoding	OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0	x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)	x	x	0	0	0	0	0	1	Delay Outputs by +(1/64) * t _{CK}	x	x	0	0	0	0	1	0	Delay Outputs by +(2/64) * t _{CK}	x	x	...							...	x	x	1	1	1	1	0	1	Delay Outputs by +(61/64) * t _{CK}	x	x	1	1	1	1	1	0	Delay Outputs by +(62/64) * t _{CK}	x	x	1	1	1	1	1	1	Delay Outputs by +(63/64) * t _{CK}	x	0	x	x	x	x	x	x	Reserved	Reserved	x	1	x	x	x	x	x	x	Reserved	0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled	1	x	x	x	x	x	x	x	Feature Enabled ³
Setting								Definition	Encoding																																																																																																																	
OP 7	OP 6	OP 5	OP 4	OP 3	OP 2	OP 1	OP 0																																																																																																																			
x	x	0	0	0	0	0	0	Output Delay Control for QACK_t/ QACK_c Output Signals ^{1,2}	Delay Outputs by +(0/64) * t _{CK} (Same as Default)																																																																																																																	
x	x	0	0	0	0	0	1		Delay Outputs by +(1/64) * t _{CK}																																																																																																																	
x	x	0	0	0	0	1	0		Delay Outputs by +(2/64) * t _{CK}																																																																																																																	
x	x	...							...																																																																																																																	
x	x	1	1	1	1	0	1		Delay Outputs by +(61/64) * t _{CK}																																																																																																																	
x	x	1	1	1	1	1	0		Delay Outputs by +(62/64) * t _{CK}																																																																																																																	
x	x	1	1	1	1	1	1		Delay Outputs by +(63/64) * t _{CK}																																																																																																																	
x	0	x	x	x	x	x	x	Reserved	Reserved																																																																																																																	
x	1	x	x	x	x	x	x		Reserved																																																																																																																	
0	x	x	x	x	x	x	x	Output Delay Feature Enable for QACK_t/QACK_c ²	(Default) Feature Disabled																																																																																																																	
1	x	x	x	x	x	x	x		Feature Enabled ³																																																																																																																	
	Ex. CC-30 (JEDEC 82-513) at 150; <i>see also id.</i> at 150-156 (showing how to program the delays independently, in tables 120, 121, and 122).																																																																																																																									

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																																																									
	Table 232 — Output Clock Driver Characteristics at Application Mode (Frequency Band 1) (cont'd) <table><tr><th rowspan="2">Symbol</th><th rowspan="2">Parameter</th><th rowspan="2">Conditions</th><th colspan="2">DDR5-3200</th><th colspan="2">DDR5-3600</th><th colspan="2">DDR5-4000/ 4400/4800</th><th colspan="2">DDR5-5200/ 5600</th><th rowspan="2">Unit</th><th rowspan="2">Note</th></tr><tr><th>Min</th><th>Max</th><th>Min</th><th>Max</th><th>Min</th><th>Max</th><th>Min</th><th>Max</th></tr><tr><td>tSTAB_DBDATA</td><td>LRDIMM stabilization time for commands requiring use of data path</td><td>DCK_t/DCK_c stable</td><td>-</td><td>4.14</td><td>-</td><td>4.14</td><td>-</td><td>4.14</td><td>-</td><td>4.14</td><td>μs</td><td>4,5</td></tr><tr><td>tCKsk</td><td>Fractional Clock Output Skew</td><td></td><td>-</td><td>20</td><td>-</td><td>20</td><td>-</td><td>20</td><td>-</td><td>20</td><td>ps</td><td>6</td></tr><tr><td>tstaoff</td><td>Clock delay through the register between the input clock and output clock</td><td></td><td colspan="8">Min:(tPDM(min) + 1/2 tCK); Max: (tPDM(max) + 1/2 tCK)</td><td>ps</td><td>7,8</td></tr><tr><td>tdynoff</td><td>Maximum re-driven dynamic clock offset</td><td></td><td>-</td><td>40</td><td>-</td><td>35</td><td>-</td><td>35</td><td>-</td><td>35</td><td>ps</td><td>9,10</td></tr></table> NOTE 1 At Down-bin Data Rate (1980 MT/s ≤ f ≤ 2100 MT/s), the max value of tSTAB01 is 10μs. NOTE 2 This parameter applies to control word writes to RW05/RW06 and Bit OP0 in RW00. Since application of RW05/RW06 is postponed in the frequency change sequence defined in Section 5.4.2, a tSTAB01 waiting time is always sufficient after exit from Self Refresh with Clock Stop (i.e., tSTAB02 is not required). See timing diagrams in Figure 35 on page 48 and Figure 36 on page 49. NOTE 3 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface but does not require the use of data path (e.g., SRX, NOP, etc.). NOTE 4 Applies only to Self Refresh Exit without frequency change. NOTE 5 Measured from start of DCK after Exit from Self Refresh with Clock Stop to the first DCA Command that gets forwarded to the LRDIMM BCOM bus interface and requires the use of data path (e.g., WR, RD). NOTE 6 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCK_A_t/QCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCK_B_t/QCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only. NOTE 7 RW05/RW06 must be set to correct speed and the device must run at one of the speed nodes defined in Table 199 on page 191. See Figure 111 and Figure 112. NOTE 8 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock tPD. NOTE 9 VDD measurement DC bandwidth is limited to 20 MHz. NOTE 10 See Figure 111 and Figure 112.	Symbol	Parameter	Conditions	DDR5-3200		DDR5-3600		DDR5-4000/ 4400/4800		DDR5-5200/ 5600		Unit	Note	Min	Max	Min	Max	Min	Max	Min	Max	tSTAB_DBDATA	LRDIMM stabilization time for commands requiring use of data path	DCK_t/DCK_c stable	-	4.14	-	4.14	-	4.14	-	4.14	μs	4,5	tCKsk	Fractional Clock Output Skew		-	20	-	20	-	20	-	20	ps	6	tstaoff	Clock delay through the register between the input clock and output clock		Min:(tPDM(min) + 1/2 tCK); Max: (tPDM(max) + 1/2 tCK)								ps	7,8	tdynoff	Maximum re-driven dynamic clock offset		-	40	-	35	-	35	-	35	ps	9,10
Symbol	Parameter				Conditions	DDR5-3200		DDR5-3600		DDR5-4000/ 4400/4800		DDR5-5200/ 5600			Unit	Note																																																										
		Min	Max	Min		Max	Min	Max	Min	Max																																																																
tSTAB_DBDATA	LRDIMM stabilization time for commands requiring use of data path	DCK_t/DCK_c stable	-	4.14	-	4.14	-	4.14	-	4.14	μs	4,5																																																														
tCKsk	Fractional Clock Output Skew		-	20	-	20	-	20	-	20	ps	6																																																														
tstaoff	Clock delay through the register between the input clock and output clock		Min:(tPDM(min) + 1/2 tCK); Max: (tPDM(max) + 1/2 tCK)								ps	7,8																																																														
tdynoff	Maximum re-driven dynamic clock offset		-	40	-	35	-	35	-	35	ps	9,10																																																														
	Ex. CC-32 (JEDEC 82-512) at 240 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”). <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 250.																																																																									

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Claim Language

'407 Accused Products⁴

14.7 Output Clock Driver Characteristics (cont'd)

Table 233 — Output Clock Driver Characteristics at Test Mode (Frequency Band 2)

Symbol	Parameter	Conditions	Test Frequency (140 to 990 MHz)		Unit	Note
			Min	Max		
t _{STAB01}	RCD Stabilization time from start of DCK toggle to PLL lock	DCK_t/DCK_c stable	-	50	μs	
t _{STAB02}	RCD Stabilization time from control word write to PLL lock	DCK_t/DCK_c stable	-	50	μs	1
t _{CKsk}	Fractional Clock Output Skew		-	100	ps	2
t _{stao}	Clock delay through the register between the input clock and output clock		Min: (t _{PD} (min) + 1/2 tCK); Max: (t _{PD} (max) + 1/2 tCK)		ps	3,4
t _{dyno}	Maximum re-driven dynamic clock offset		-	500	ps	5,6

NOTE 1 This parameter applies to control word writes to [RW05](#) and Bit OP0 in [RW00](#).

NOTE 2 This skew represents the absolute output clock skew and contains the pad skew and package skew (See Figure 113). This parameter is specified for the clock pairs on each side of the register independently. The skew is applicable to left side clock pairs between QACK_A_t/QACK_A_c and QCCK_A_t/QCCK_A_c, as well as right side of the clock pairs between QACK_B_t/QACK_B_c and QCCK_B_t/QCCK_B_c. Similarly, to left side clock pairs between QBCK_A_t/QBCK_A_c and QDCK_A_t/QDCK_A_c, as well as right side of the clock pairs between QBCK_B_t/QBCK_B_c and QDCK_B_t/QDCK_B_c. This is not a tested parameter and has to be considered as a design goal only.

NOTE 3 [RW05](#) must not be set to enable PLL bypass mode and the device must run within the Test Frequency range defined in Table 199 on page 191. See Figure 111 and Figure 112.

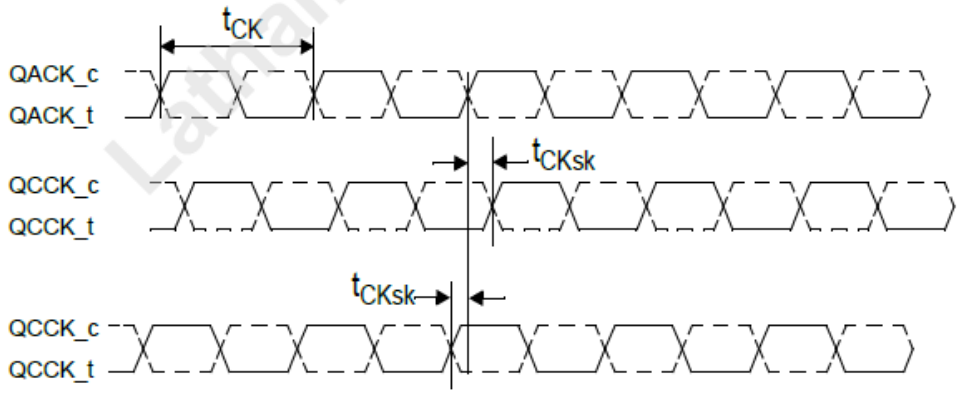
NOTE 4 This measures the delay from the rising differential input clock which samples incoming DCA to the rising differential output clock that will be used to sample the same QCA data. The parameter describes the maximum and minimum QxCK clock t_{PD}.

NOTE 5 V_{DD} measurement DC bandwidth is limited to 20 MHz.

NOTE 6 See Figure 111 and Figure 112.

Ex. CC-32 (JEDEC 82-512) at 241 (describing “output clock skew” that “is specified for the clock pairs on each side of the register independently”). *See also* Ex. CC-30 (JEDEC 82-513) at 251.

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Claim Language	'407 Accused Products ⁴
	 The diagram shows three sets of clock signals. The top set shows QACK_c (dashed line) and QACK_t (solid line) with a period t_{CK}. The middle set shows QCCK_c (dashed line) and QCCK_t (solid line) with a skew t_{CKsk} between them. The bottom set shows another instance of QCCK_c and QCCK_t with a skew t_{CKsk}. A vertical line marks a specific time point across all signals. Figure 113 — Clock Output (QxCK) Skew Ex. CC-32 (JEDEC 82-512) at 242. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 254.
[6.c.4] output the module C/A signals to the memory devices; and	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to output the module C/A signals to the memory devices. For example: <i>See supra</i> limitation [1.c.4].
[6.c.5] output the local clocks to the memory devices,	The '407 Accused Products comprise circuitry mounted on the PCB and configurable to output the local clocks to the memory devices. For example: The DDR5 RCD circuitry is configurable to output local clocks (QACK, QBCK, QCCK, and QDCK) to the memory devices.

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Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1.

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Claim Language	'407 Accused Products ⁴
	Host Side DRAM Side Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (showing generation of local clocks). To reduce the electrical load on the host memory controller's command, address, and control bus, Micron's RDIMMs and LRDIMMs utilize a DDR5 registering clock driver (RCD). The RCD presents a single load to the controller while redriving signals to the DDR5 SDRAM devices, which helps enable higher densities and increase signal integrity. The RCD also provides a low-jitter, low-skew PLL that redistributes a differential clock pair to multiple differential pairs of clock outputs. Ex. CC-03 (DDR5 RDIMM Datasheet) at 12 (stating the RCD provides multiple clock outputs).

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Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

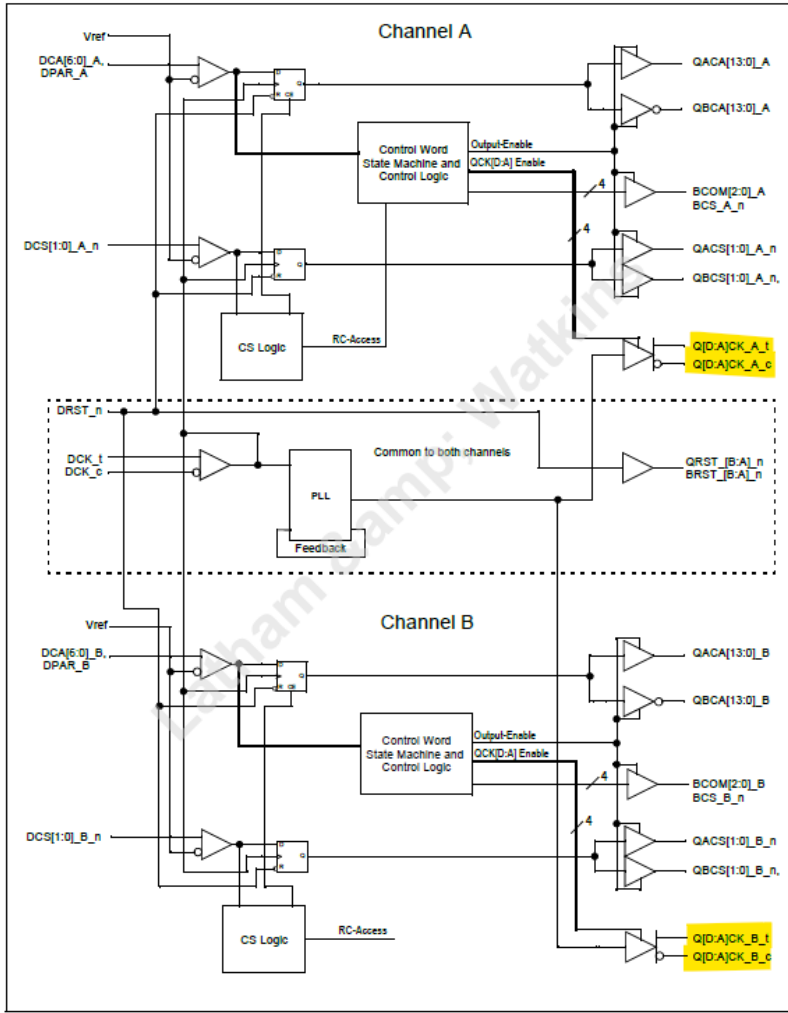
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴								
	Block Diagram The diagram shows a central rectangular block labeled 'RCD'. On the left side, there are four output arrows pointing left: 'BCOM_A BUS', 'Q[D:A]CK_A_t/c' (highlighted in yellow), 'QACA[13:0]_A', and 'QBCA[13:0]_A'. On the right side, there are four output arrows pointing right: 'BCOM_B BUS', 'Q[D:A]CK_B_t/c' (highlighted in yellow), 'QACA[13:0]_B', and 'QBCA[13:0]_B'. At the bottom, there are three input arrows pointing up into the block: 'DCA[6:0]_A', 'DCK_t/c', and 'DCA[6:0]_B'. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. <table><tr><td>Clock outputs</td><td>Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c</td><td>POD differential</td><td>cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.</td></tr><tr><td>Feedback</td><td>OT RD</td><td>POD</td><td>Feedback outputs to Host</td></tr></table> Ex. CC-32 (JEDEC 82-512) at 4 (showing the RCD includes clock outputs for the memory devices).	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.	Feedback	OT RD	POD	Feedback outputs to Host
Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	cycles (i.e., Deselect). Clock outputs to the DRAMs. Four copies per channel.						
Feedback	OT RD	POD	Feedback outputs to Host						

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Claim Language	'407 Accused Products ⁴
	3.1 Description The DDR5RCD02 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-32 (JEDEC 82-512) at 6.

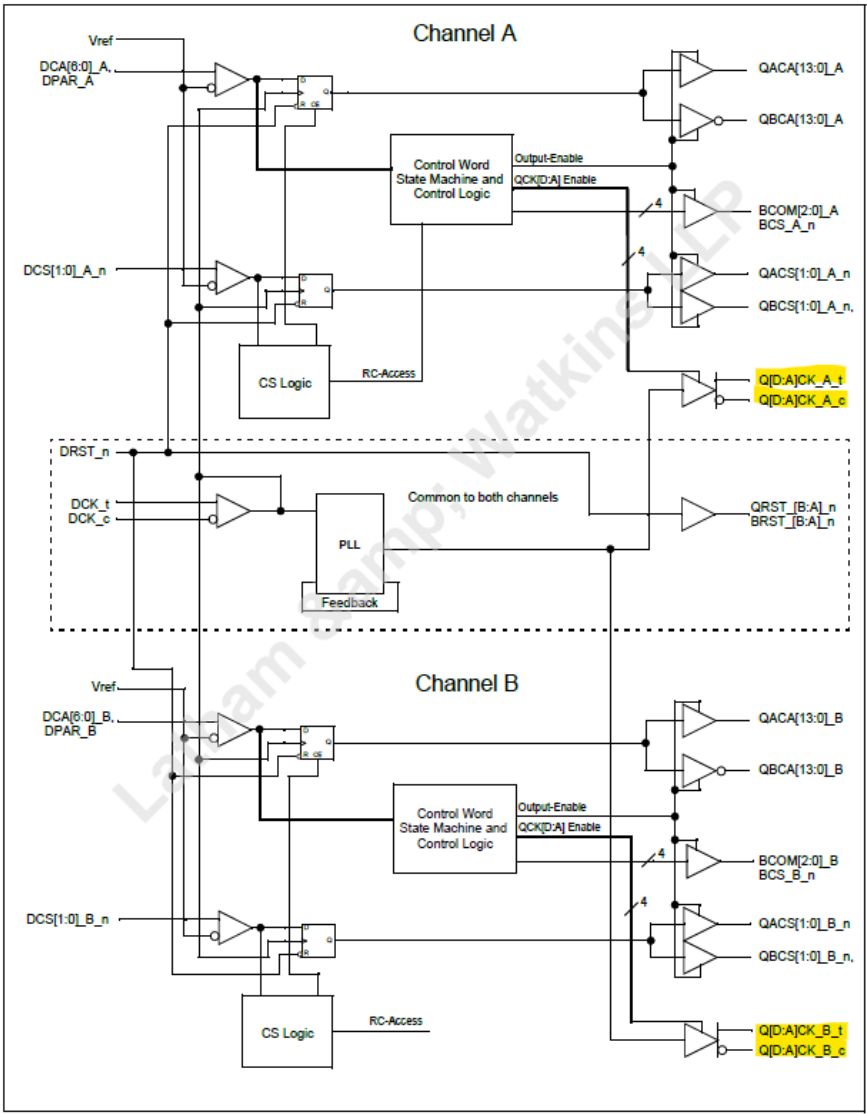
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	3.2 Logic Diagram Positive Logic  Ex. CC-32 (JEDEC 82-512) at 7.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴			
	Clock outputs	Q[D:A]CK_[B:A]_t Q[D:A]CK_[B:A]_c	POD differential	Clock outputs to the DRAMs. Four copies per channel.
	Ex. CC-30 (JEDEC 82-513) at 4 (showing the RCD includes clock outputs for the memory devices). 3.1 Description The DDR5RCD03 is a registering clock driver used on DDR5 RDIMMs and LRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the host controller and the DRAMs. It also creates a BCOM bus which controls the data buffers for LRDIMMs. It contains two separate channels which have some common logic such as clocking, but otherwise operate independently of each other. Each channel has a 7-bit double data rate CA bus input, a single parity input, two chip select inputs, and produces two copies of 14-bit single data rate CA bus outputs, and two copies of the chip select outputs. The RCD has a common clock input and PLL, but produces separate clock outputs to the DRAM channels. Ex. CC-30 (JEDEC 82-513) at 6.			

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Claim Language	'407 Accused Products ⁴
	 The diagram illustrates a dual-channel PLL system, specifically Channel A and Channel B. Channel A includes inputs for Vref, DCA[6:0]_A, DPAR_A, DCS[1:0]_A_n, and DCK_t/DCK_c. It features a PLL block with feedback, CS Logic, and a Control Word State Machine and Control Logic block. The outputs for Channel A include QACA[13:0]_A, QBCA[13:0]_A, BCOM[2:0]_A, BCS_A_n, QACS[1:0]_A_n, QBCS[1:0]_A_n, and Q[D:A]CK_A_t/Q[D:A]CK_A_c. Channel B has similar inputs and outputs, with Vref, DCA[6:0]_B, DPAR_B, DCS[1:0]_B_n, and DCK_t/DCK_c. The outputs for Channel B include QACA[13:0]_B, QBCA[13:0]_B, BCOM[2:0]_B, BCS_B_n, QACS[1:0]_B_n, QBCS[1:0]_B_n, and Q[D:A]CK_B_t/Q[D:A]CK_B_c. A common section for both channels includes DRST_n, DCK_t/DCK_c, and a PLL block with feedback, leading to QNST[B:A]_n and BRST[B:A]_n outputs. The diagram is labeled 'Ex. CC-30 (JEDEC 82-513) at 7.'

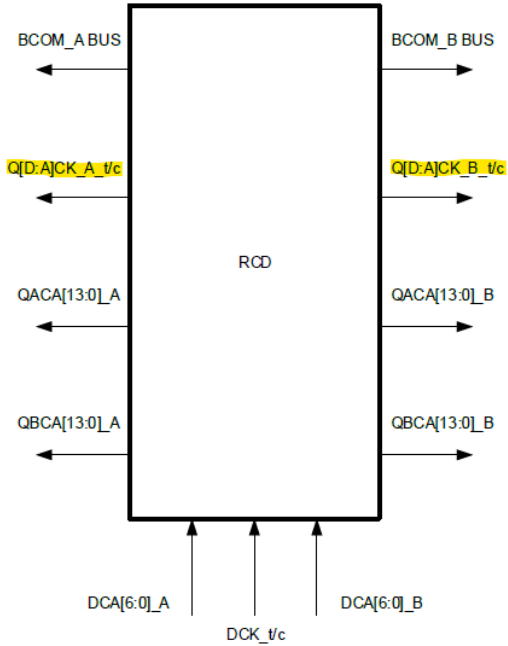
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
[6.c.5.i] wherein the first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, the second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices, and	In the '407 Accused Products, a first local clock is output to the first memory devices and not to any of the memory devices other than the first memory devices, and a second local clock is output to the second memory devices and not to any of the memory devices other than the second memory devices. For example: A first local clock generated by the RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 6.c.3.i and 6.c.3.ii above) is output to first memory devices in a rank and not to any other memory devices. A second local clock generated by the RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 6.c.3.i and 6.c.3.ii above) is output to second memory devices in a rank and not to any other memory devices. <i>See, e.g.,</i>
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1 (showing the RCD circuitry outputs each local clock to a specific group of devices and not to any others).

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Description The DDR5 MRCD is a registering clock driver used on DDR5 MRDIMMs. Its primary function is to buffer the Command/Address (CA) bus, chip selects, and clock between the Host Controller and the DRAMs. It also creates a BCOM bus which control the data buffers. The RG5R188B0AIGBY contains two separate channels with some common logic such as clocking, but which otherwise operate independently of each other. For each channel, the Host interface is a 7-bit + parity CA bus with two chip-select inputs. It is double data rate, carries two multiplexed pseudo-channels which can operate in the same direction simultaneously, and runs at twice the frequency of the DRAM interfaces. The MRCD demultiplexes the pseudo-channels and transmits each to a separate DRAM output interface. Each DRAM interface is a 14-bit single data rate CA bus output with a chip-select. The MRCD also transcodes commands from the CA bus onto a per-channel BCOM bus for Data Buffer control. The RG5R188B0AIGBY has a common clock input and PLL, but produces 4 separate clock pairs to the DRAM channels.

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	Block Diagram  The diagram shows a central rectangular block labeled 'RCD'. It has several input and output lines. On the left side, from top to bottom, there are: an output line labeled 'BCOM_A BUS' with an arrow pointing left; an input line labeled 'Q[0:A]CK_A_1/c' with an arrow pointing right; an input line labeled 'QACA[13:0]_A' with an arrow pointing right; an input line labeled 'QBCA[13:0]_A' with an arrow pointing right; and three input lines at the bottom labeled 'DCA[6:0]_A', 'DCK_1/c', and 'DCA[6:0]_B' with arrows pointing up. On the right side, from top to bottom, there are: an output line labeled 'BCOM_B BUS' with an arrow pointing right; an output line labeled 'Q[0:A]CK_B_1/c' with an arrow pointing right; an output line labeled 'QACA[13:0]_B' with an arrow pointing right; an output line labeled 'QBCA[13:0]_B' with an arrow pointing right; and no lines at the bottom. Ex. CC-21, RG5R188B01AIGBY Short-Form Datasheet, Renesas, at 1. The JEDEC DDR5 RDIMM standard further demonstrates the local clocks generated by the Accused Products' RCD with a programmable phase relationship with the system clock (<i>see</i> limitations 1.c.3, 1.c.3.i, and 1.c.3.ii above) are output to a corresponding group of the memory devices and not to any other group of the memory devices.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

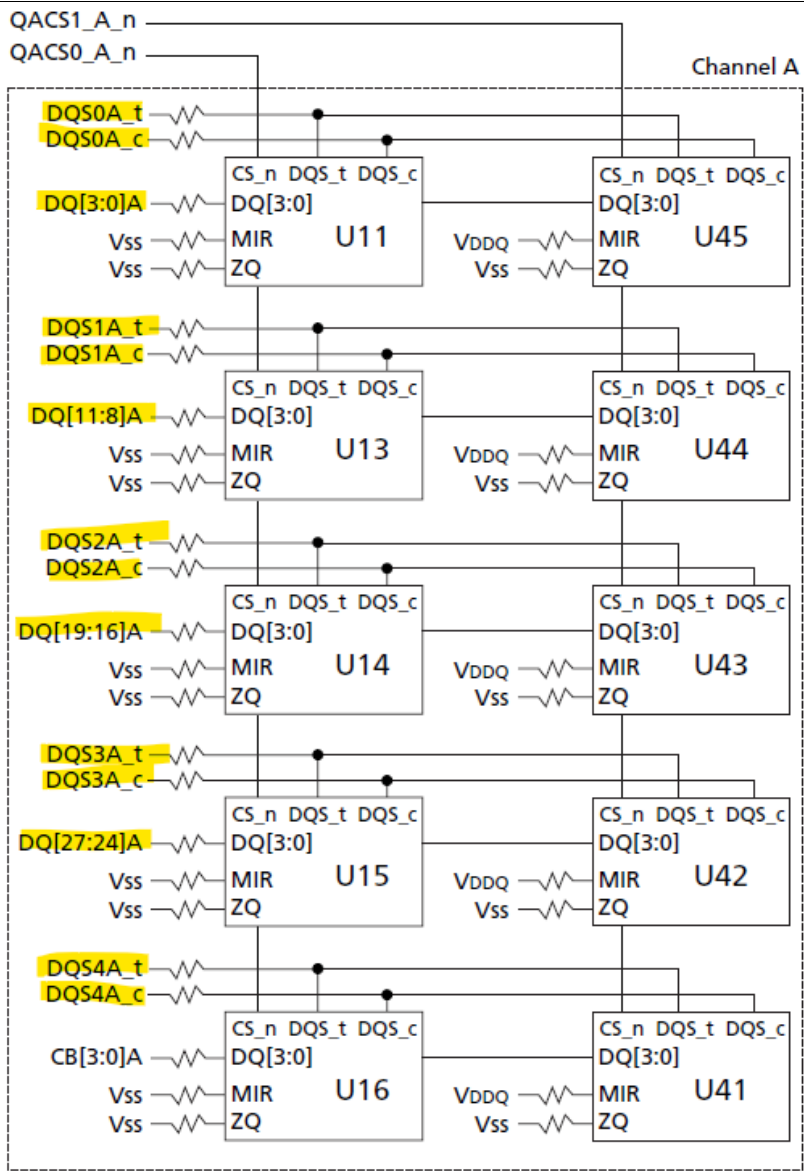
2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

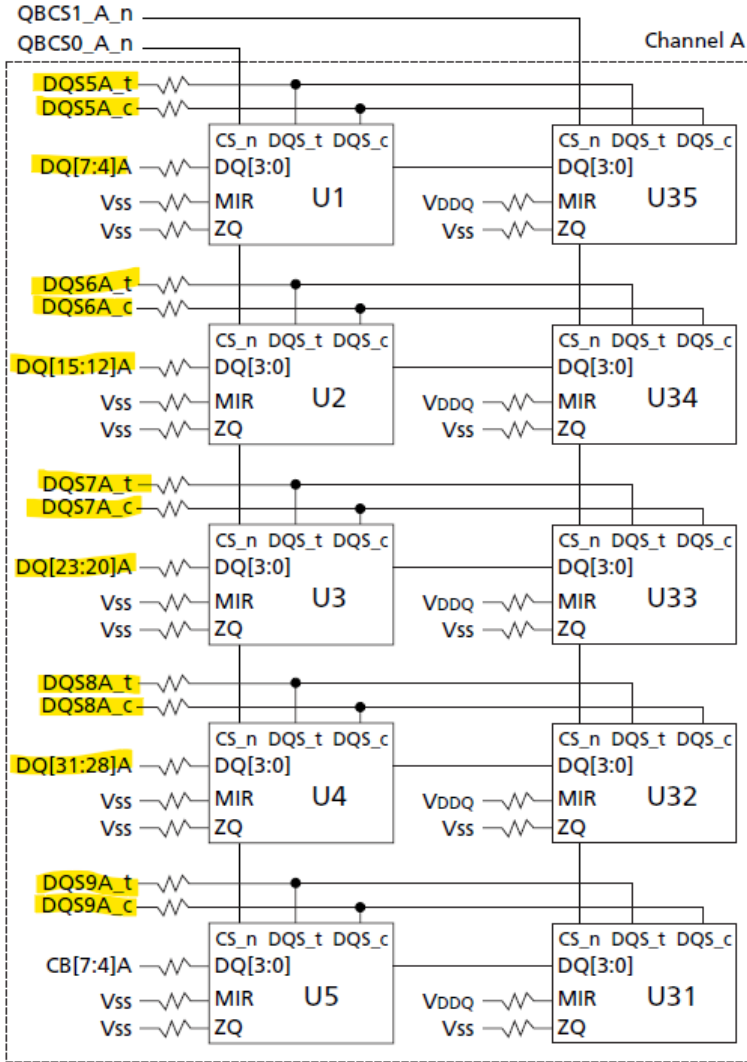
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴																																																																																																																																																														
	6.9.1 DIMM Connectivity Wiring (cont'd) <table><tr><th colspan="6">1Rx8 (less than 8000 Mbps)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QACK</td><td>QACS0</td><td>QACA</td><td>0</td></tr></table><table><tr><th colspan="6">1Rx8 (8000 Mbps and Faster)</th></tr><tr><th>DIMM Side</th><th>Row</th><th>Clock</th><th>CS</th><th>CA</th><th>Rank</th></tr><tr><td>Top</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr><tr><td>Bottom</td><td>Lower</td><td>QCCK</td><td>QACS0</td><td>QACA</td><td>0</td></tr></table><table><tr><th colspan="11">1Rx8 Sub Channel 0 (Channel A)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td colspan="2"></td><td colspan="2">[15:8]</td><td colspan="2"></td><td colspan="2">[31:24]</td><td colspan="2"></td></tr><tr><td>Bottom</td><td colspan="2">[7:0]</td><td colspan="2"></td><td colspan="2">[23:16]</td><td colspan="2"></td><td colspan="2">CB[7:0]</td></tr></table><table><tr><th colspan="11">1Rx8 Sub Channel 1 (Channel B)</th></tr><tr><th>Side</th><th>DQS0</th><th>DQS5</th><th>DQS1</th><th>DQS6</th><th>DQS2</th><th>DQS7</th><th>DQS3</th><th>DQS8</th><th>DQS4</th><th>DQS9</th></tr><tr><td></td><td>Nib 0</td><td>Nib 1</td><td>Nib 2</td><td>Nib 3</td><td>Nib 4</td><td>Nib 5</td><td>Nib 6</td><td>Nib 7</td><td>Nib 8</td><td>Nib 9</td></tr><tr><td>Top</td><td colspan="2">[7:0]</td><td colspan="2"></td><td colspan="2">[23:16]</td><td colspan="2"></td><td colspan="2"></td></tr><tr><td>Bottom</td><td colspan="2"></td><td colspan="2">[15:8]</td><td colspan="2"></td><td colspan="2">[31:24]</td><td colspan="2">CB[7:0]</td></tr></table> Figure 12 — DIMM Connectivity Wiring Ex. CC-88 (JEDEC 305B) at 31-32 (showing local clocks, e.g., QACK, QBCK, QCCK, QDCK, being output to certain groups of memory devices but not others).	1Rx8 (less than 8000 Mbps)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QACK	QACS0	QACA	0	Bottom	Lower	QACK	QACS0	QACA	0	1Rx8 (8000 Mbps and Faster)						DIMM Side	Row	Clock	CS	CA	Rank	Top	Lower	QCCK	QACS0	QACA	0	Bottom	Lower	QCCK	QACS0	QACA	0	1Rx8 Sub Channel 0 (Channel A)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top			[15:8]				[31:24]				Bottom	[7:0]				[23:16]				CB[7:0]		1Rx8 Sub Channel 1 (Channel B)											Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9		Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9	Top	[7:0]				[23:16]						Bottom			[15:8]				[31:24]		CB[7:0]	
1Rx8 (less than 8000 Mbps)																																																																																																																																																															
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Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																					
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																					
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Bottom	[7:0]				[23:16]				CB[7:0]																																																																																																																																																						
1Rx8 Sub Channel 1 (Channel B)																																																																																																																																																															
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9																																																																																																																																																					
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9																																																																																																																																																					
Top	[7:0]				[23:16]																																																																																																																																																										
Bottom			[15:8]				[31:24]		CB[7:0]																																																																																																																																																						
[6.c.5.ii] wherein the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines. For example:	In the '407 Accused Products, the first memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the first local clock via a first subset of the data/strobe signal lines, and the second subset of the memory devices are configurable to perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines. For example: For example, first memory devices associated with channel A are configurable to perform memory read and write operations by communicating data/strobe signals via a subset of data/strobe signal lines (e.g., DQ[]A, DQS[]_A_t, DQS[]_A_c). See, e.g.:																																																																																																																																																														

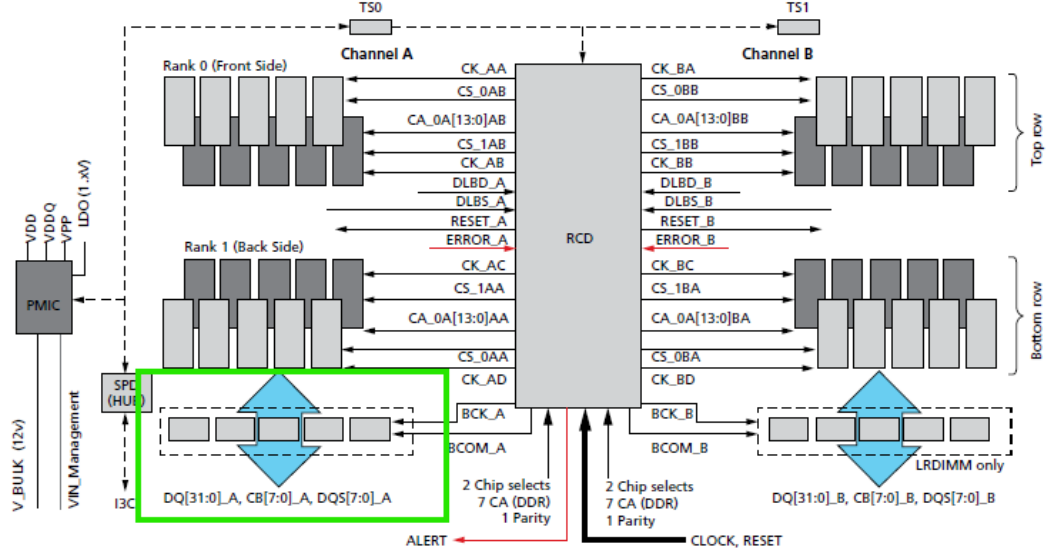
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
perform memory read or write operations by communicating data/strobe signals in accordance with the second local clock via a second subset of the data/strobe signal lines.	 The diagram illustrates the internal structure of Channel A, which is a memory array. It shows four banks of memory cells, each connected to a local clock (DQS) and data/strobe signals (DQ). The memory cells are organized into four banks, each with a local clock (DQS) and data/strobe signals (DQ). The sense amplifiers (U41-U45) are connected to the data/strobe signals and provide feedback to the memory array. The diagram also shows the connection of the second local clock (DQS) to the memory array and sense amplifiers.

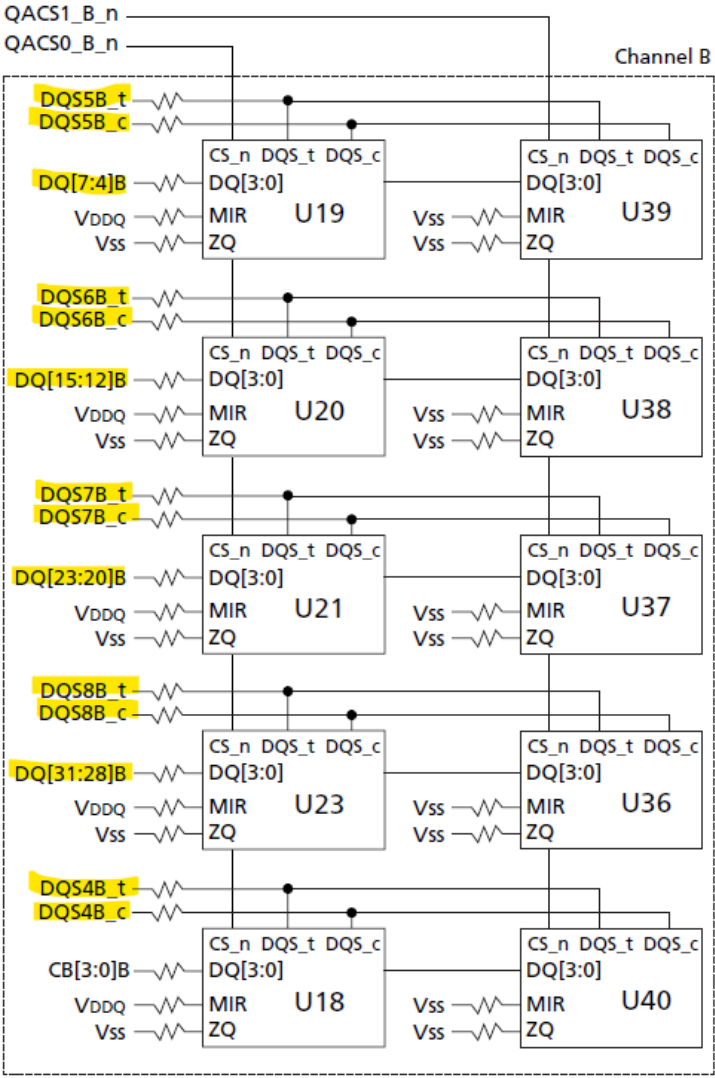
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 The diagram illustrates the internal structure of Channel A, which is part of a memory module. It shows five pairs of memory modules, labeled U1 through U5 on the left and U35 through U31 on the right. Each module is connected to a common data bus. The data signals are labeled DQ[7:4]A, DQ[15:12]A, DQ[23:20]A, DQ[31:28]A, and DQ[39:36]A. Each module has a chip select (CS_n), data strobe (DQS_t, DQS_c), data (DQ[3:0]), mirror (MIR), and zero-queue (ZQ) pins. Power pins Vss and VDDQ are also shown. The modules are connected to a common data bus through a series of multiplexers or switches. Ex. CC-07 (DDR5 RDIMM Addendum) at 8 (also showing similar for data signals DQ[JB]).

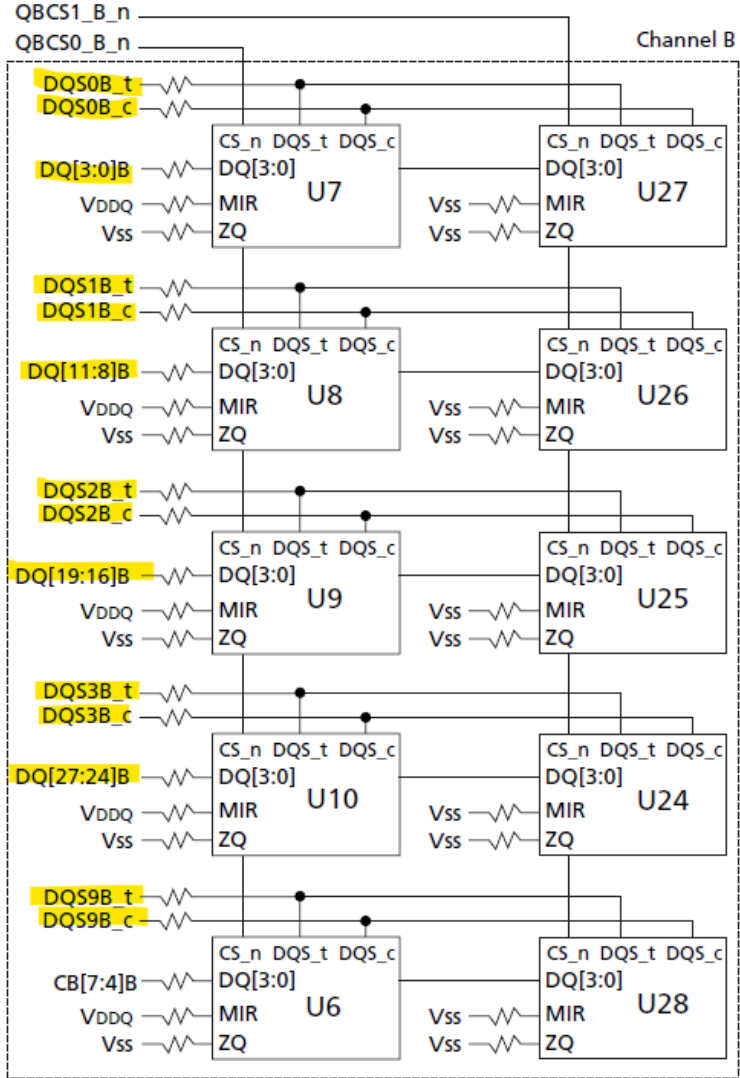
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram  Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. Furthermore, second memory devices associated with channel B are configurable to perform memory read and write operations by communicating data/strobe signals via a subset of data/strobe signal lines (e.g., DQ[]B, DQS[]_B_t, DQS[]_B_c). <i>See, e.g.:</i>

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Claim Language	'407 Accused Products ⁴
	 The diagram illustrates the internal structure of Channel B, which consists of five pairs of memory blocks (U19/U39, U20/U38, U21/U37, U23/U36, U18/U40). Each block is connected to a common data bus (DQ[3:0]) and has its own control signals (CS_n, DQS_t, DQS_c, MIR, ZQ, VDDQ, Vss). The blocks are organized into five groups, each receiving a different set of data signals (DQ[7:4]B, DQ[15:12]B, DQ[23:20]B, DQ[31:28]B, CB[3:0]B) and a common clock signal (DQS5B_t/c, DQS6B_t/c, DQS7B_t/c, DQS8B_t/c, DQS4B_t/c). The diagram also shows the connection of the blocks to the QACS1_B_n and QACS0_B_n signals.

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Claim Language	'407 Accused Products ⁴
	 Channel B QBCS1_B_n QBCS0_B_n DQS0B_t DQS0B_c DQ[3:0]B VDDQ Vss CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U7 CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U27 DQS1B_t DQS1B_c DQ[11:8]B VDDQ Vss CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U8 CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U26 DQS2B_t DQS2B_c DQ[19:16]B VDDQ Vss CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U9 CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U25 DQS3B_t DQS3B_c DQ[27:24]B VDDQ Vss CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U10 CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U24 DQS9B_t DQS9B_c CB[7:4]B VDDQ Vss CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U6 CS_n DQS_t DQS_c DQ[3:0] MIR ZQ U28 Ex. CC-07 (DDR5 RDIMM Addendum) at 8.

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Claim Language	'407 Accused Products ⁴
	Figure 1: DDR5 LRDIMM/RDIMM Functional Block Diagram Note: 1. The above illustrates a dual-rank x80 LRDIMM/RDIMM with DRAM, PMIC, RCD, SPD, temp sensors and data buffers. Ex. CC-03 (DDR5 RDIMM Datasheet) at 1. Furthermore, data is read or written in memory devices in accordance with the respective local clock. The JEDEC DDR5 RDIMM standard further demonstrates groups of memory devices perform memory read and write operations by communicating data signals with the memory controller via corresponding subsets of data signal lines (e.g., DQS[]).

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Claim Language

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6.9.1 DIMM Connectivity Wiring

2Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Top	Upper	QBCK	QBCS0	QBCA	0
Bottom	Lower	QCCK	QACS1	QACA	1
Bottom	Upper	QDCK	QBCS1	QBCA	1

2Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Top		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]
Bottom	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

2Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Top	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]
Bottom		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

1Rx4					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QBCS0	QACA	0

1Rx4 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top	[3:0]		[11:8]		[19:16]		[27:24]		CB[3:0]	
Bottom		[7:4]		[15:12]		[23:20]		[31:28]		CB[7:4]

1Rx4 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:4]		[15:12]		[23:20]		[31:28]	CB[3:0]	
Bottom	[3:0]		[11:8]		[19:16]		[27:24]			CB[7:4]

2Rx8					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS1	QACA	1

2Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

2Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]
Bottom		[7:0]		[15:8]		[23:16]		[31:24]		CB[7:0]

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Claim Language

'407 Accused Products⁴

6.9.1 DIMM Connectivity Wiring (cont'd)

1Rx8 (less than 8000 Mbps)					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QACK	QACS0	QACA	0
Bottom	Lower	QACK	QACS0	QACA	0

1Rx8 (8000 Mbps and Faster)					
DIMM Side	Row	Clock	CS	CA	Rank
Top	Lower	QCCK	QACS0	QACA	0
Bottom	Lower	QCCK	QACS0	QACA	0

1Rx8 Sub Channel 0 (Channel A)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top				[15:8]				[31:24]		
Bottom		[7:0]			[23:16]					CB[7:0]

1Rx8 Sub Channel 1 (Channel B)										
Side	DQS0	DQS5	DQS1	DQS6	DQS2	DQS7	DQS3	DQS8	DQS4	DQS9
	Nib 0	Nib 1	Nib 2	Nib 3	Nib 4	Nib 5	Nib 6	Nib 7	Nib 8	Nib 9
Top		[7:0]			[23:16]			[31:24]		
Bottom			[15:8]							CB[7:0]

Figure 12 — DIMM Connectivity Wiring

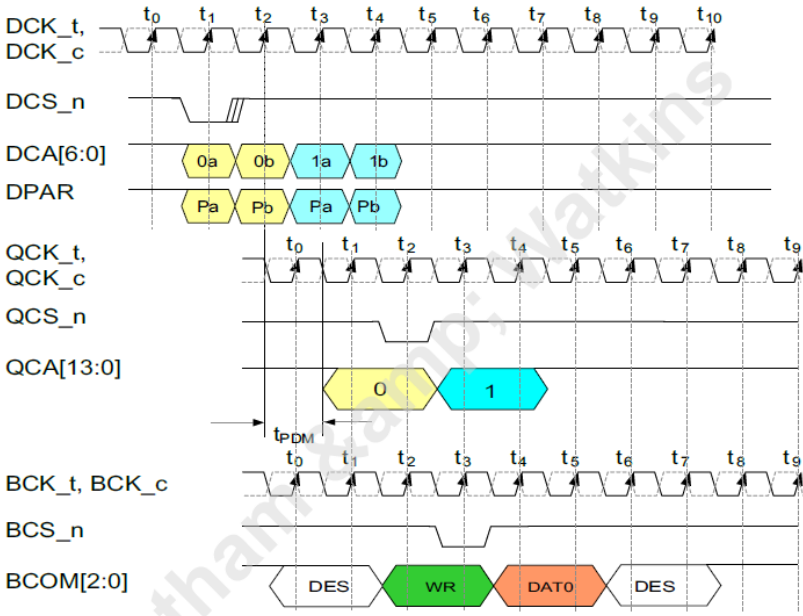
Ex. CC-88 (JEDEC 305B) at 31-32 (showing memory devices receiving data from the memory controller via subsets of data signal lines, e.g., [3:0], [11:8], [15:12]).

The JEDEC DDR5 RCD definition further demonstrates the chip select signal (QCS) and command and address signal (QCA) received by each DDR memory device is synchronized with the local clock (QCK):

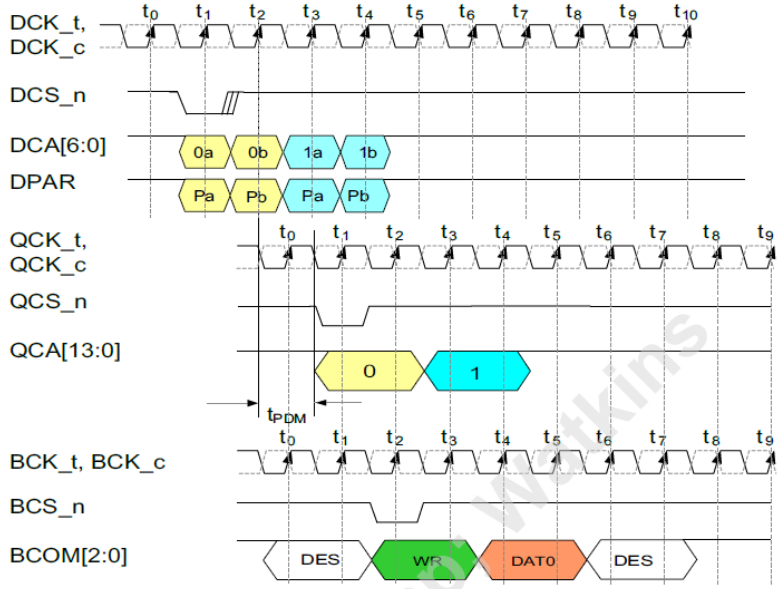
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	3.4.1 Double Data Rate DCA (cont'd) NOTE Timing Diagram for 0 nCK command latency adder. Figure 7 — One UI DRAM Command Timing Diagram NOTE Timing Diagram for 0 nCK command latency adder. Figure 8 — Two UI DRAM Command Timing Diagram Ex. CC-32 (JEDEC 82-512) at 16. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 16.

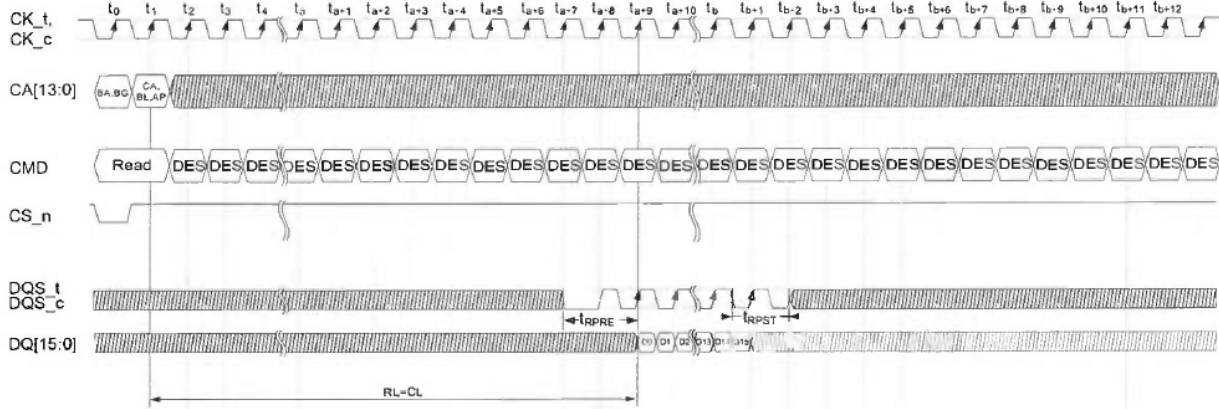
U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	The JEDEC DDR5 RCD definition explains the Accused Products are configurable to perform READ and WRITE operations synchronized with the local clock QCK. <i>See, e.g.,</i>:  Figure 11 — SDR1 Mode Example of DRAM WR Command with RW00[1] = 0

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	 Figure 12 — SDR2 Mode Example of DRAM WR Command with RW00[1] = 1 Ex. CC-32 (JEDEC 82-512) at 18-19. <i>See also</i> Ex. CC-30 (JEDEC 82-513) at 18-19. Further, the timing diagrams for read and write operations show that DQ signals appear after a set latency in terms of clock cycles after the module C/A signals are received. As another example, DQ is output/received in relation to DQS, and DQS is aligned with clock signals in predetermined manner. So the DQ output/input is in accordance with local clock signals.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	4.7 Read Operation The Read Operation causes the DRAM to retrieve and output data stored in its array. The Read Operation is initiated by the Read Command during which the beginning column address and bank/group address for the data to be retrieved from the array is provided. The data is driven by the DRAM on its DQ pins RL (CL) cycles after the Read Command along with the proper waveform on the DQS inputs. Read Latency (RL or CL) is defined from the Read command to data and is not affected by the Read DQS offset timing (MR40 OP[2:0]). 4.7.1 READ Burst Operation During a READ or WRITE command, DDR5 shall support BC8, BL16, BL32 (optional) and BL32 OTF (optional) during the READ or WRITE. MR0[1:0] is used to select burst operation mode.  NOTES: 1. BL=16, Preamble = 2tCK - 0010 Pattern Preamble, 1.5tCK Postamble 2. DES commands are shown for ease of illustration; other commands may be valid at these times. 3. In this example, Read DQS Offset Timing is set to 0 Clocks. Figure 35 — READ Burst Operation (BL16) Ex. CC-05 (JESD79-5) at 135.

U.S. Patent No. 12,675,407 Infringement Claim Chart

Claim Language	'407 Accused Products ⁴
	4.8.2 Write Burst Operation The following write timing diagrams are to help understand the meaning of each write parameter; the diagrams are just examples. The details of each parameter are defined separately. In these write timing diagrams, for clarity of illustration, CK and DQS are shown aligned. As well, DQS and DQ are shown center-aligned. Offset between CK and DQS, and between DQS and DQ may be appropriate. NOTES: 1. BL=16, 2tCK Preamble, 1.5tCK Postamble 2. DES commands are shown for ease of illustration; other commands may be valid at these times. 3. The DQ signal is shown as "Don't Care" before the first Write data bit indicating DFE is disabled. When DFE is enabled, the DQ signal shall be high for a minimum of 4UI prior to the first Write data bit for proper DFE synchronization. Figure 49 — WRITE Burst Operation (BL16)

Ex. CC-05 (JESD79-5) at 147.